From Perception to Action

World Models Learning for Generalist Agents

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CMSC 848n

What Makes Good Foundation Models for Robotics?

A Wishlist

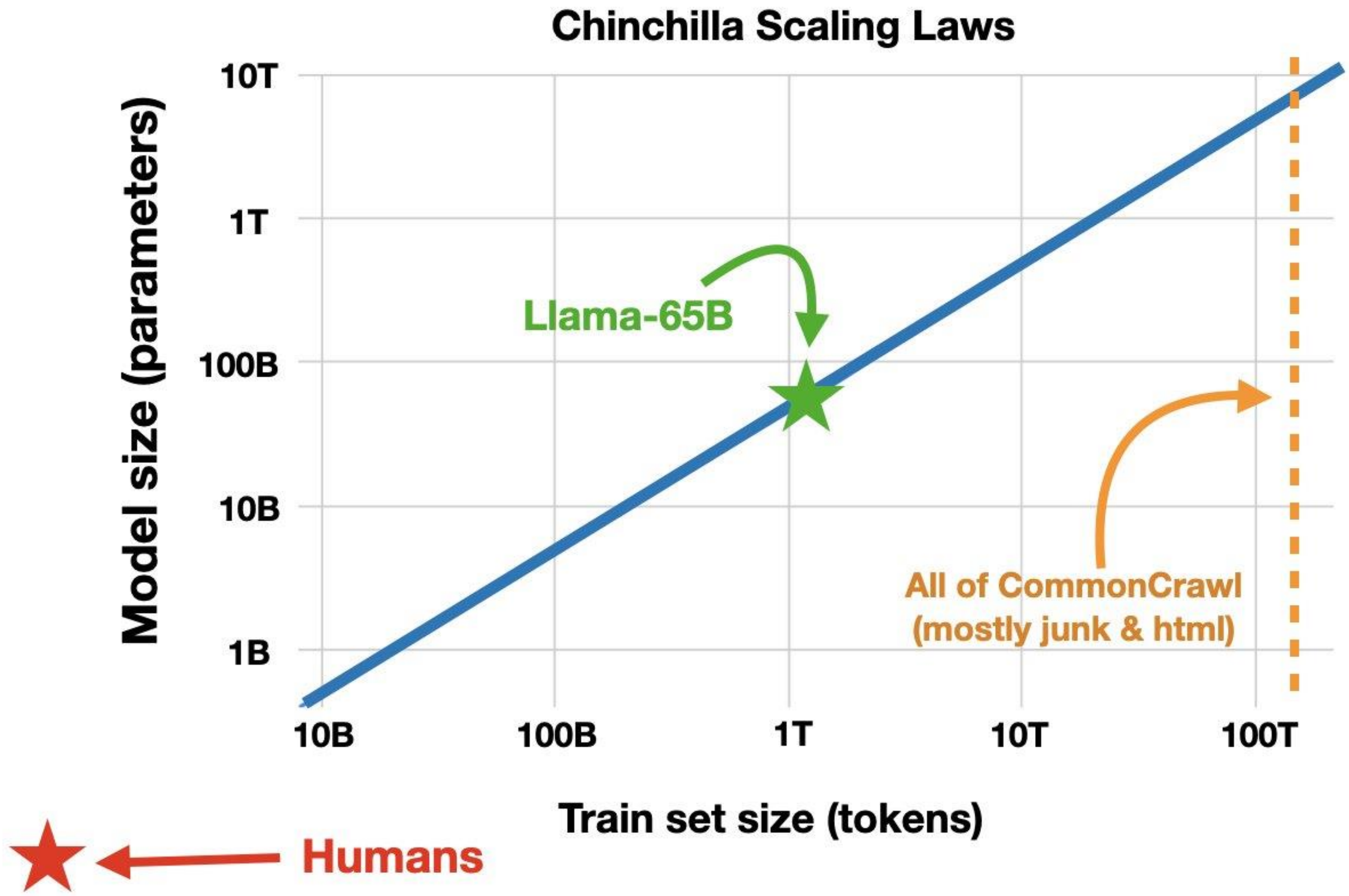
- **Versatile**: Diverse downstream tasks & algorithms ?
- **Generalizable**: **Unseen** tasks & domains ?
- **Resilient**: **Low-quality** training data ?

Do you have enough data?

Is next-token prediction all you need?

Is scaling up all you need?

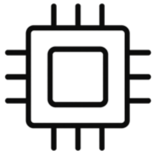
Data Are Not Efficiently Used



Llama 2 was trained on 1.4 trillion tokens, **3,800 times** more tokens than a human has verbally exchanged at age 50 (370M words).

Will We Ever Run Out of Data?

Pre-training at Internet Scale Is Hitting a Data Wall

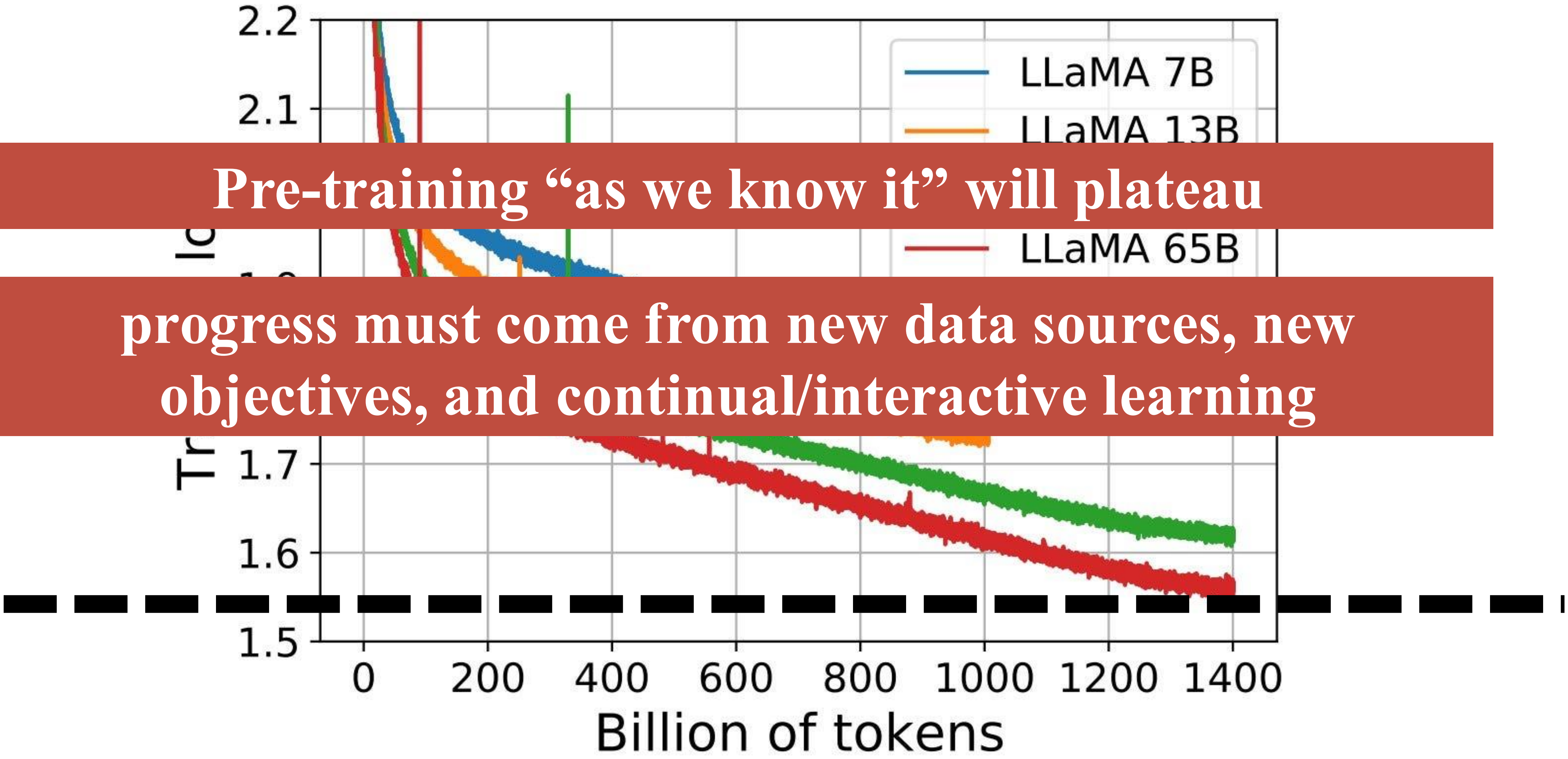


Compute keeps scaling

- Better hardware
- Better algorithms
- Larger clusters

Data is not

- We still have one public internet
- The internet is the **fossil fuel** of LLMs — finite and increasingly mined



Pre-training "as we know it" will plateau

progress must come from new data sources, new objectives, and continual/interactive learning

Will We Ever Run Out of Data?

Source: Mark Cummins on X

Recent LLM	Training Set (Tokens)	Relative Size (Llama 3 =1)
Llama 3	15T	1
GPT-4	6.5T	0.5

Web data	Training Set (Tokens)	Relative Size (Llama 3 =1)
FineWeb	15T	1
Non-English web data	18T	1

It is indeed happening now!

Embodied AI's Core Bottleneck: Human Fuel

<u>Human fuel</u> doesn't scale: < 24 hours of supervision per robot per person per day

Credit to Lu et al. Mobile-TeleVision

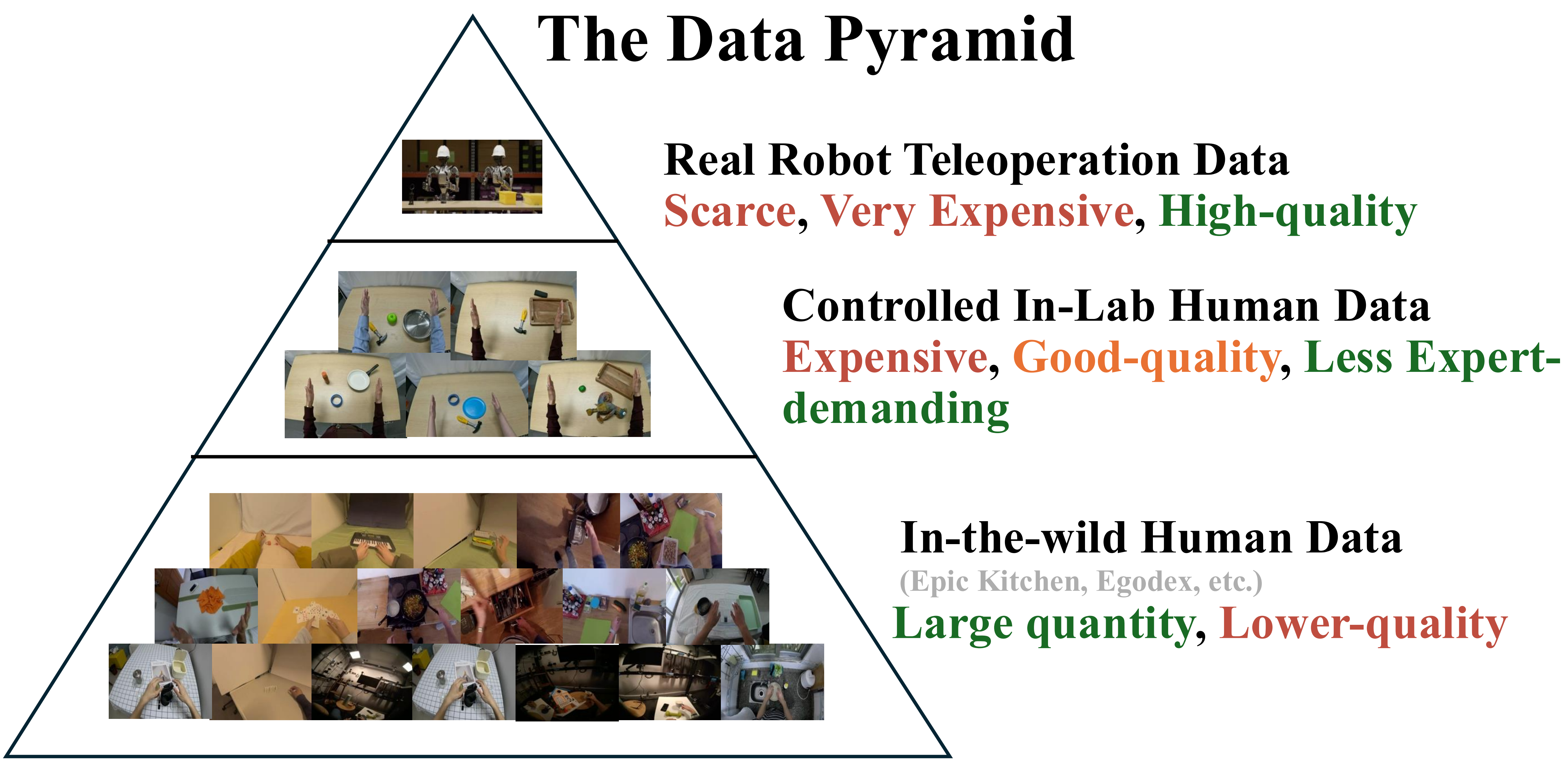
Credit to Li et al. AMO

- Teleoperation + corrective feedback is the dominant data source today
- Collection and annotation are bounded by human time, attention, & safety

Challenge in Embodied AI / Robotics

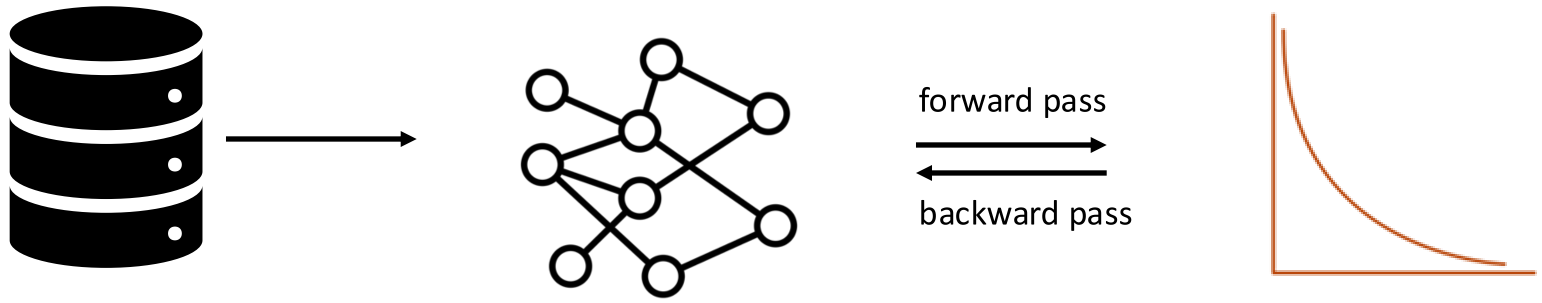
Cross-Embodiment Learning

We are still in a <u>Small Data Regime</u> for Robotics, given so many variations


The Data Pyramid
Real Robot Teleoperation Data
Scarce, Very Expensive, High-quality
Controlled In-Lab Human Data
Expensive, Good-quality, Less Expert-demanding
In-the-wild Human Data
(Epic Kitchen, Egodex, etc.)
Large quantity, Lower-quality

Foundation Models for Robotics

➢ **Pretraining/Learning Objective?**



Data Collected from **Diverse** Tasks & Domains

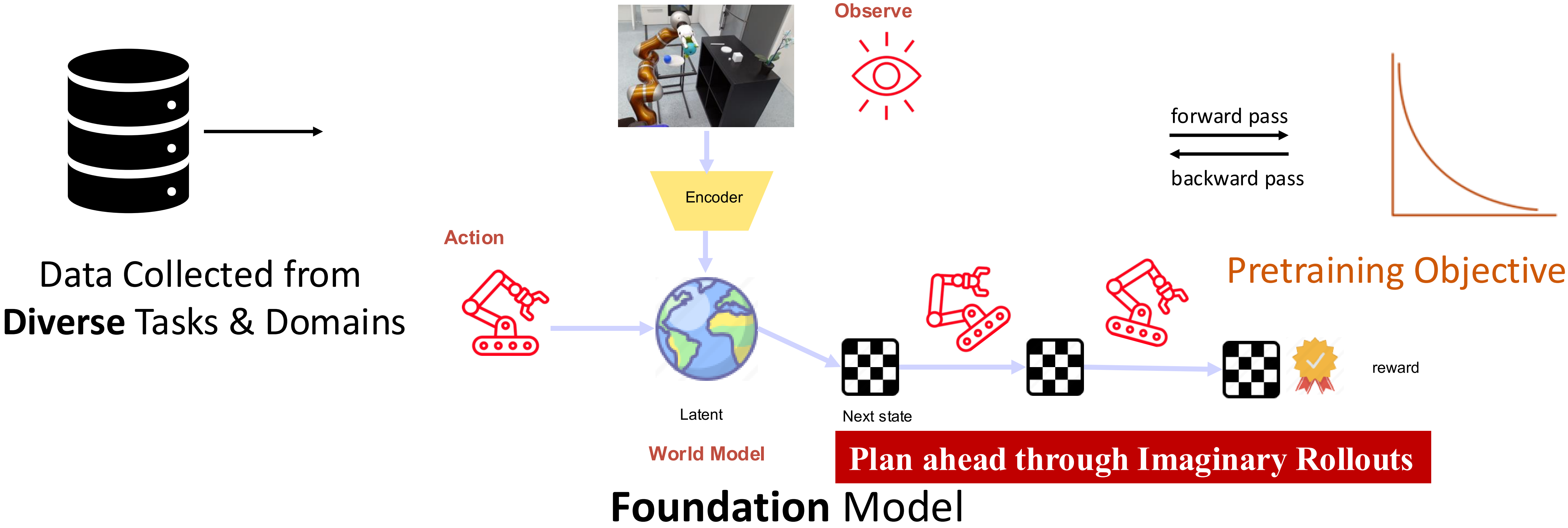
Foundation Model

Pretraining Objective

Foundation Models for Robotics

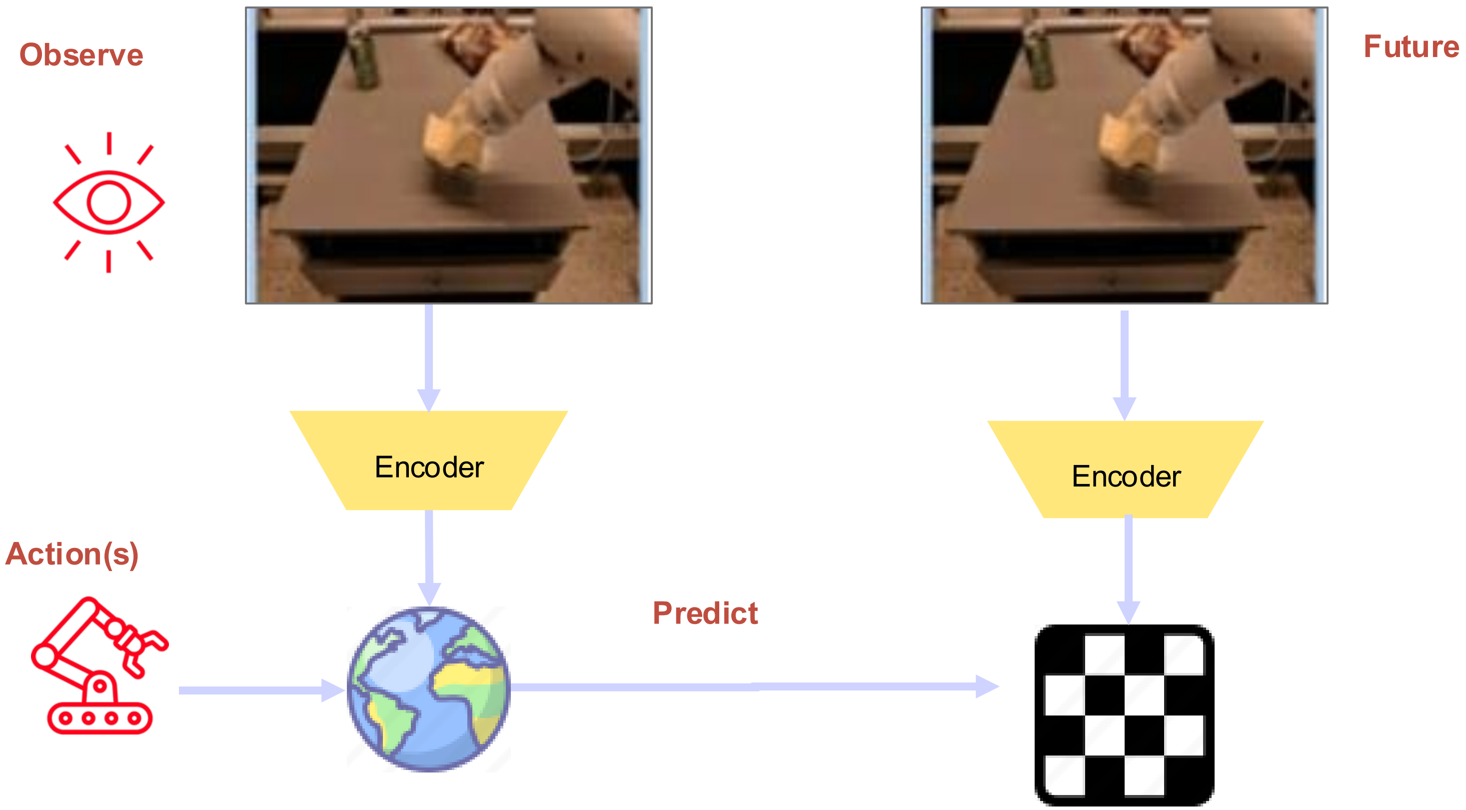
➢ Pretraining/Learning Objective?

Training Objective that Learns a Good Latent World-Models!

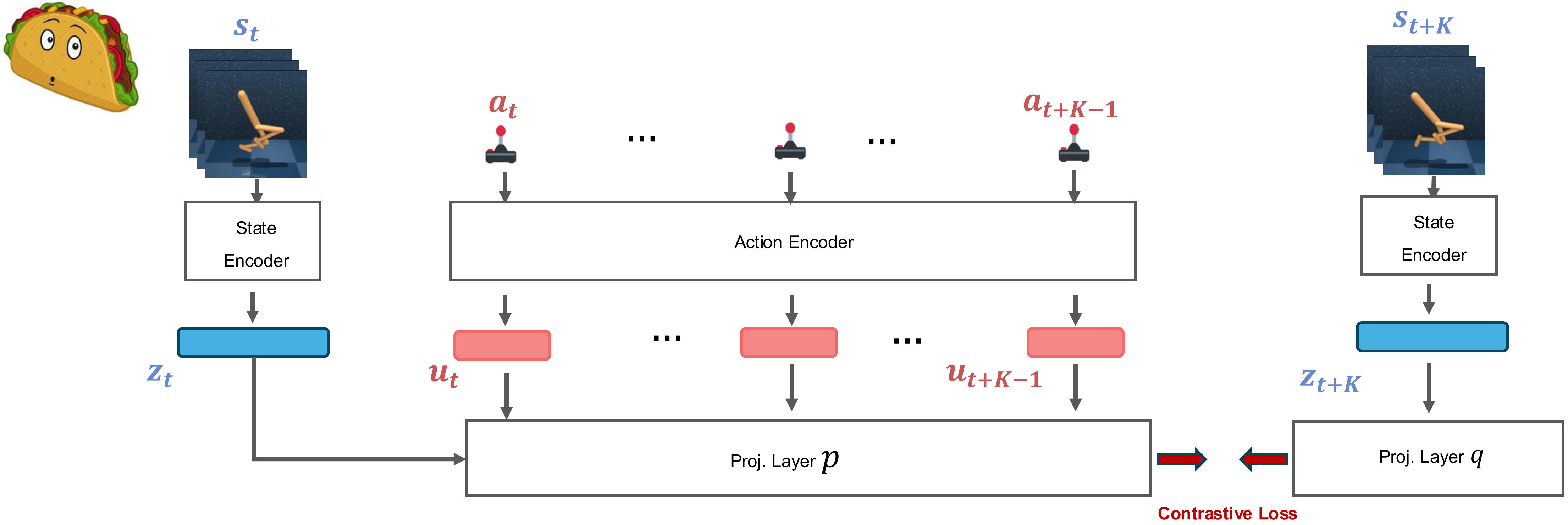


Foundation Model for Robotics

(Implicit) Latent World Model



TACO: Temporal Latent Action-Driven Contrastive Loss



Predicting observation latent K steps into the future
via contrastive loss

Zheng, Wang, Sun, Ma, Zhao, Xu, Daume, H.. "TACO: Temporal Latent Action-Driven Contrastive Loss for Visual Reinforcement Learning." NeurIPS 2023.

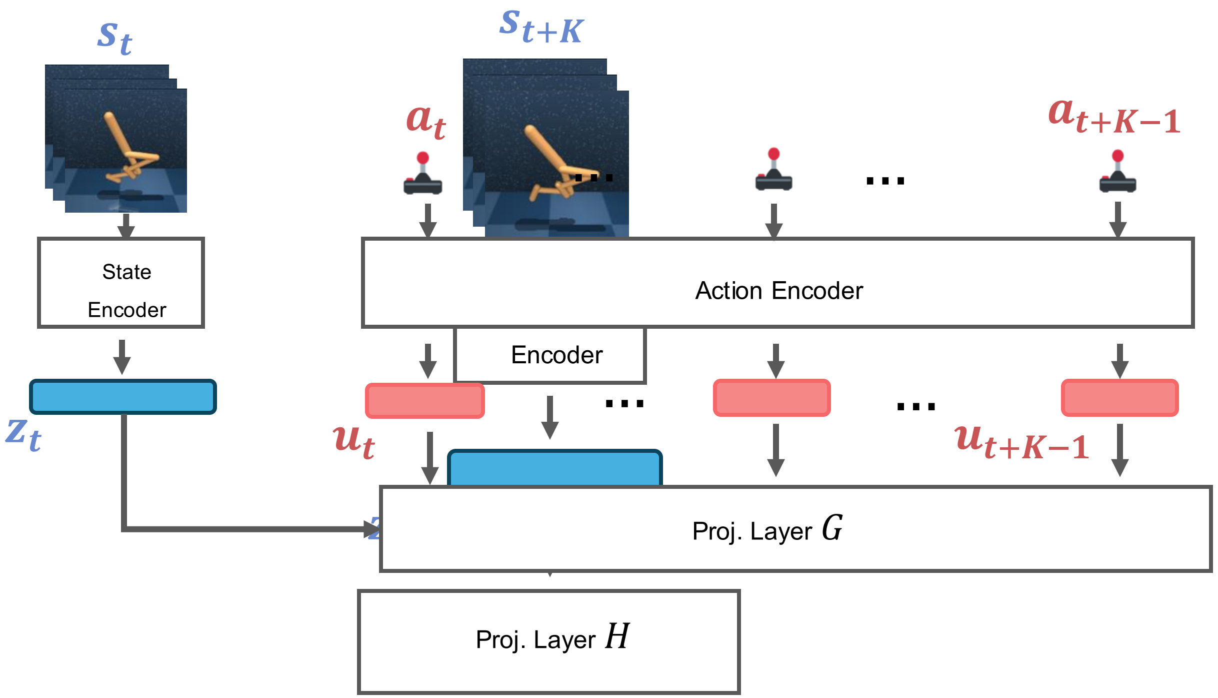
TACO: Temporal Latent Action-Driven Contrastive Loss

Embedding of current latent state
+ action sequence

$$g_t = G_\theta([z_t, (u_t, \dots, u_{t+K-1})])$$

Embedding of future latent state

$$h_t = H_\theta(z_{t+K})$$



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$$\mathcal{L}_{TACO} = \sum_{i=1}^{N} log \frac{{g_{t_i}^{(i)}}^T W h_{t_i}^{(i)}}{\sum_{j=1}^{N} {g_{t_i}^{(i)}}^T W h_{t_j}^{(j)}}$$

	$h_{t_N}^{(N)}$	$h_{t_2}^{(2)}$	$\cdots$	$h_{t_N}^{(N)}$
$g_{t_1}^{(1)}$	${g_{t_1}^{(1)}}^T W h_{t_1}^{(1)}$	${g_{t_1}^{(1)}}^T W h_{t_1}^{(2)}$	$\cdots$	${g_{t_1}^{(1)}}^T W h_{t_N}^{(N)}$
$g_{t_2}^{(2)}$	${g_{t_2}^{(2)}}^T W h_{t_1}^{(1)}$	${g_{t_2}^{(2)}}^T W h_{t_1}^{(2)}$	$\cdots$	${g_{t_2}^{(2)}}^T W h_{t_N}^{(N)}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$g_{t_N}^{(N)}$	${g_{t_N}^{(N)}}^T W h_{t_1}^{(1)}$	${g_{t_N}^{(N)}}^T W h_{t_1}^{(2)}$	$\cdots$	${g_{t_N}^{(N)}}^T W h_{t_N}^{(N)}$

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TACO: Temporal Latent Action-Driven Contrastive Loss

Why contrastive loss?

$$\mathcal{L}_{TACO} = \sum_{i=1}^{N} log \frac{{g_{t_i}^{(i)}}^T W h_{t_i}^{(i)}}{\sum_{j=1}^{N} {g_{t_i}^{(i)}}^T W h_{t_j}^{(j)}}$$

vs.

$$\mathcal{L}_{dyn} = \sum_{i=1}^{N} \mathcal{L}_{MSE}(g_{t_i}^{(i)}, h_{t_i}^{(i)})$$

Introduction of negative example

$$\Downarrow$$

Less prone to representation collapse, better overall performance

$$\begin{array}{c|cccc}
 & h_{t_N}^{(N)} & h_{t_2}^{(2)} & \cdots & h_{t_N}^{(N)} \\
\hline
g_{t_1}^{(1)} & {g_{t_1}^{(1)}}^T W h_{t_1}^{(1)} & {g_{t_1}^{(1)}}^T W h_{t_1}^{(2)} & \cdots & {g_{t_1}^{(1)}}^T W h_{t_N}^{(N)} \\
g_{t_2}^{(2)} & {g_{t_2}^{(2)}}^T W h_{t_1}^{(1)} & {g_{t_2}^{(2)}}^T W h_{t_1}^{(2)} & \cdots & {g_{t_2}^{(2)}}^T W h_{t_N}^{(N)} \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
g_{t_N}^{(N)} & {g_{t_N}^{(N)}}^T W h_{t_1}^{(1)} & {g_{t_N}^{(N)}}^T W h_{t_1}^{(2)} & \cdots & {g_{t_N}^{(N)}}^T W h_{t_N}^{(N)}
\end{array}$$

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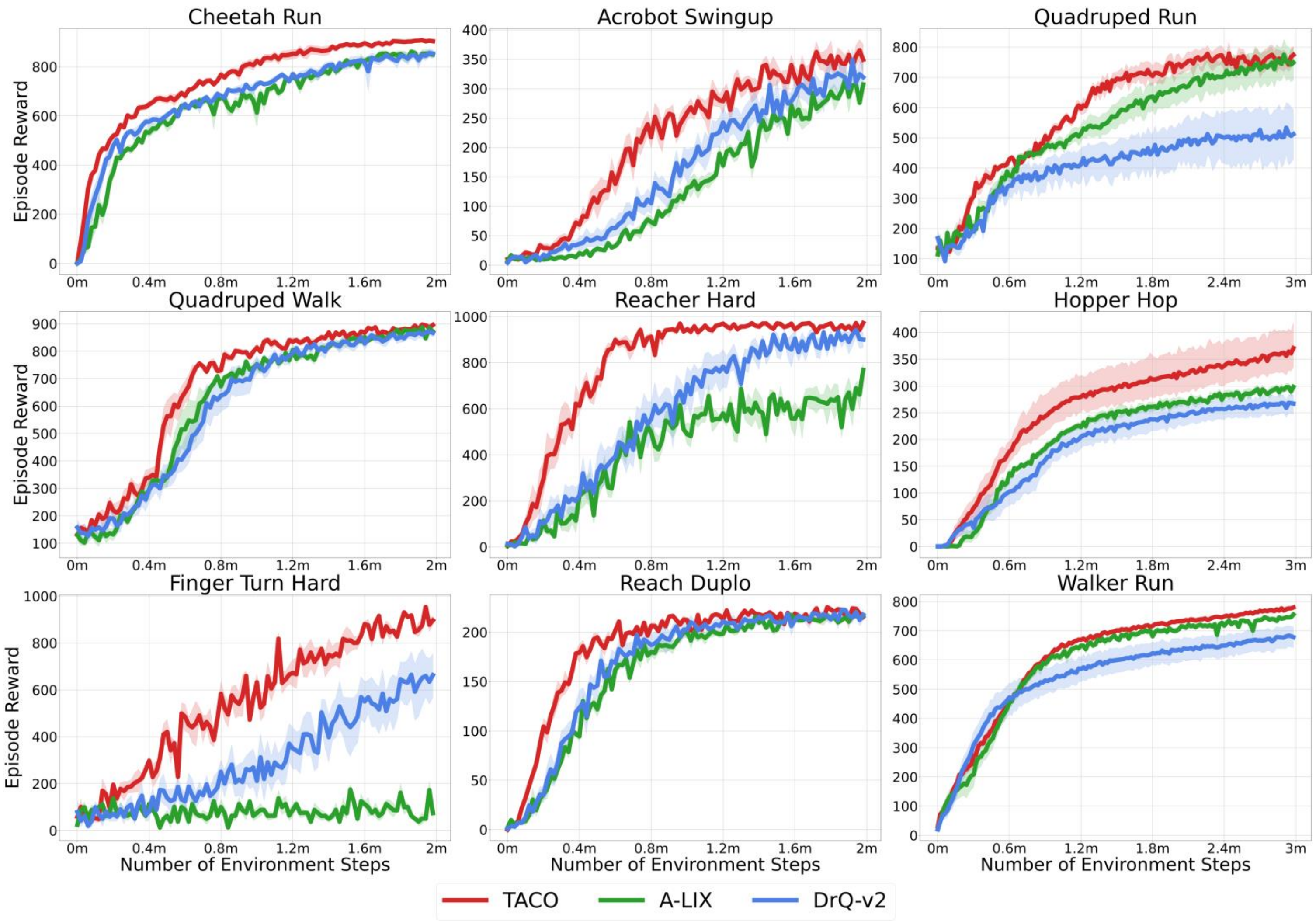
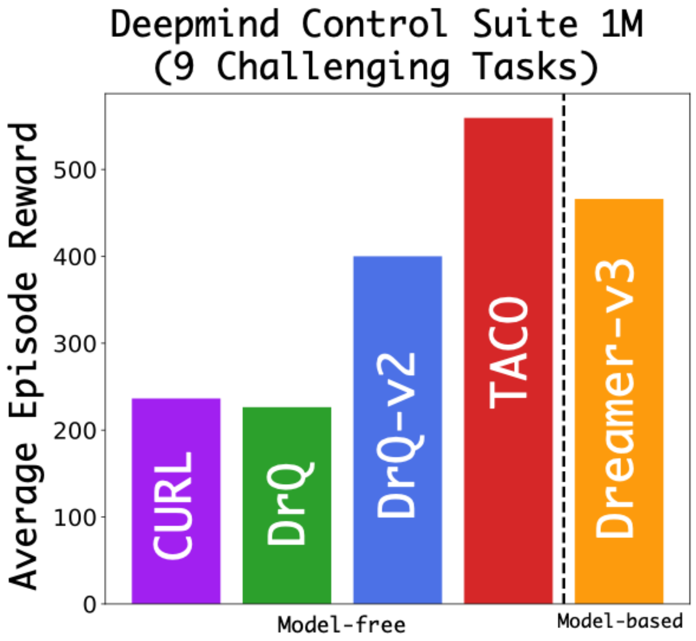
TACO for Online Reinforcement Learning

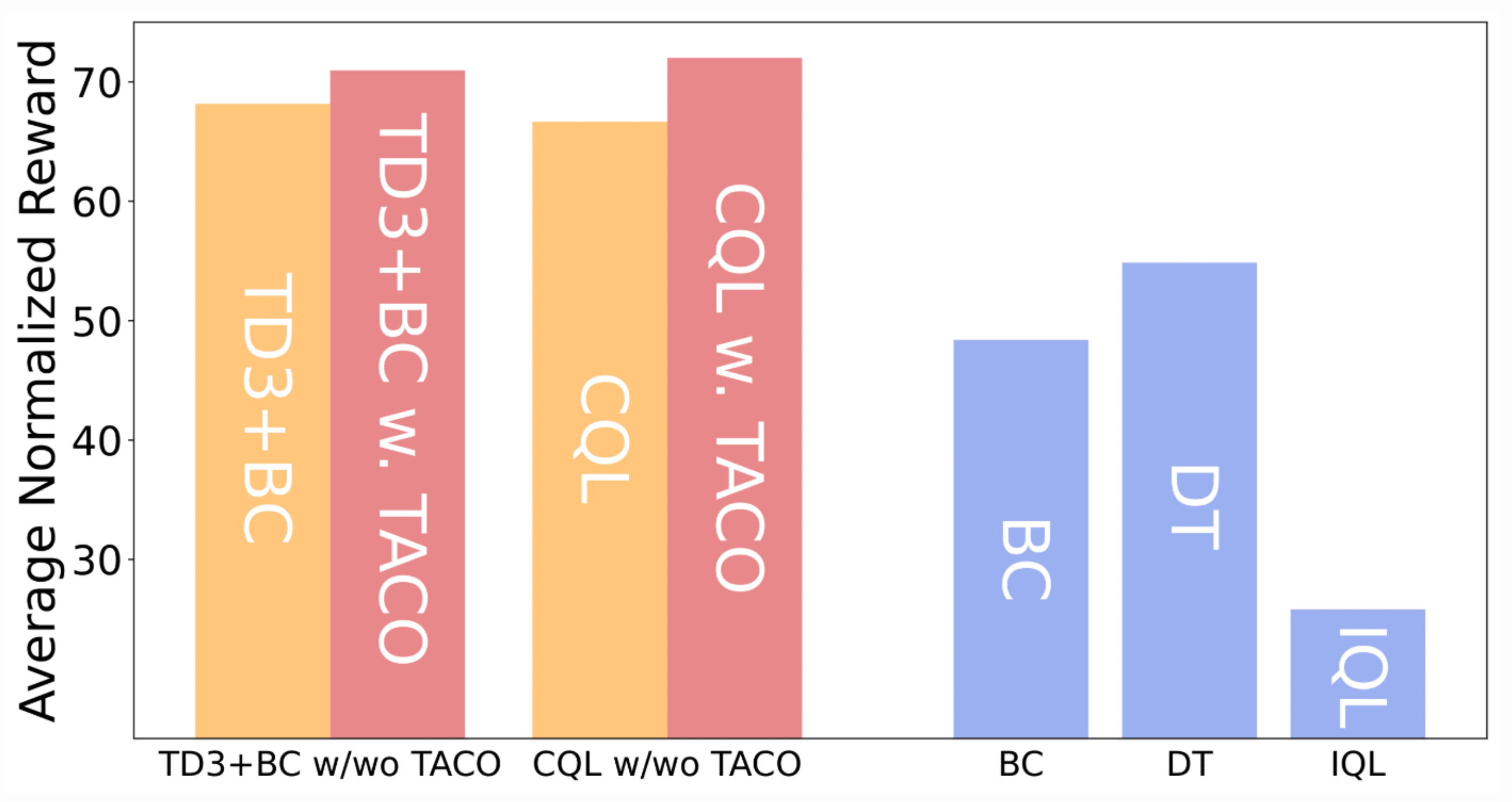




SoTA Sample Efficiency and Performance !!!

Zheng, Wang, Sun, Ma, Zhao, Xu, Daume, H.. "TACO: Temporal Latent Action-Driven Contrastive Loss for Visual Reinforcement Learning." NeurIPS 2023.

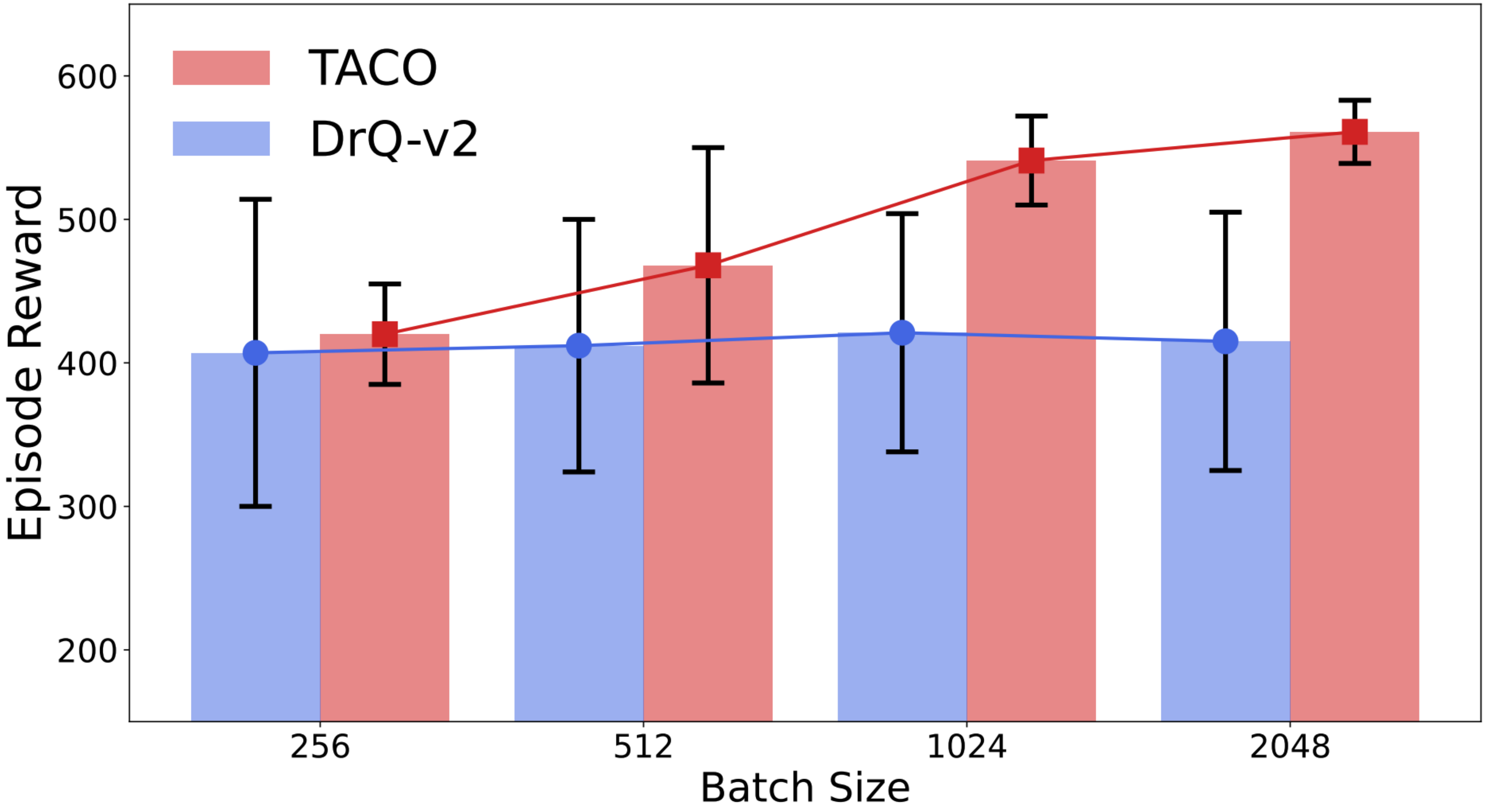
TACO for Offline Reinforcement Learning



Plug-and-play module for boosting offline RL performance

Zheng, Wang, Sun, Ma, Zhao, Xu, Daume, H.. "TACO: Temporal Latent Action-Driven Contrastive Loss for Visual Reinforcement Learning." NeurIPS 2023.

Limitation of **TACO**

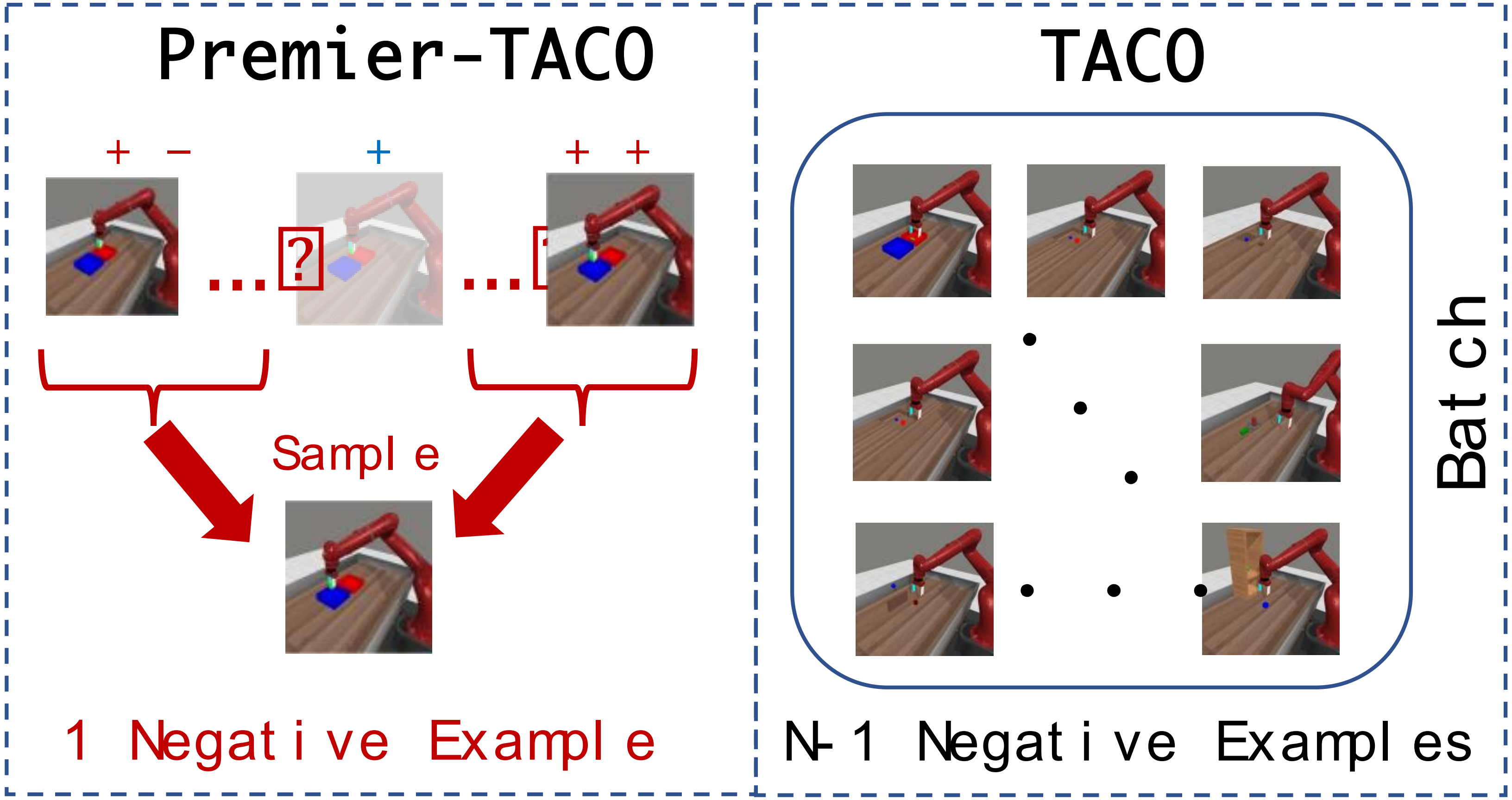


Large Batch Size is required for TACO (even single-task)

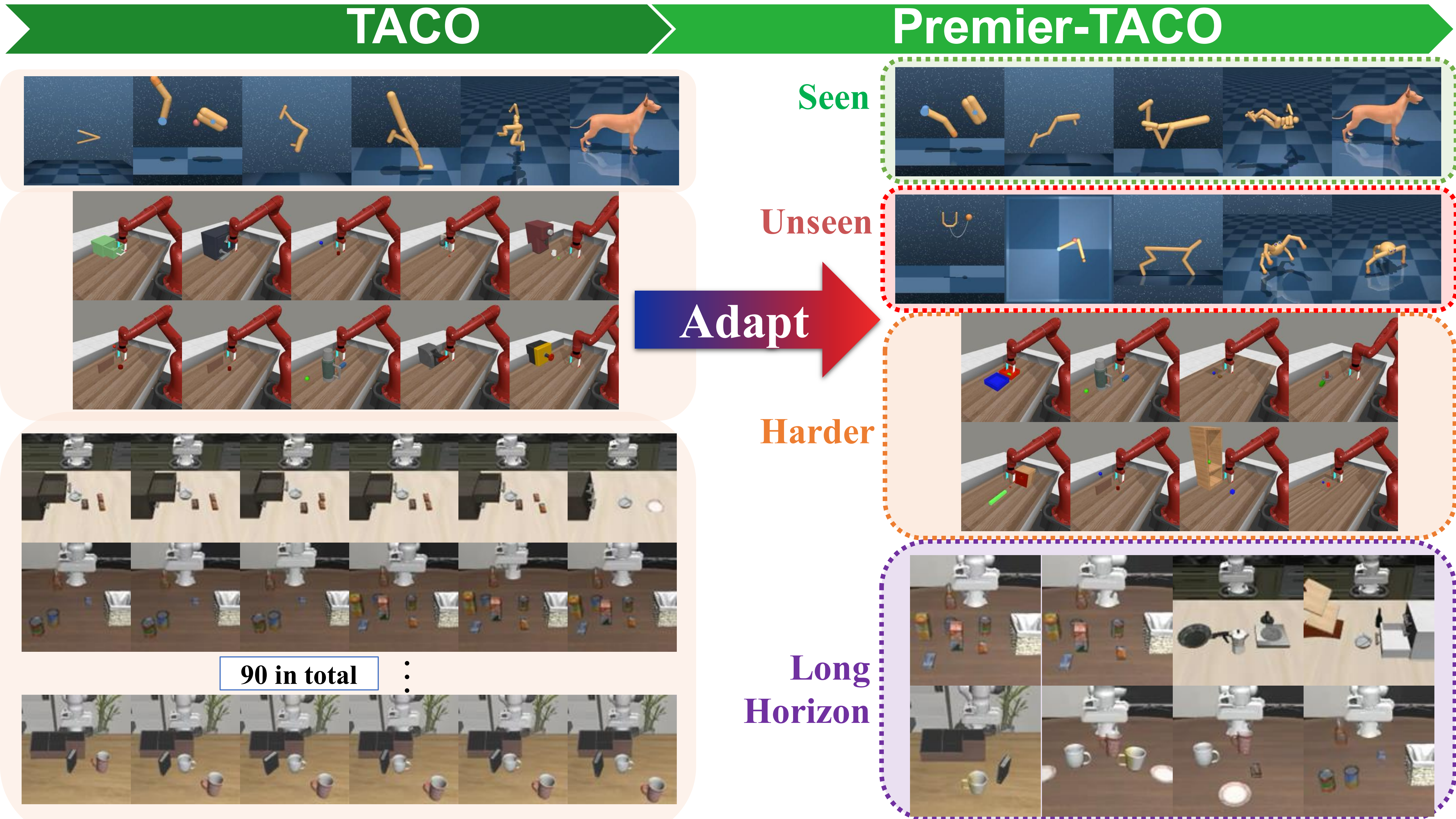
Zheng, Wang, Sun, Ma, Zhao, Xu, Daume, H.. "TACO: Temporal Latent Action-Driven Contrastive Loss for Visual Reinforcement Learning." NeurIPS 2023.

TACO Premier-TACO

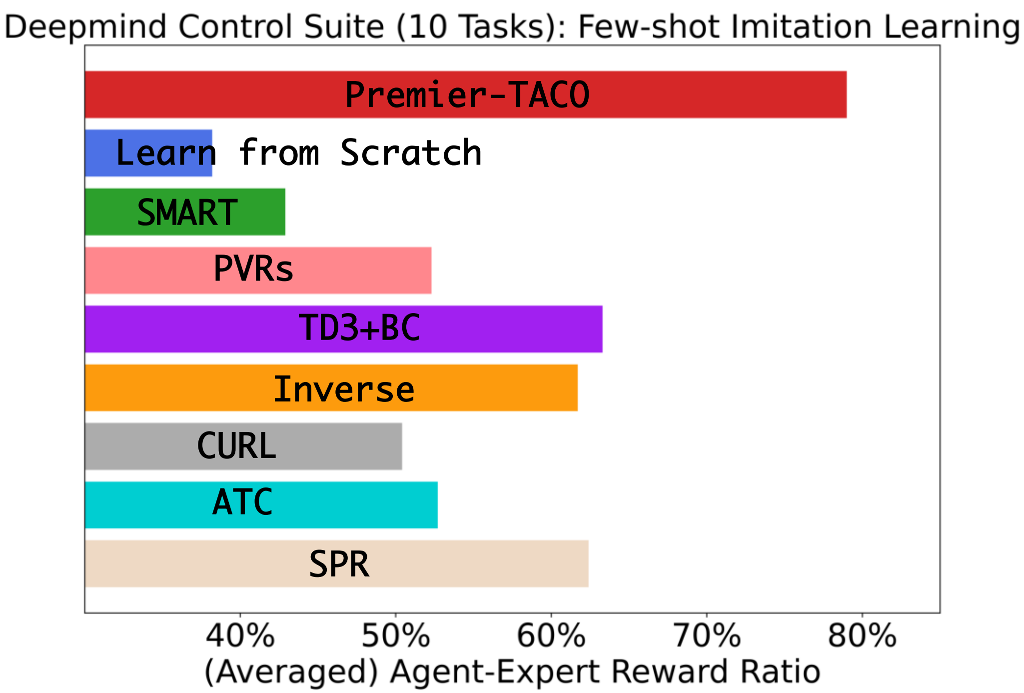
Premier-TACO v.s. TACO

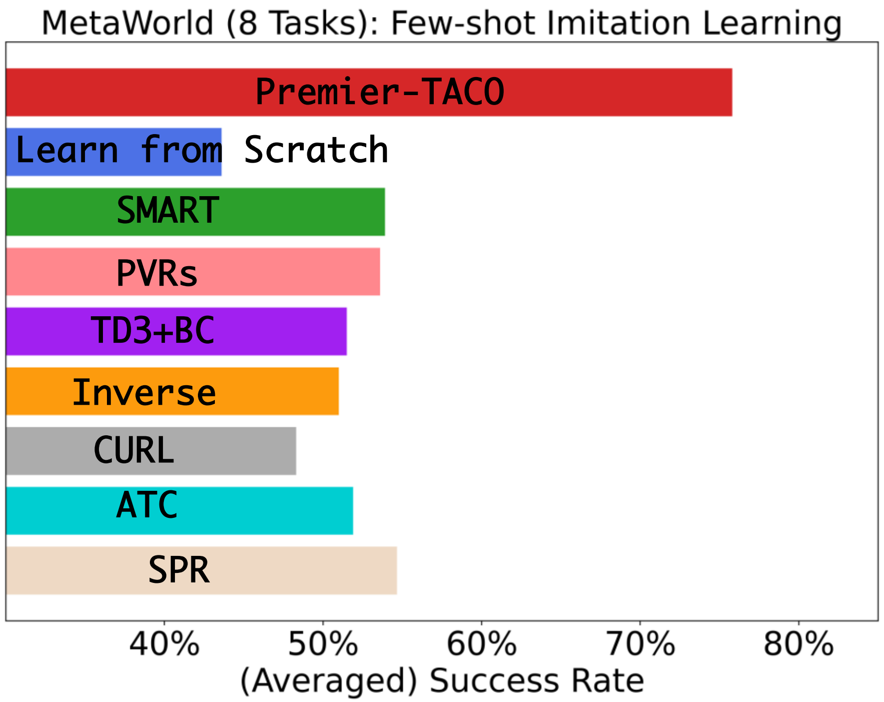


Key for Scalability

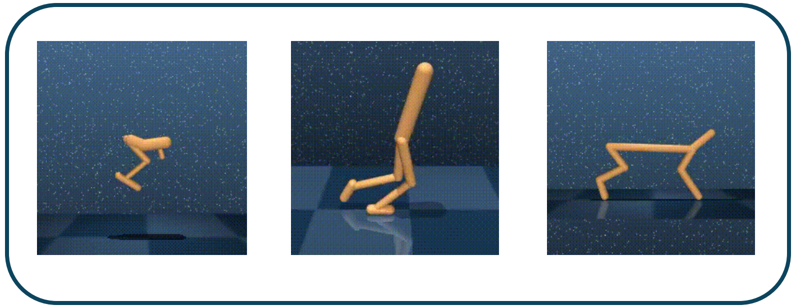

TACO
Premier-TACO
Seen
Unseen
Adapt
Harder
Long
Horizon
90 in total

Few-shot Behavior Cloning of Unseen Tasks

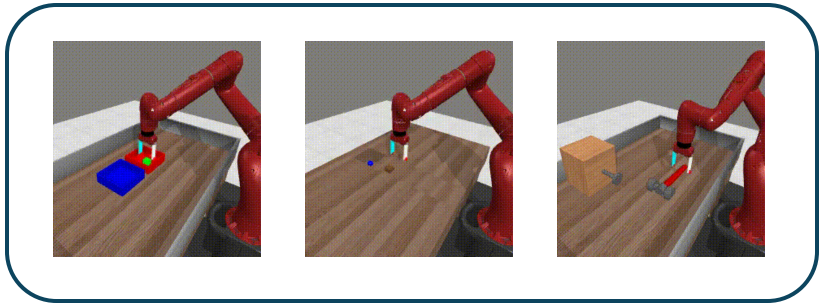




20-shots, except for dog and humanoid 100-shots



5-shots



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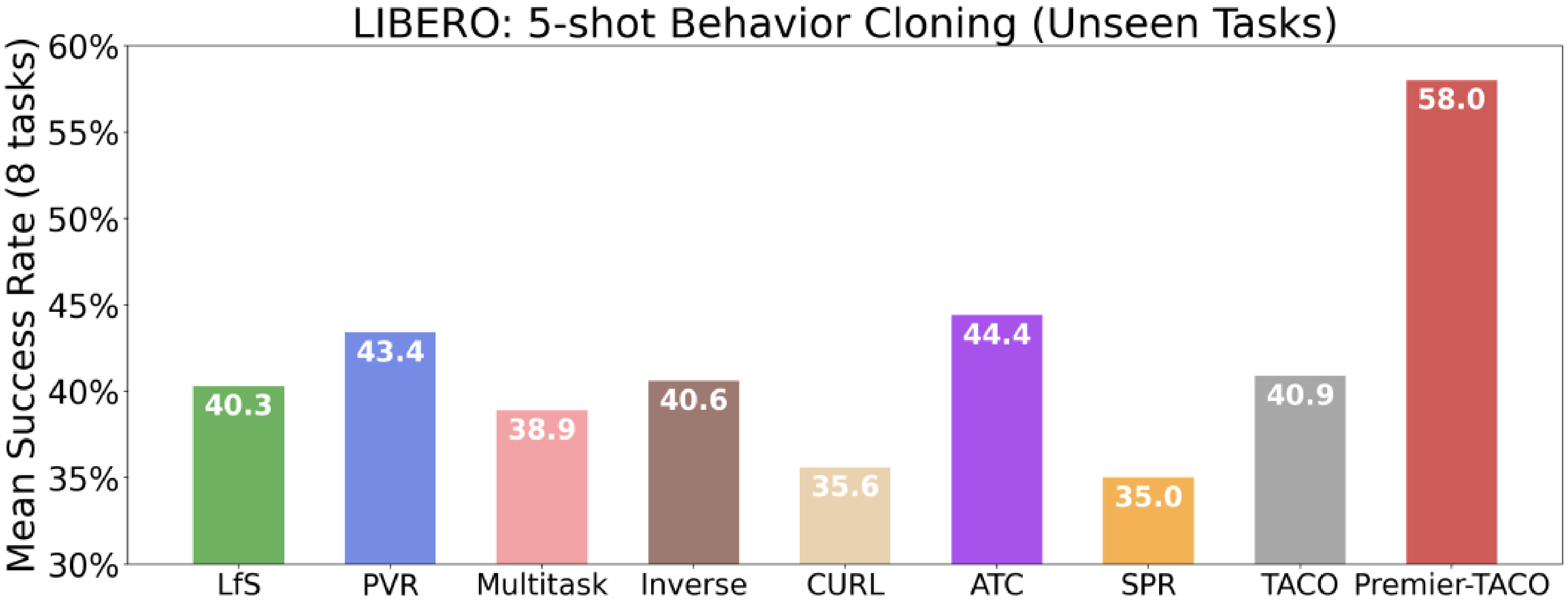
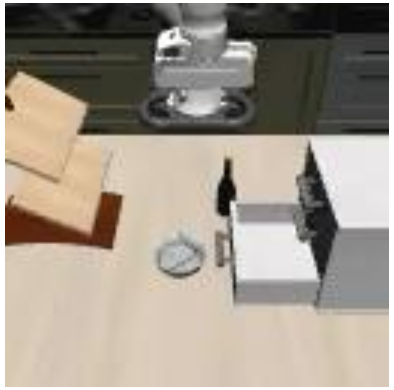
Few-shot Behavior Cloning of Unseen Tasks





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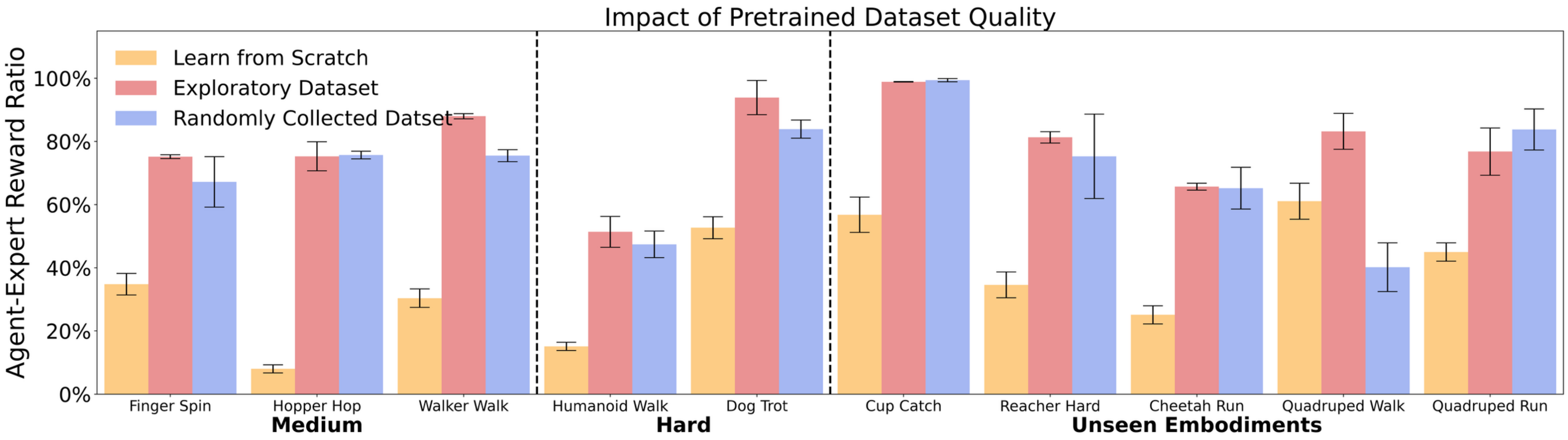
Premier-TACO is robust to low-quality pretraining data



(Uniformly) randomly collected dataset
works well!

Zheng, Liang, Wang, Ma, Daume, Xu, Langford, Palanisamy, Basu, H.. "Premier-TACO is a Few-Shot Policy Learner: Pretraining Multitask Representation via Temporal Action-Driven Contrastive Loss." ICML 2024.

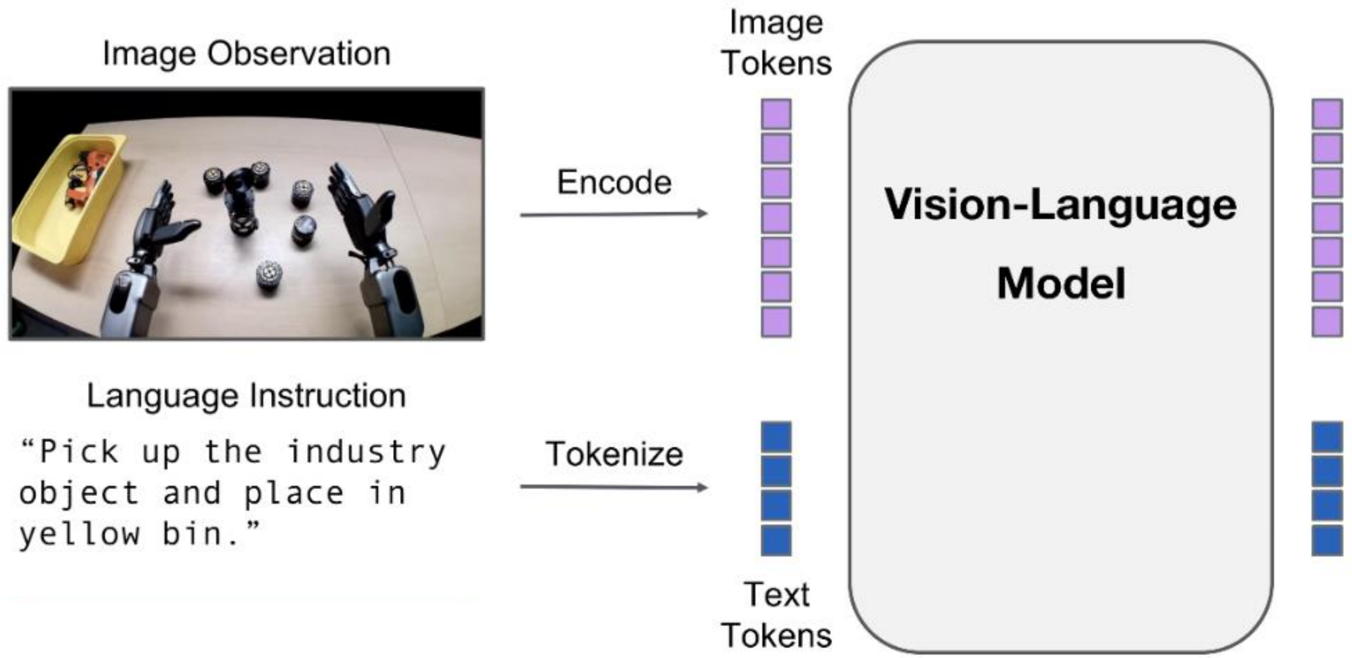
Reuse Compute

Maintain VLMs' perception in the wild

Infuse robotics (planning) capability

Reuse Data

Regular VLA Architecture (GR00t-N1, π_0)



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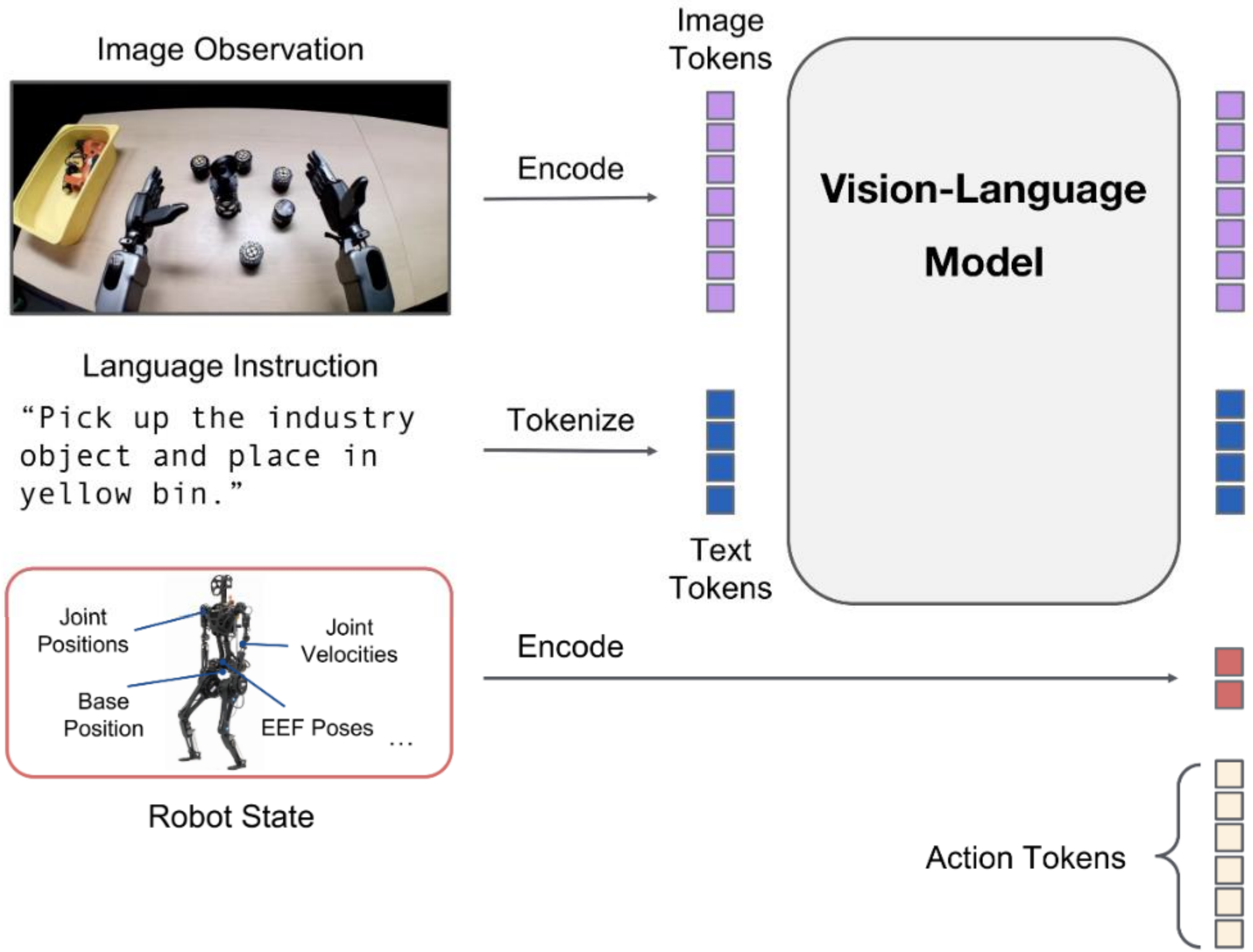

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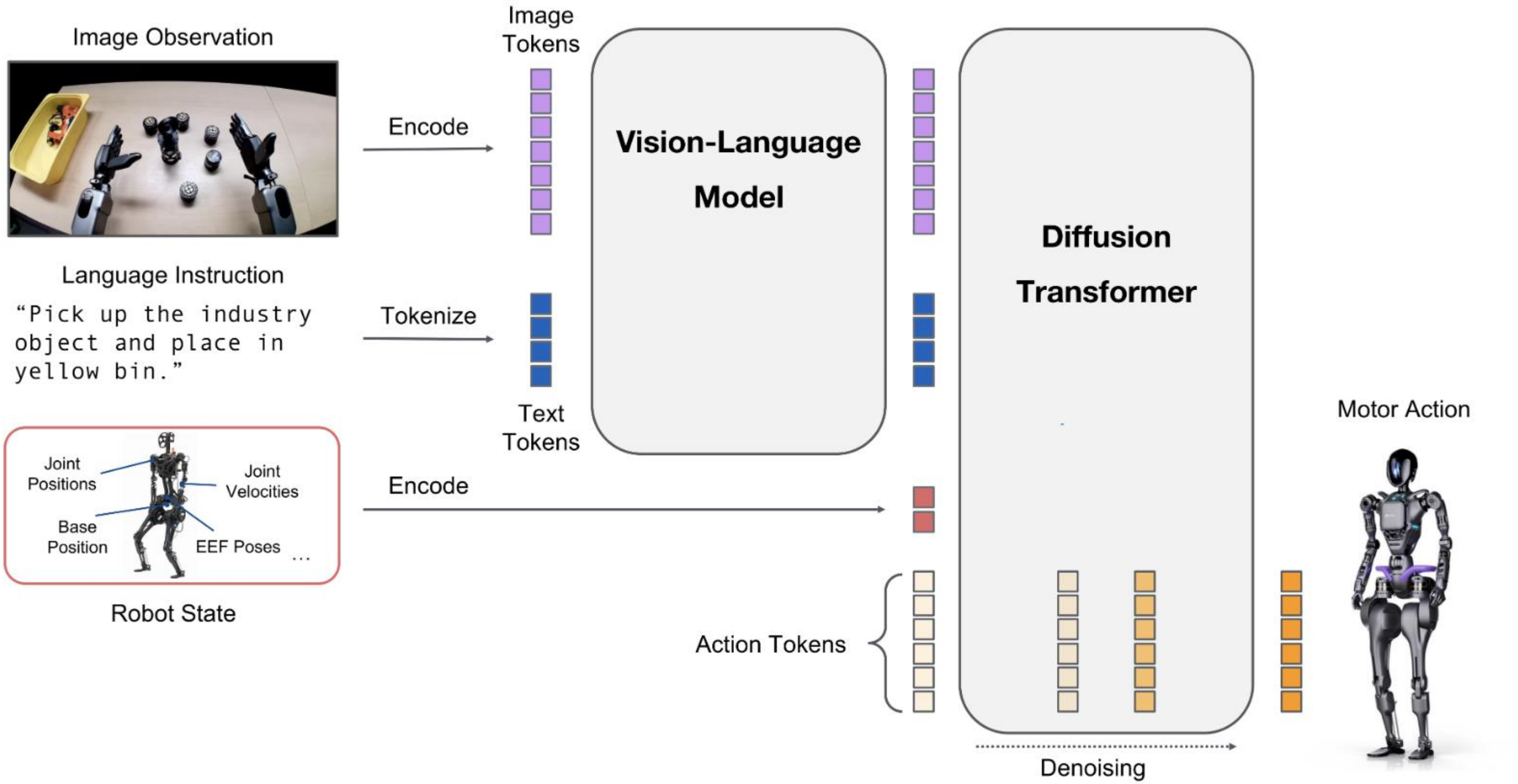

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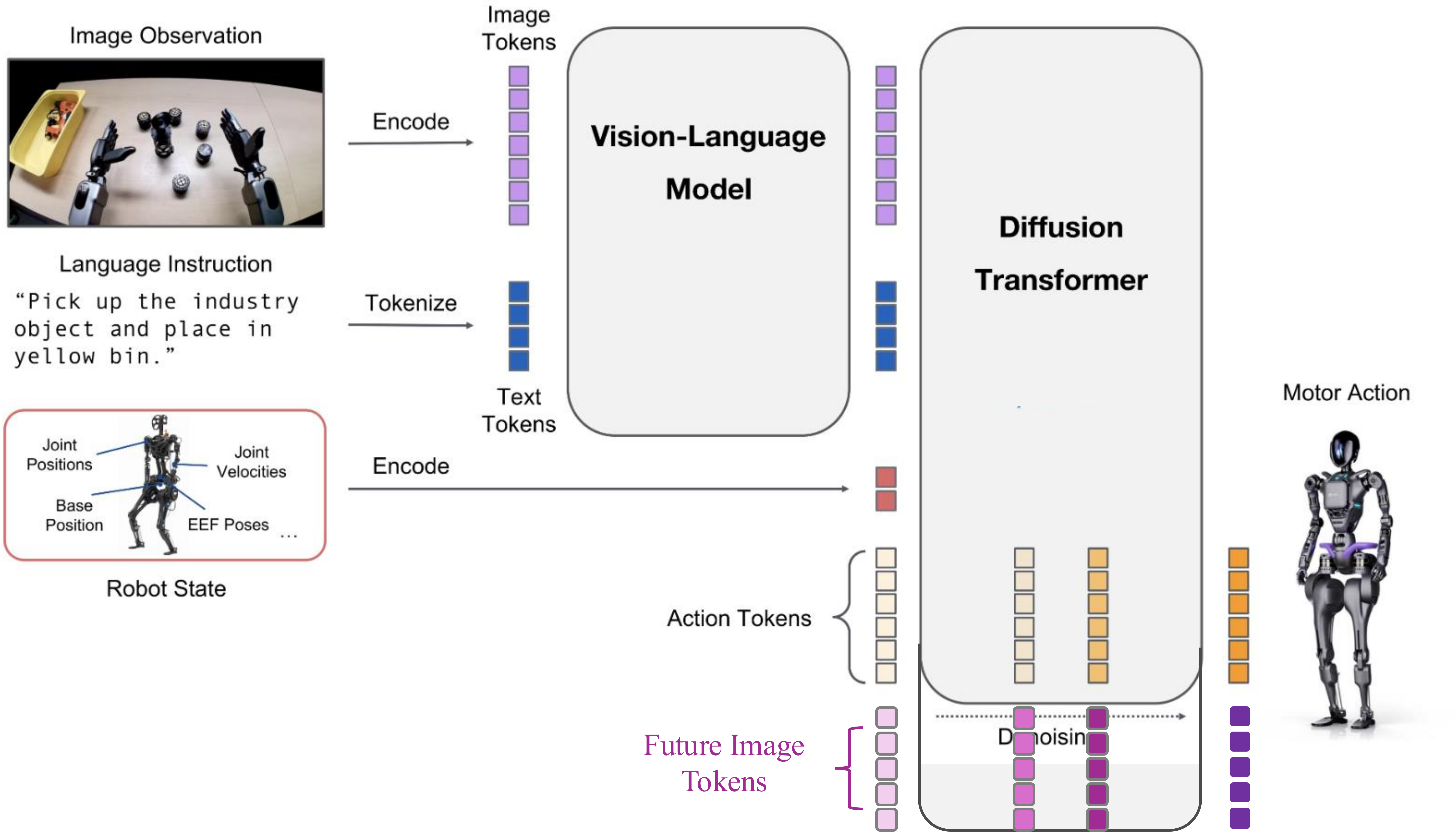

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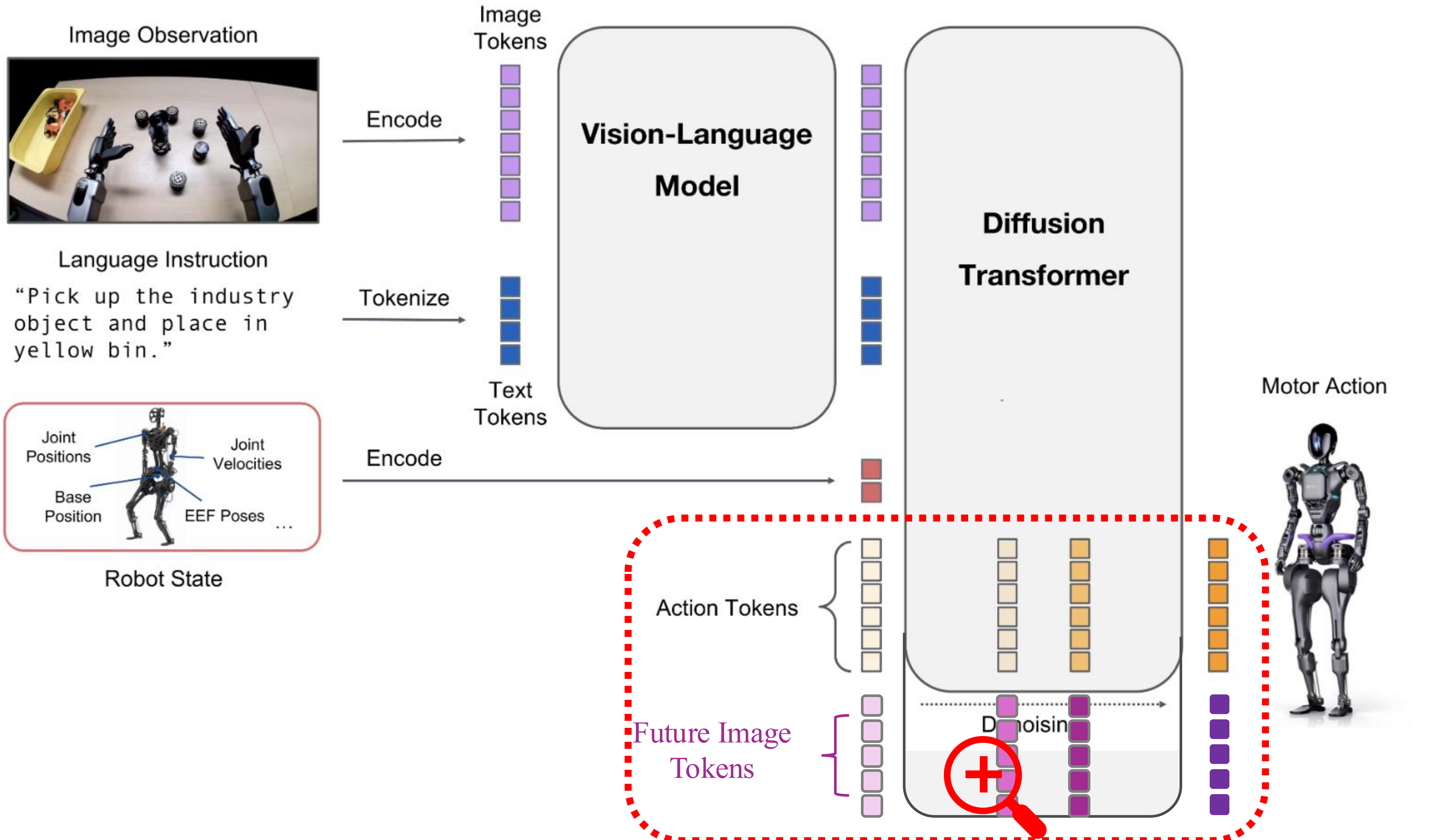
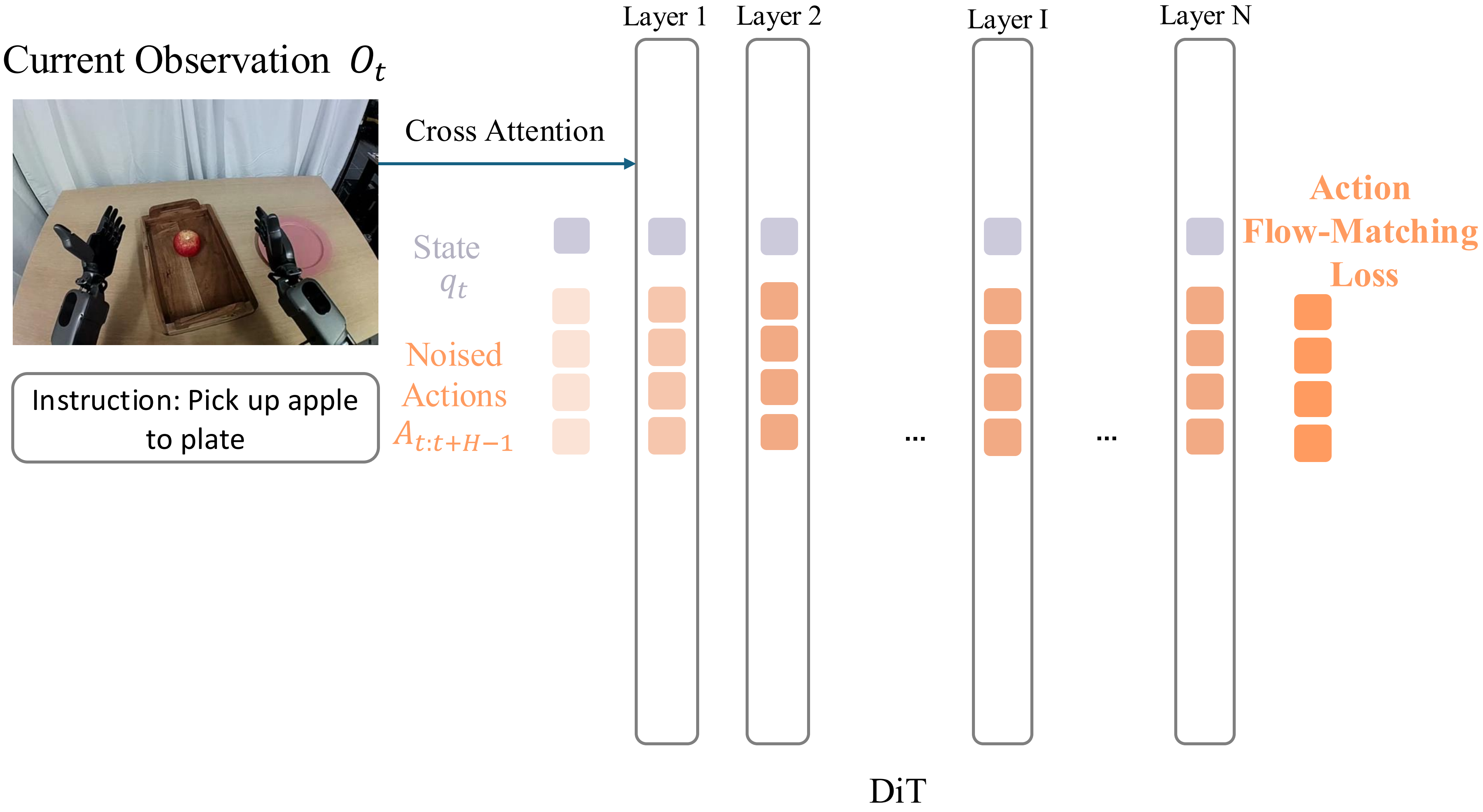


Regular VLA Architecture (GR00t-N1, π_0) → FLARE Architecture



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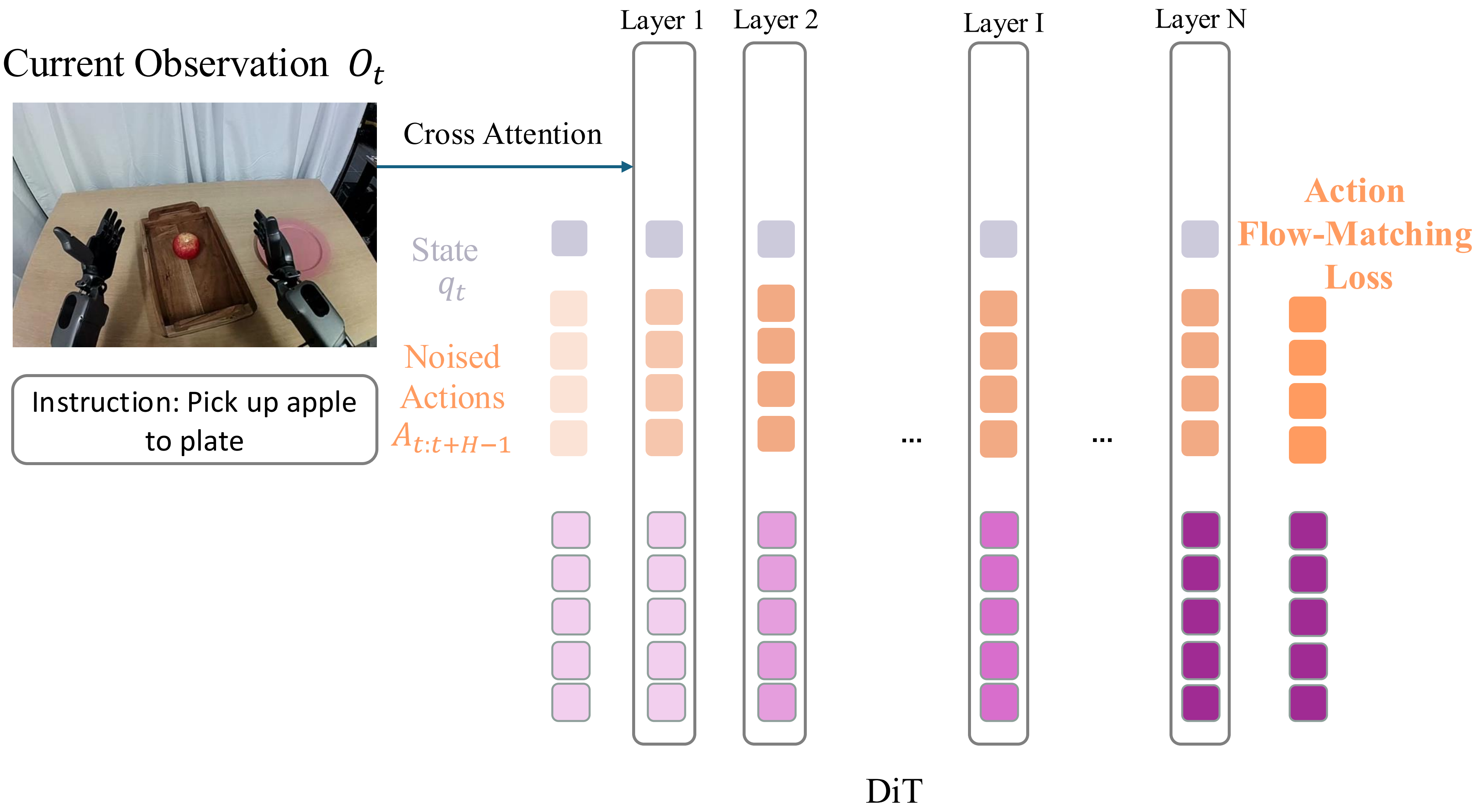
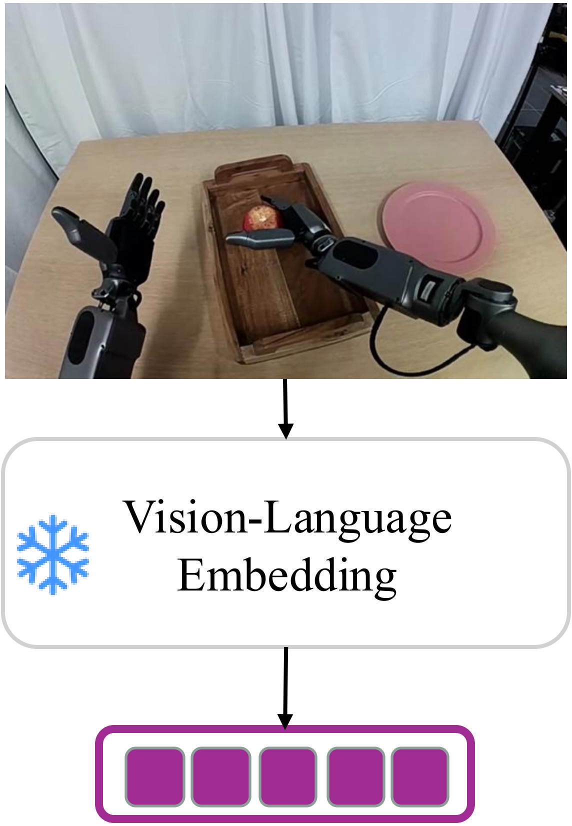


What is a good vision-language encoder?

Future Observation O_{t+H}



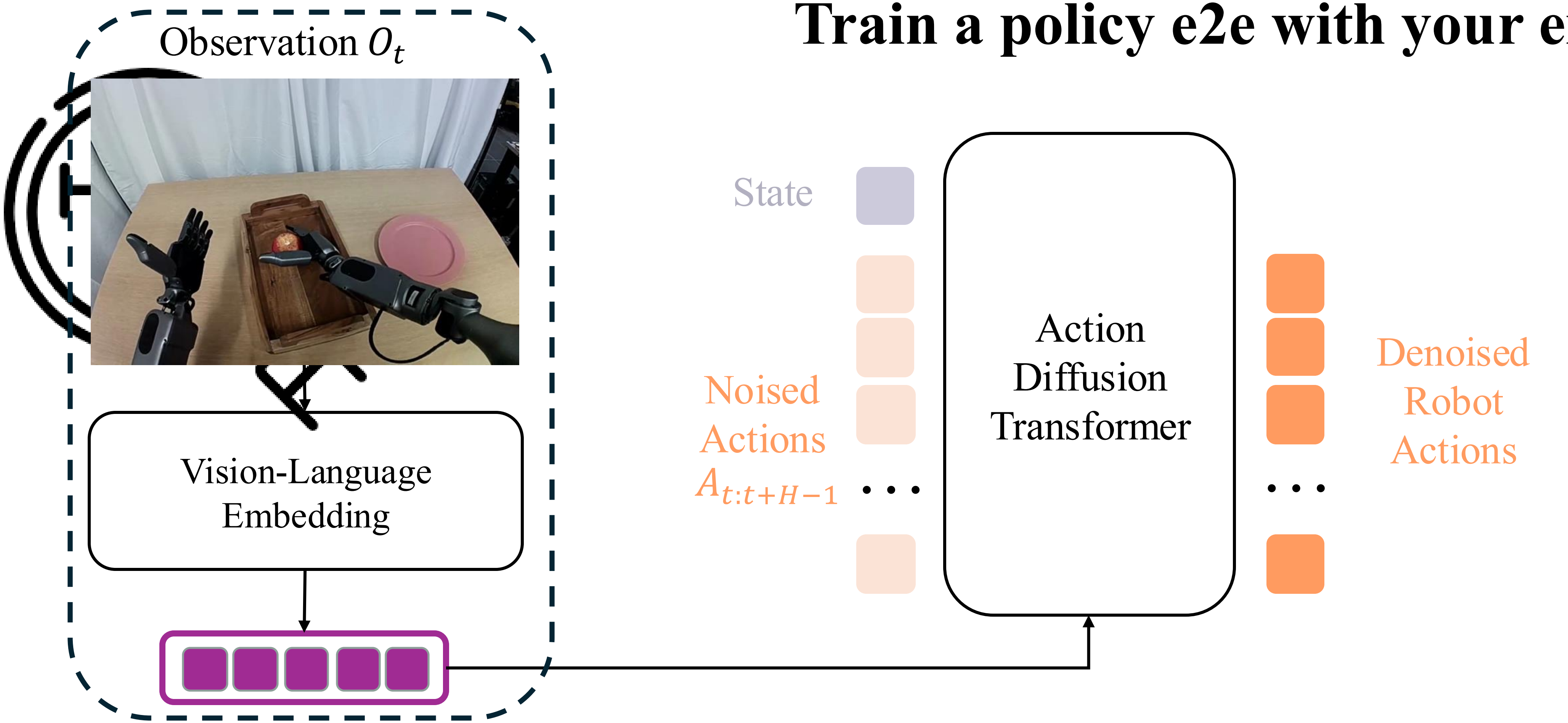
1. It must capture all relevant information for control tasks

2. It must be sufficiently compact
   - Easy to learn
   - Add minimal overhead to policy inference latency

What is a good vision-language encoder?

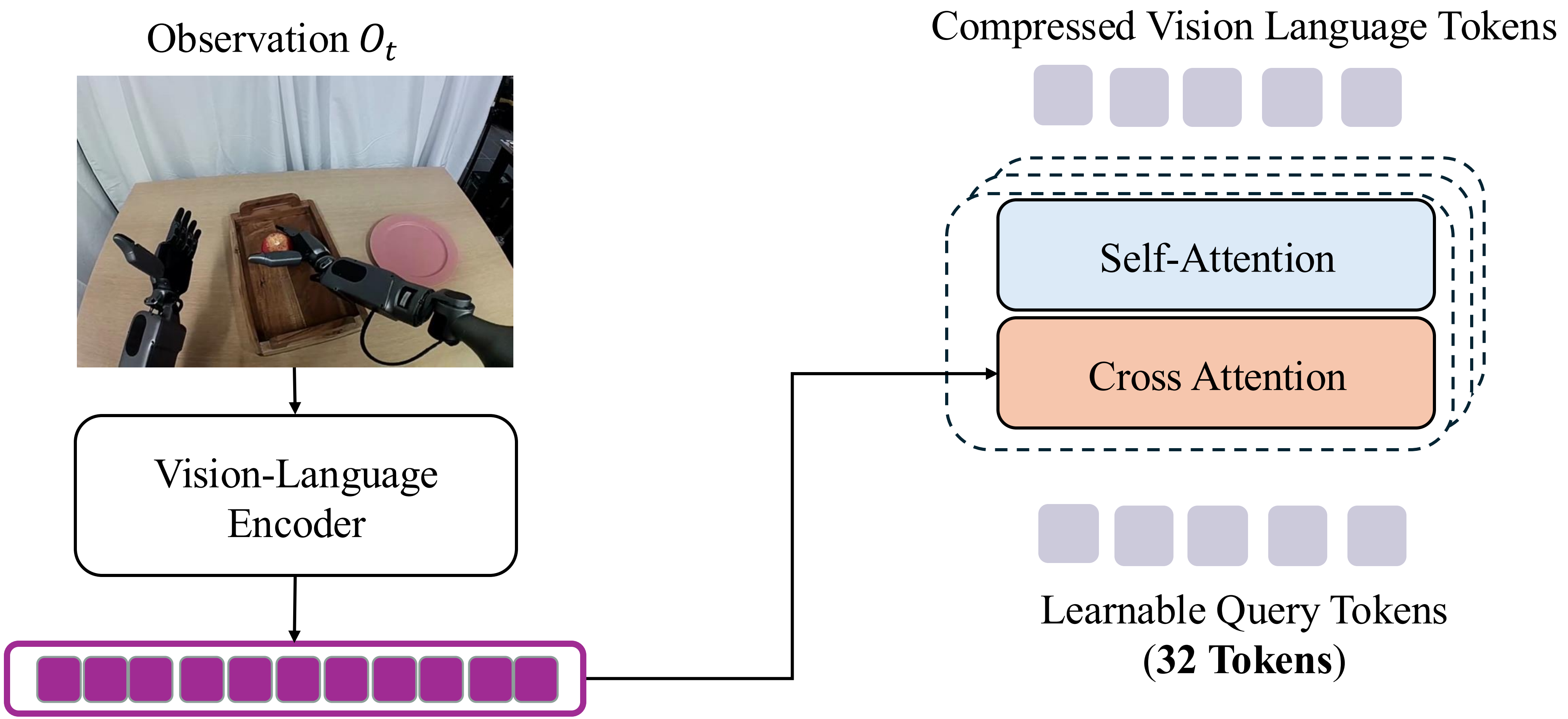
A good vision-language encoder must capture all relevant information for control tasks

Train a policy e2e with your encoder



What is a good vision-language encoder?

A good vision-language encoder must also be sufficiently compact



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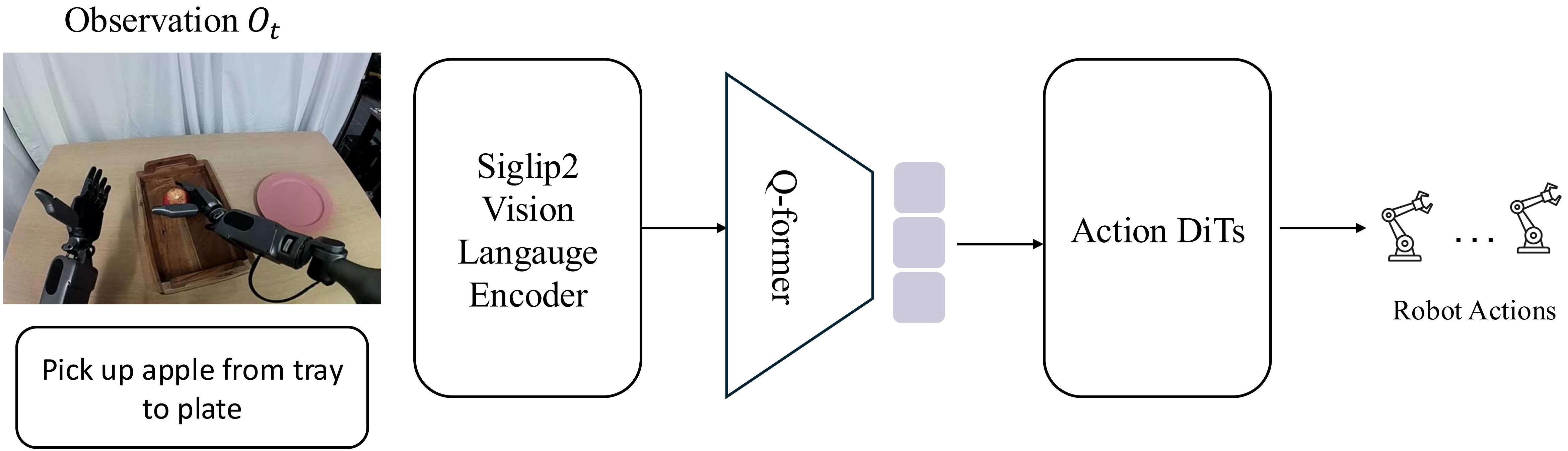

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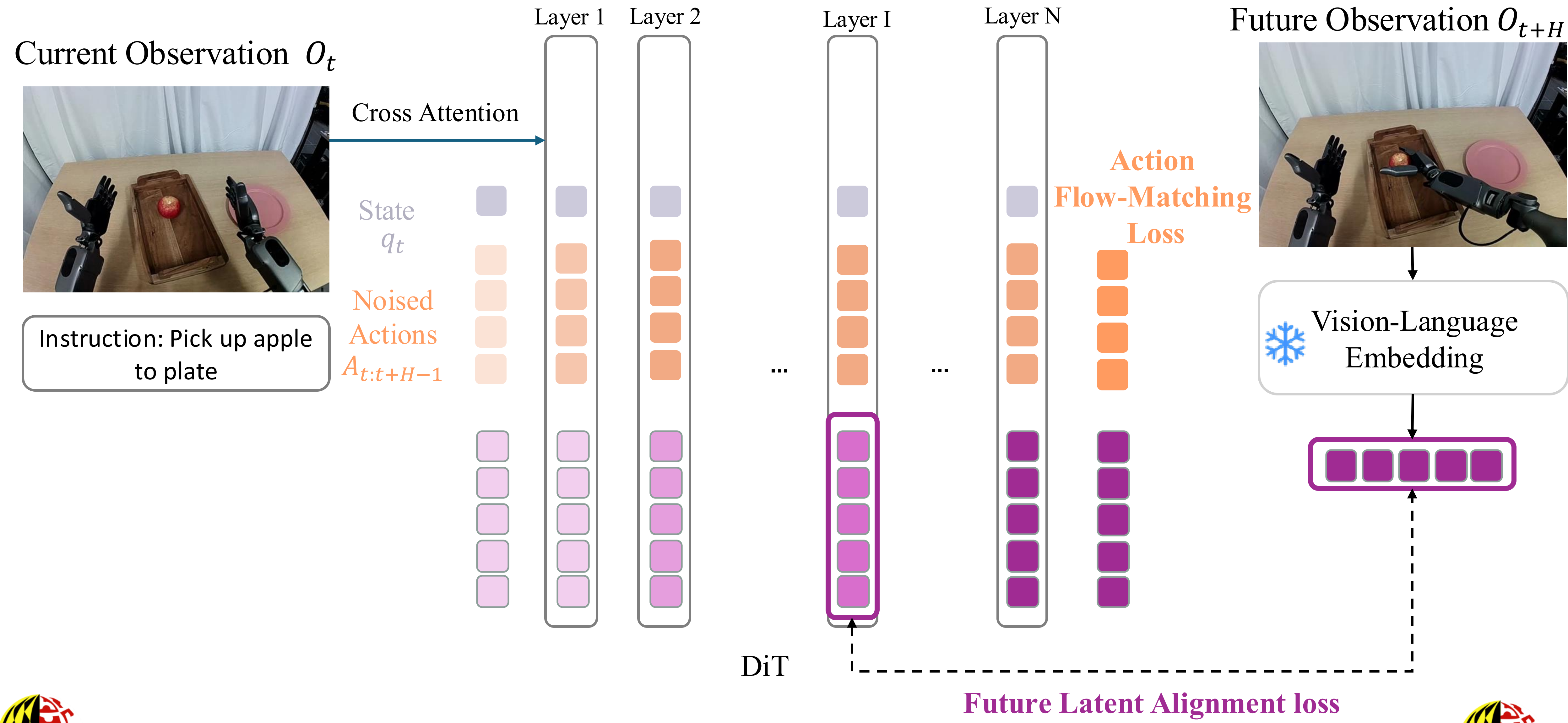
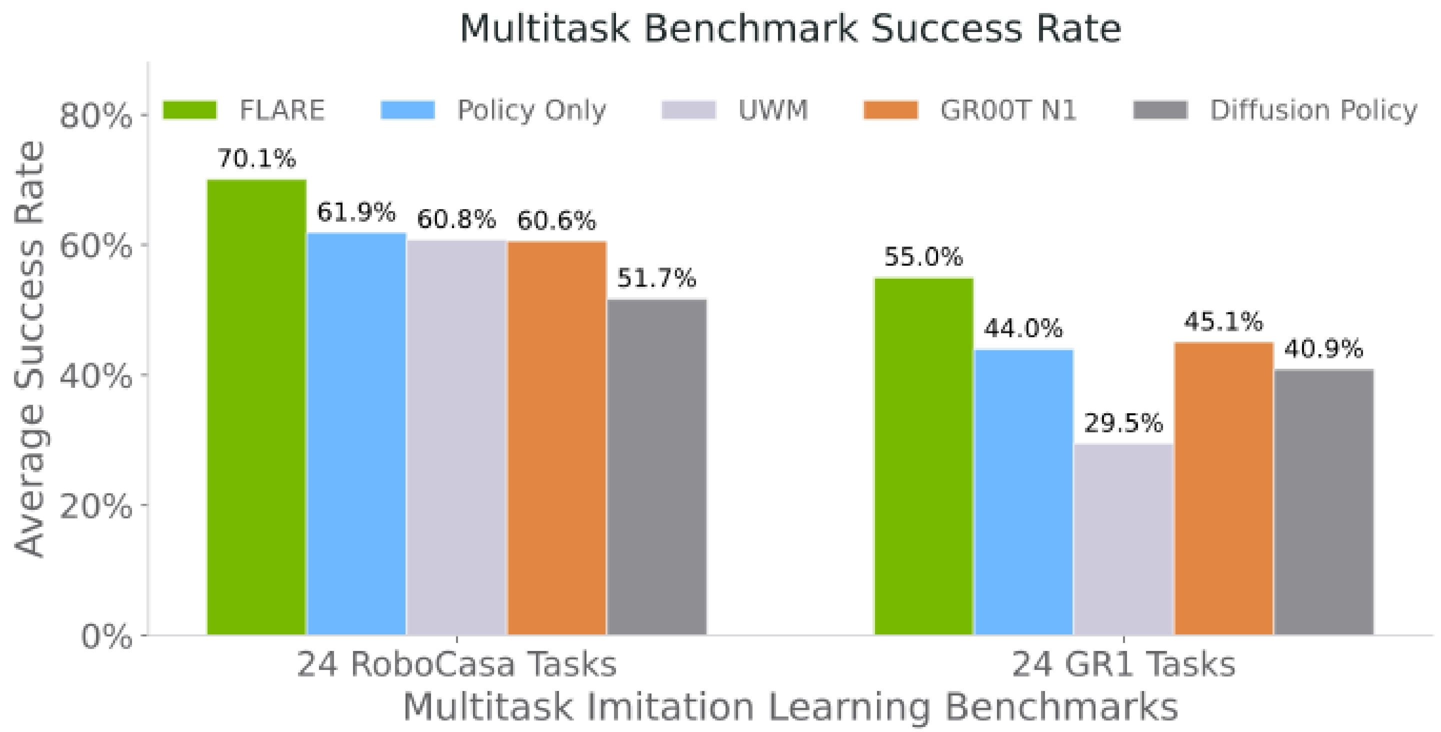
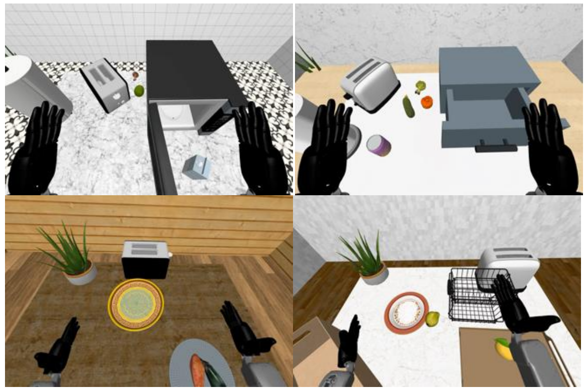
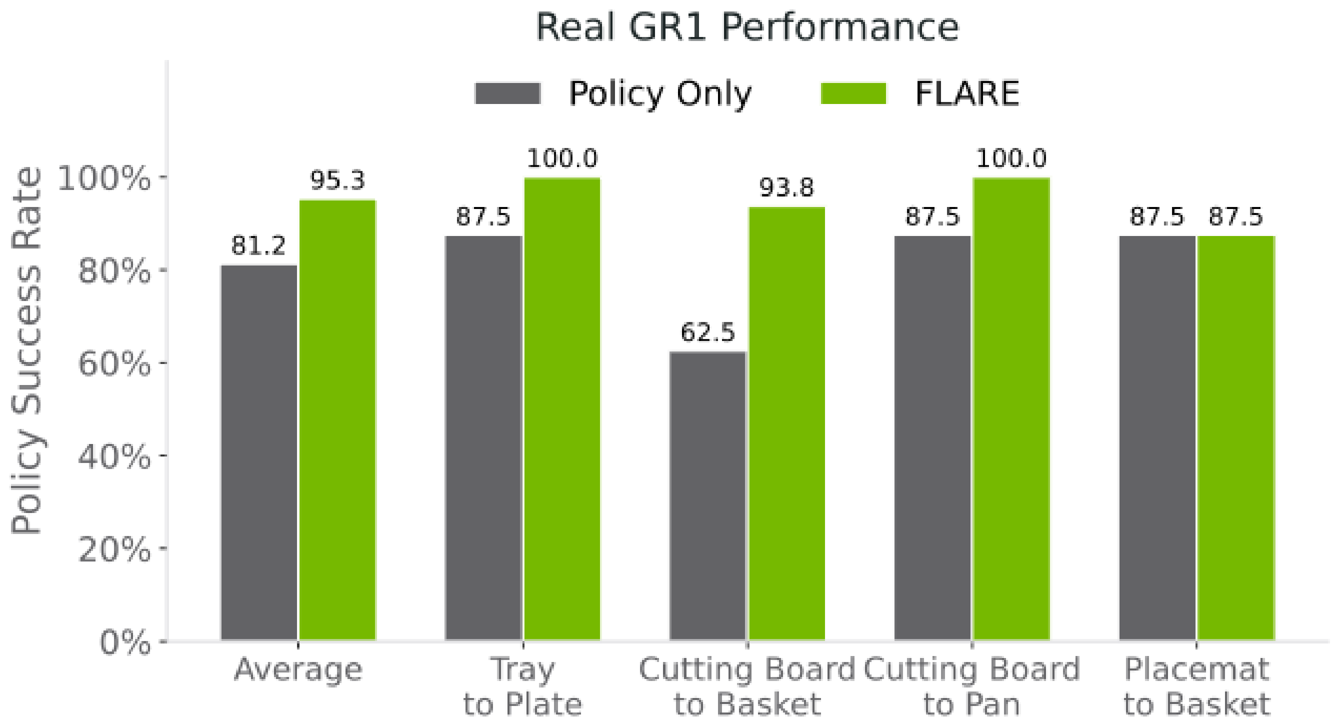


SoTA Multitask Imitation Learning Performance

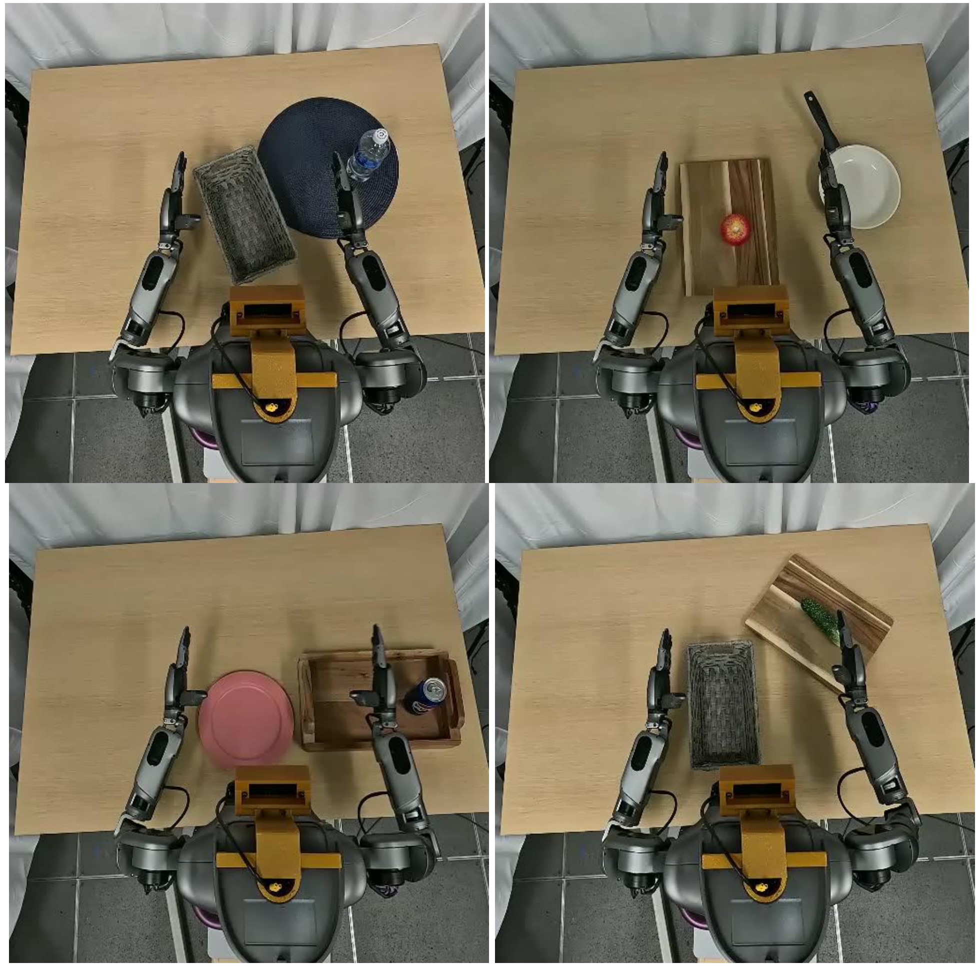




Real-world GR1 Humanoid Tabletop Manipulation



100 trajectories per task



Policy Rollout (1x speed)

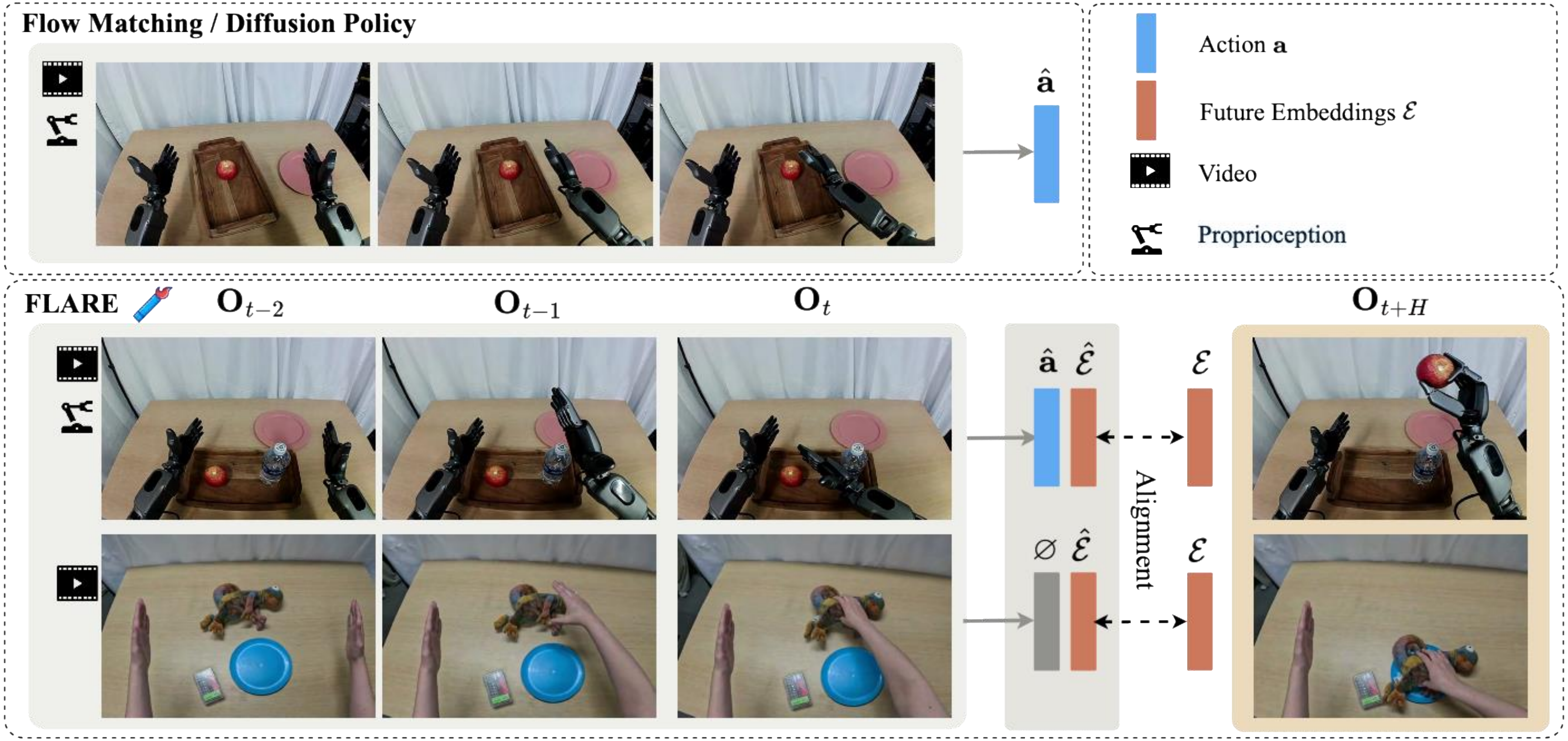
Joint Training on Human Egocentric Video Data

Human Egocentric Data
(Cheap to collect)

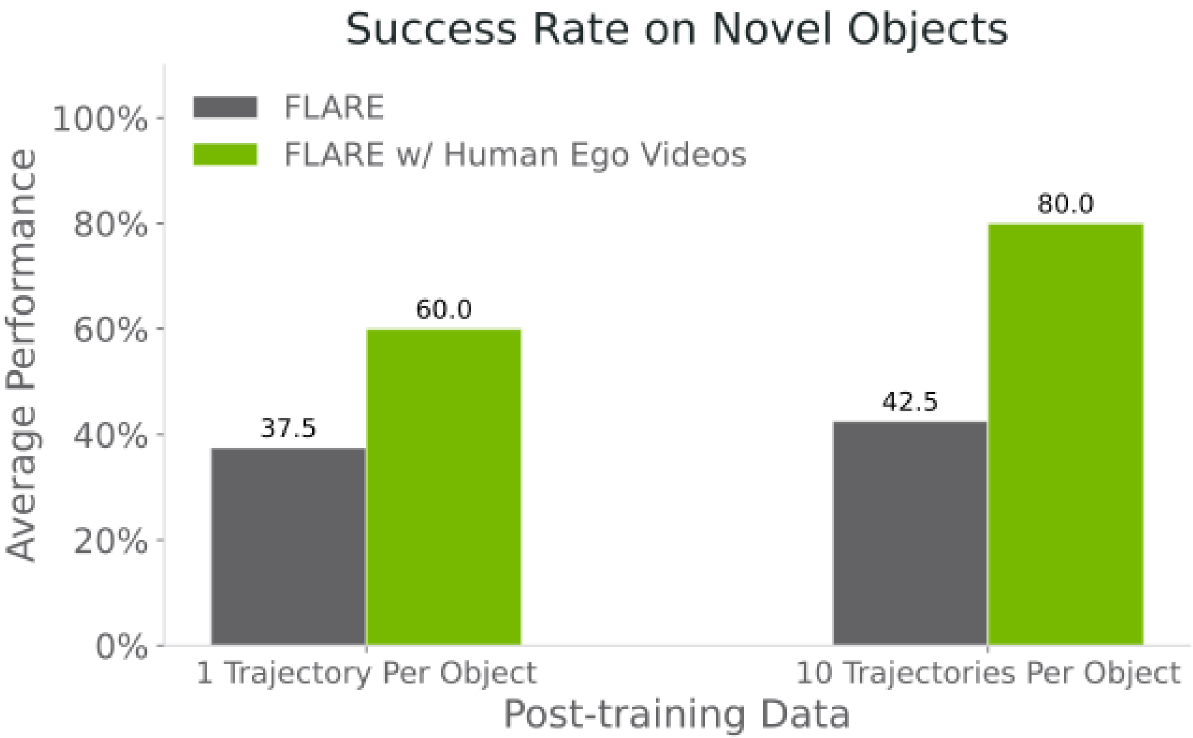
Robot Teleoperation Data
(Much more expensive)

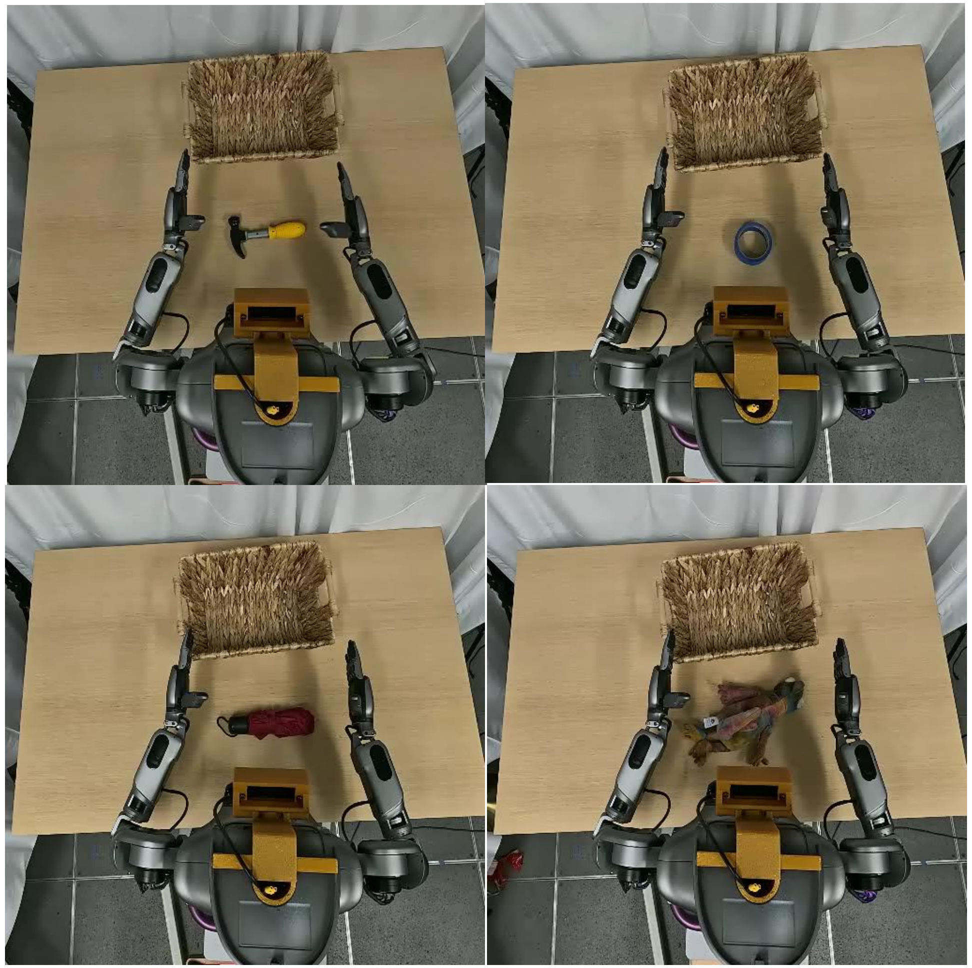
Different camera views but similar tasks

Joint Training on Human Egocentric Video Data

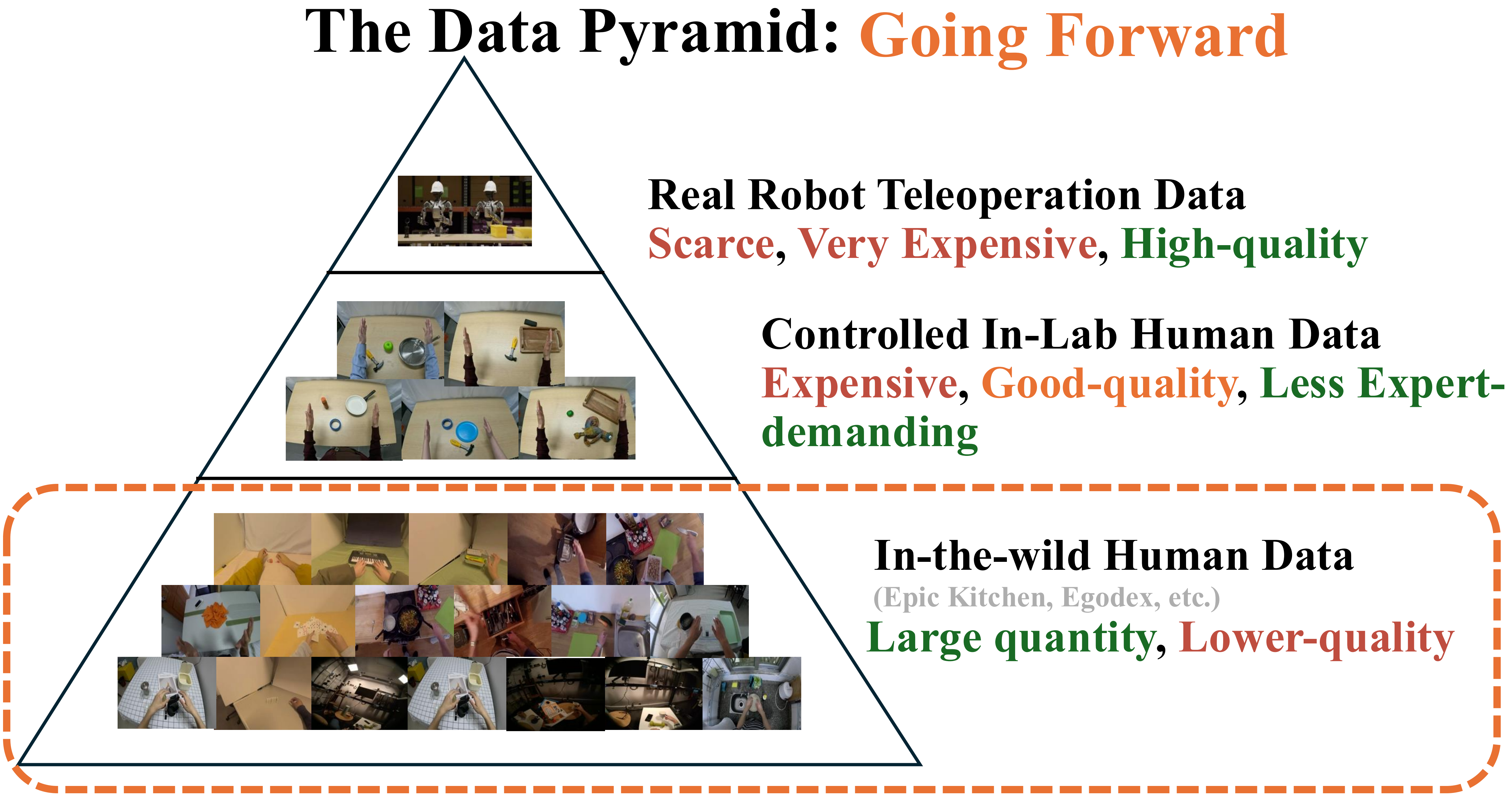


Joint Training on Human Egocentric Video Data





Policy Rollout (1x speed)
1 Demonstration Trajectory


The Data Pyramid: Going Forward
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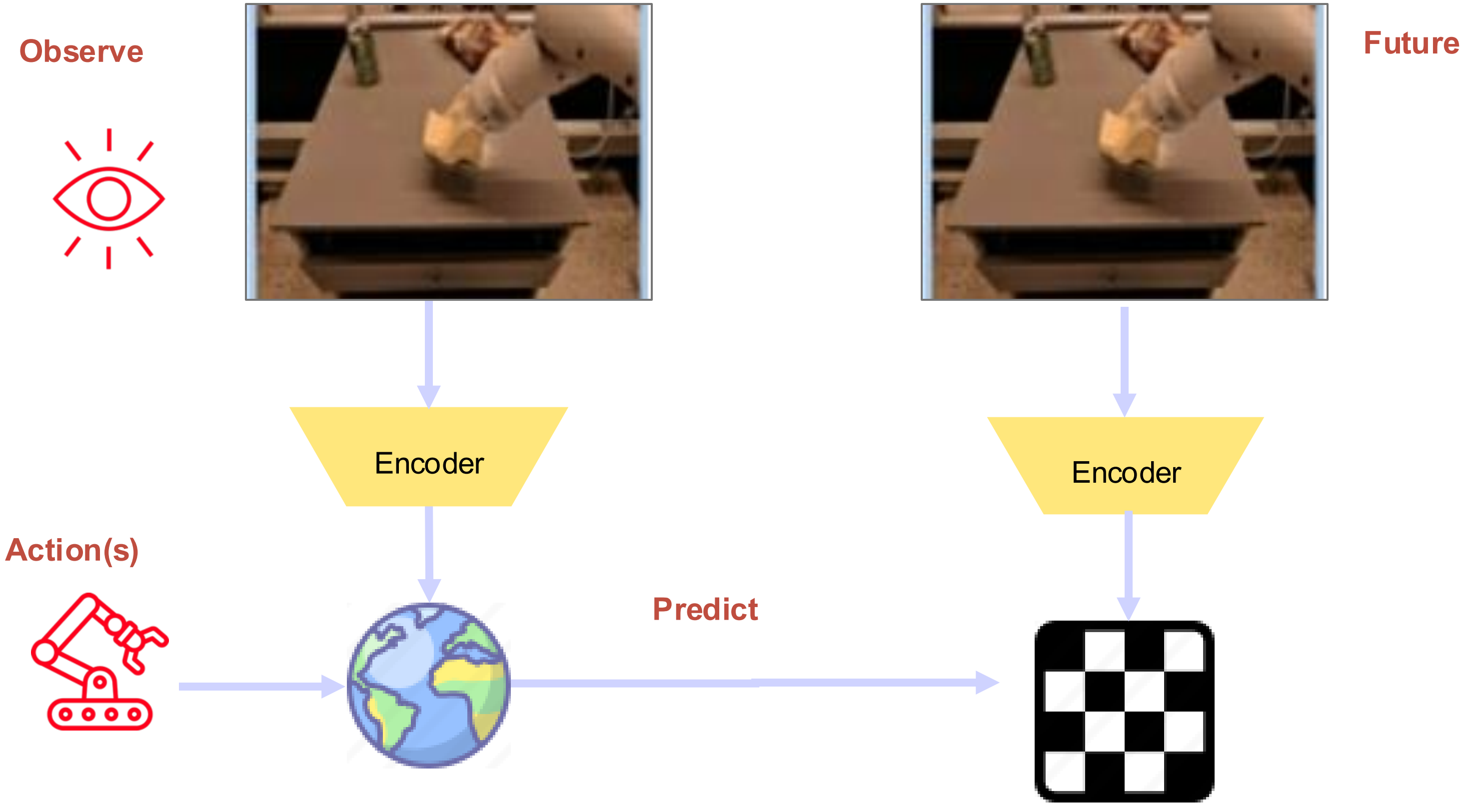
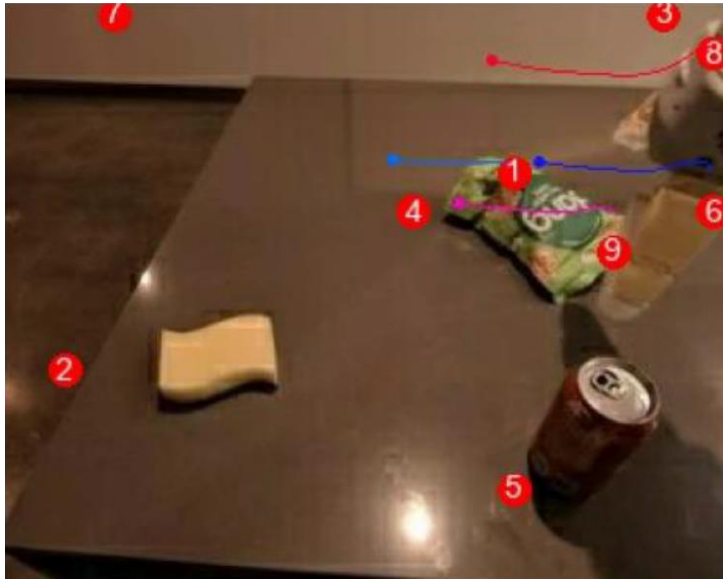
Scarce, Very Expensive, High-quality
Controlled In-Lab Human Data
Expensive, Good-quality, Less Expert-demanding
In-the-wild Human Data
(Epic Kitchen, Egodex, etc.)
Large quantity, Lower-quality

Foundation Model for Robotics

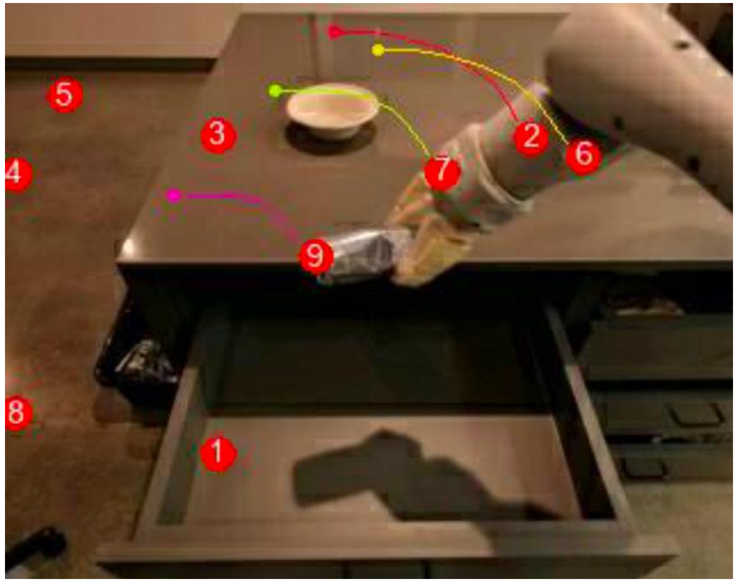
Alternative I: Latent World Model



Robotics is all about DOING SOMETHING



Push chip bag to the left



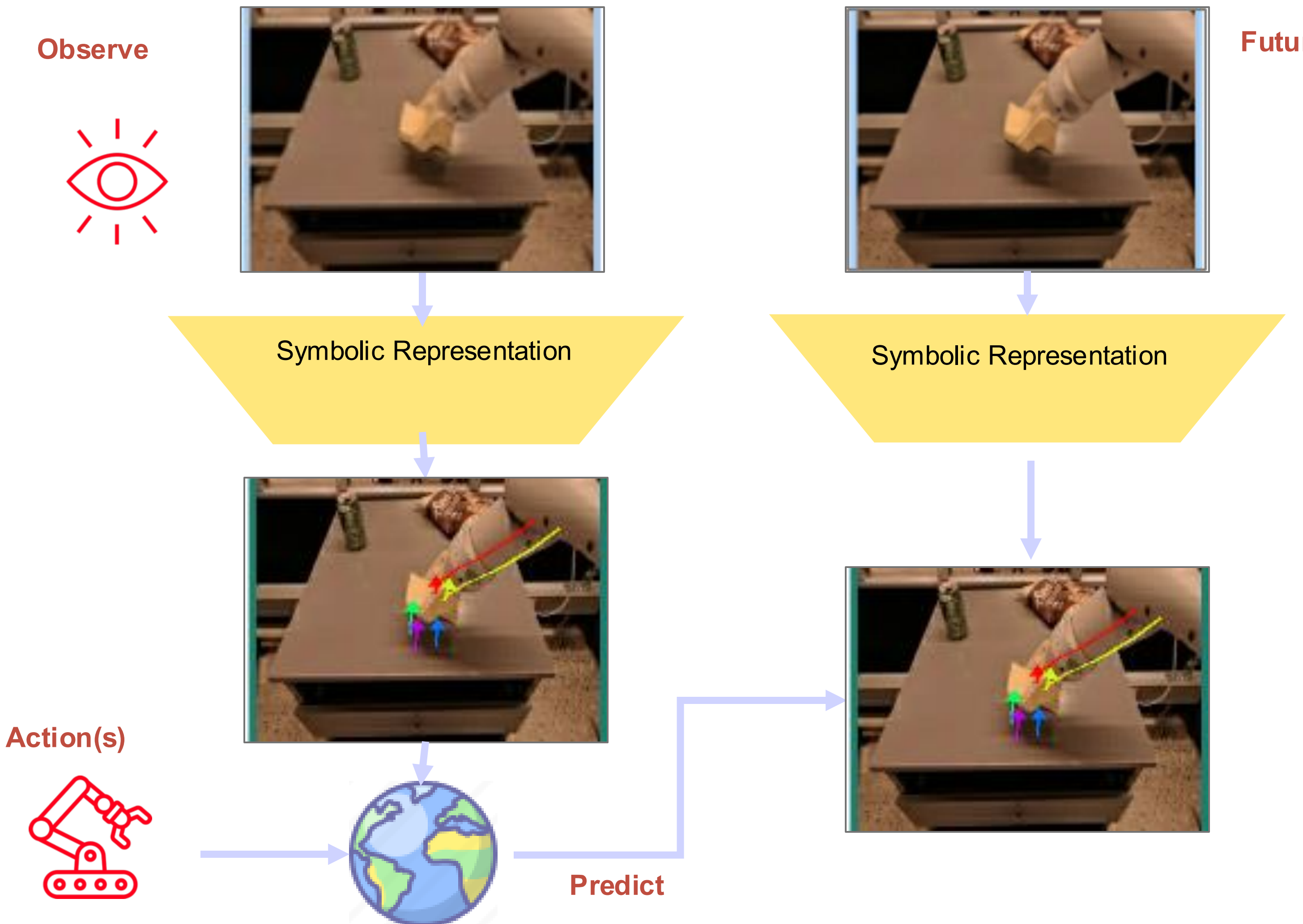
Put can onto table

Stir Food

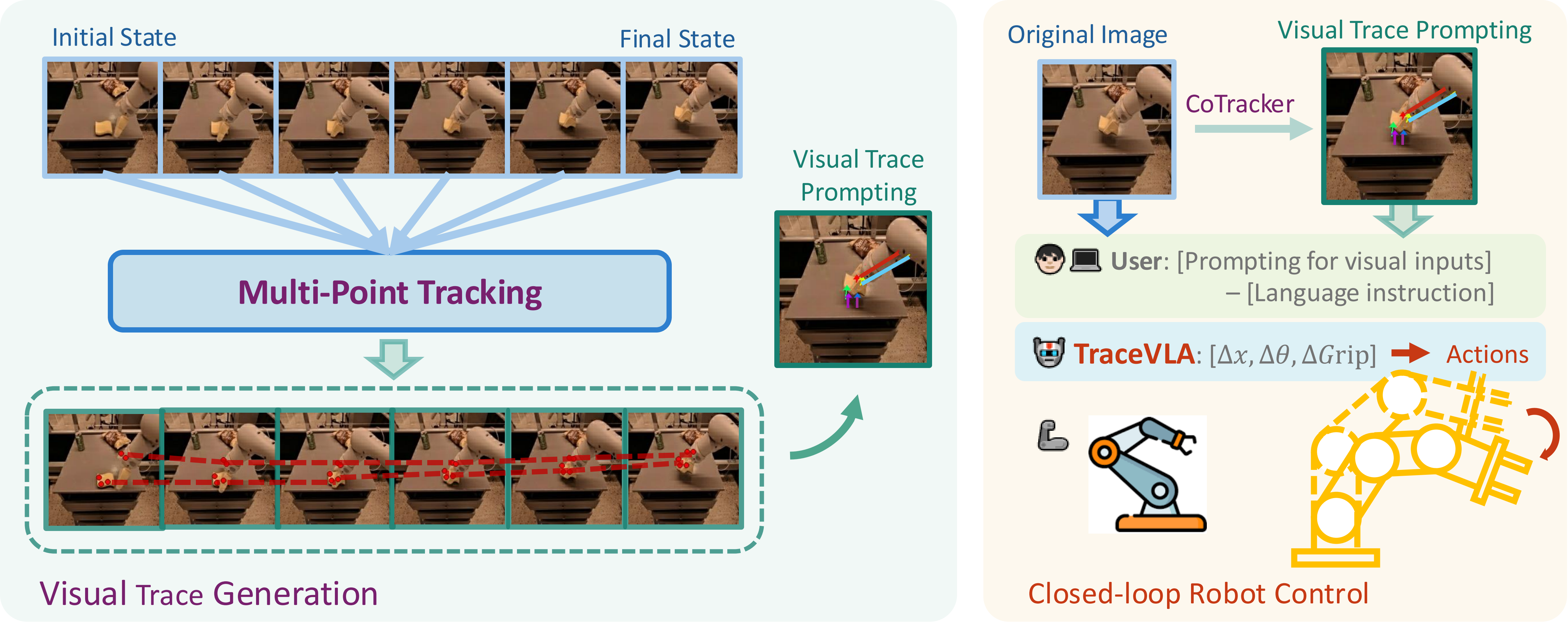
Key-point based symbolic representation

Foundation Model for Robotics

Alternative II: Symbolic Representation

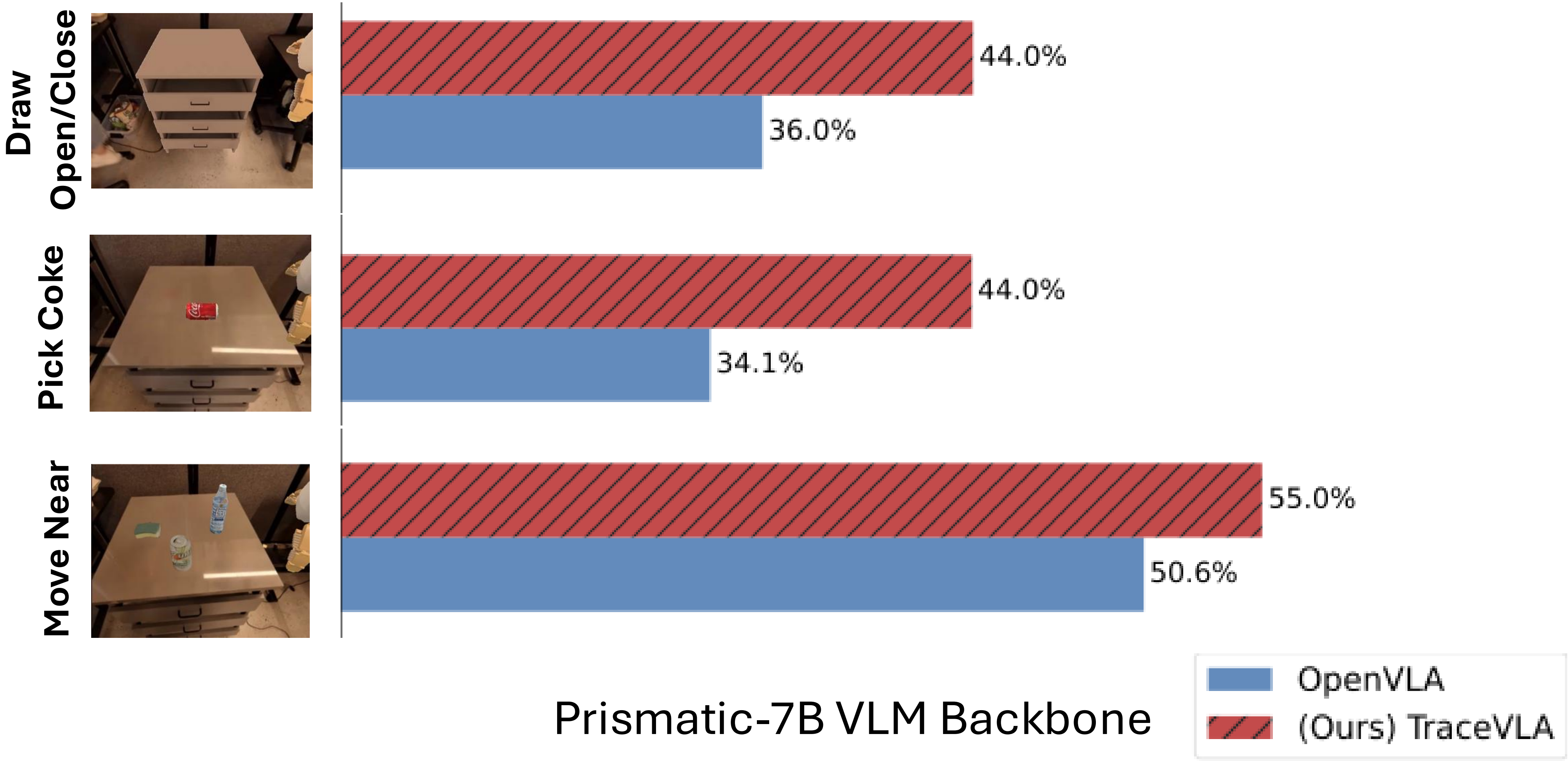


Improving VLAs



Zheng, Liang, Huang, Gao, Daumé III, Kolobov, H., Yang. “TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies.” ICLR 2025.

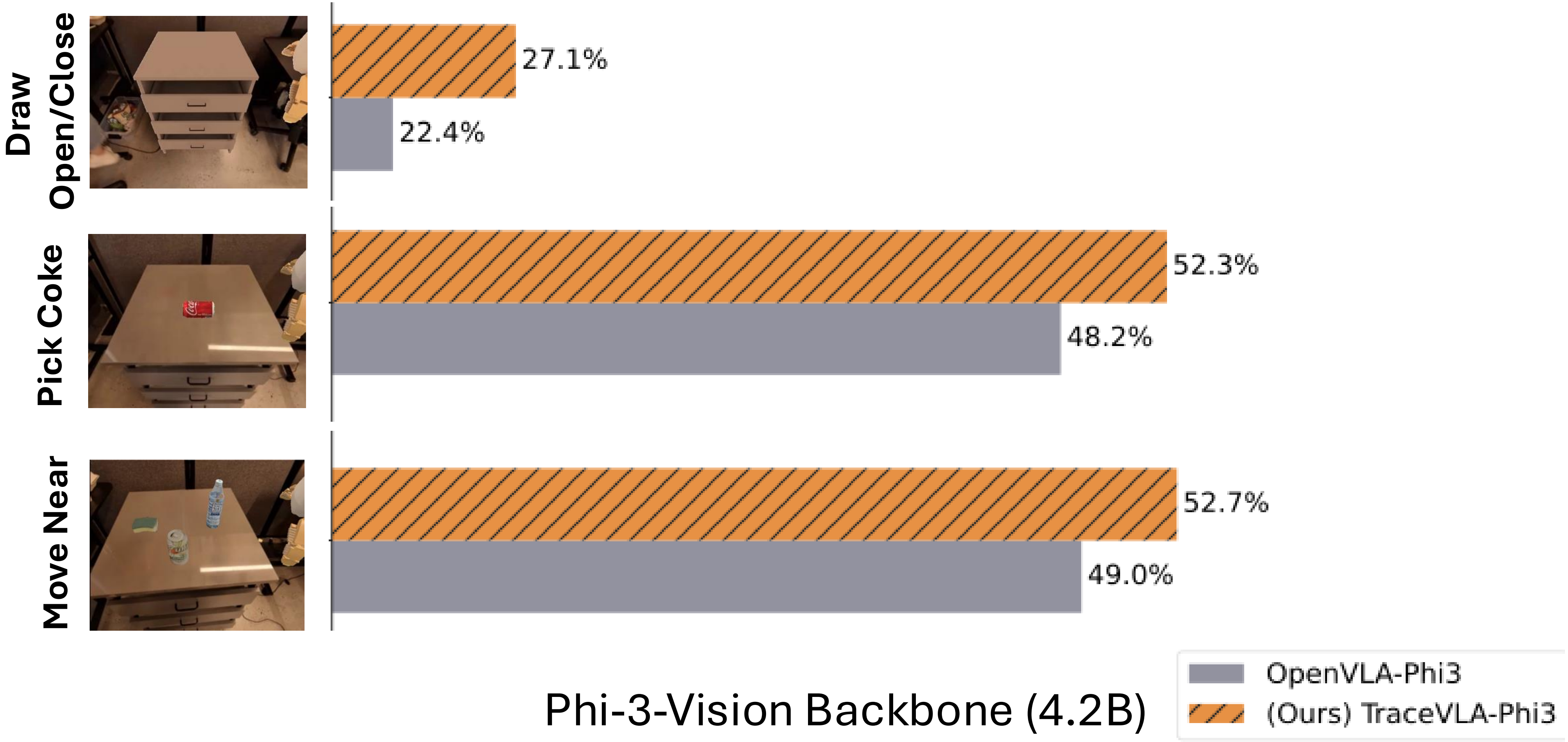
TraceVLA Improves OpenVLA Big Model



Prismatic-7B VLM Backbone

Zheng, Liang, Huang, Gao, Daumé III, Kolobov, H., Yang. "TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies." ICLR 2025.

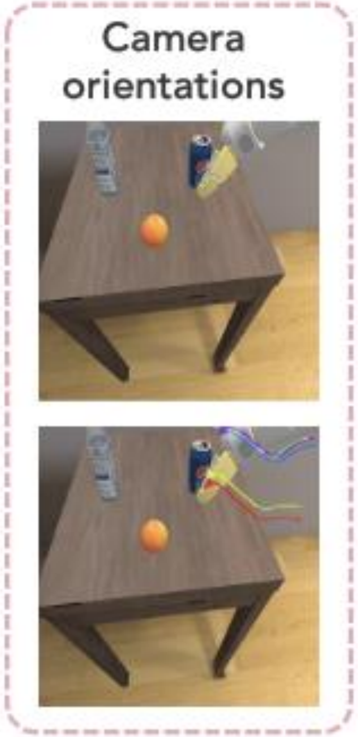
TraceVLA Improves OpenVLA Small Model



Phi-3-Vision Backbone (4.2B)

Zheng, Liang, Huang, Gao, Daumé III, Kolobov, H., Yang. “TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies.” ICLR 2025.

TraceVLA Improves OpenVLA Robustness/Generalization



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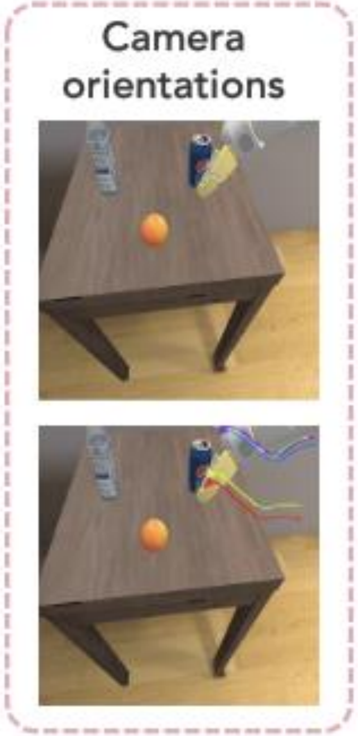
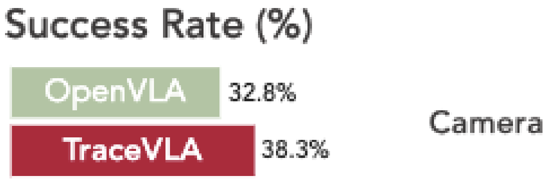
TraceVLA Improves OpenVLA Robustness/Generalization





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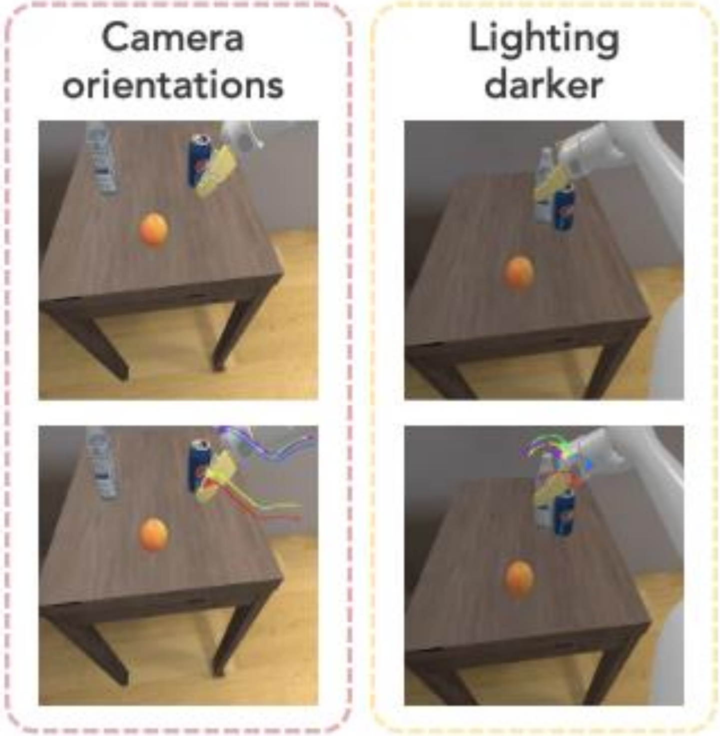

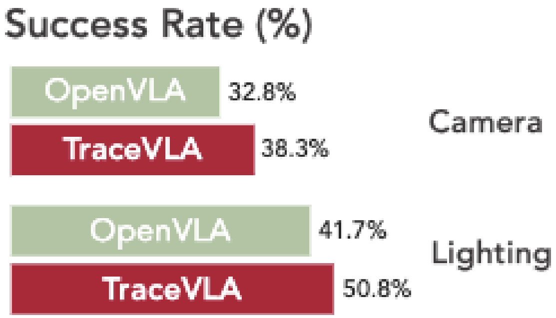

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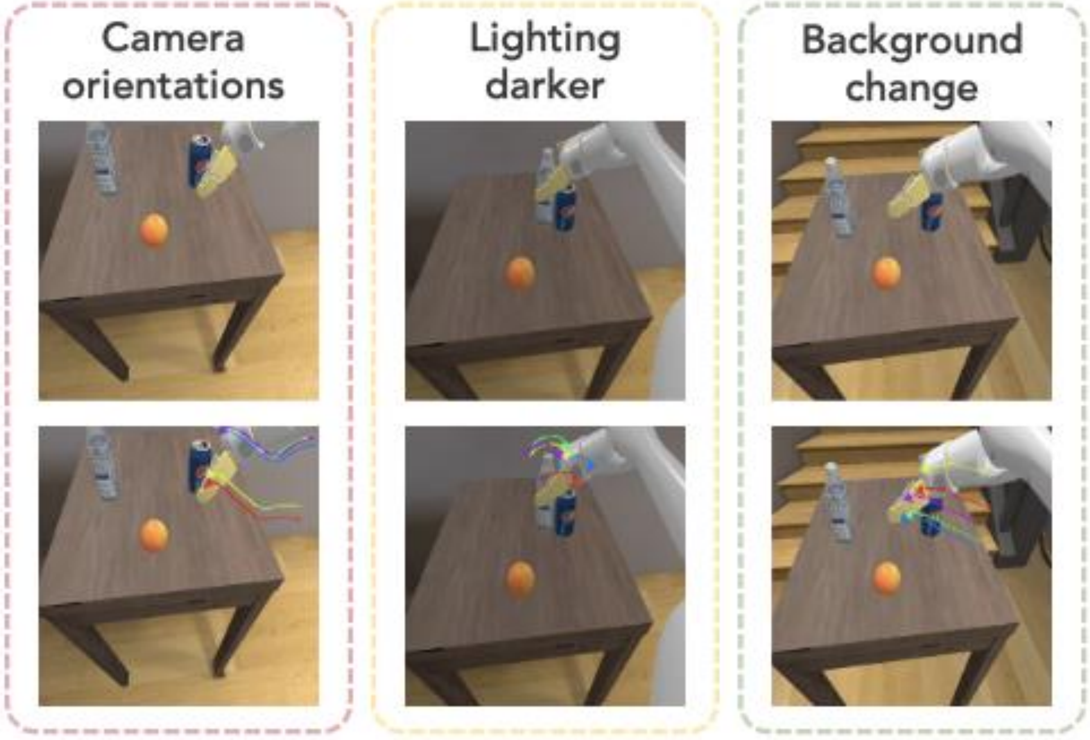
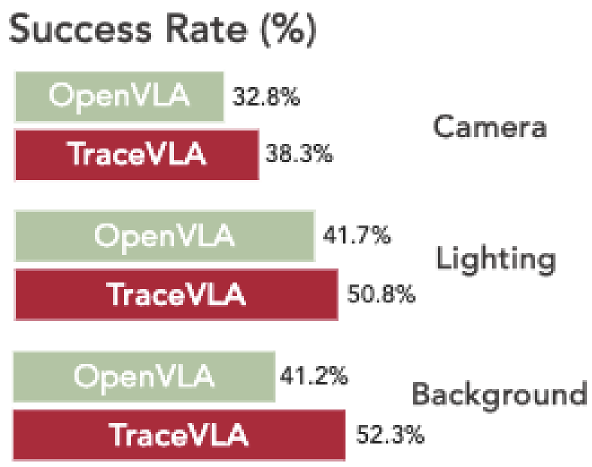
TraceVLA Improves OpenVLA Robustness/Generalization





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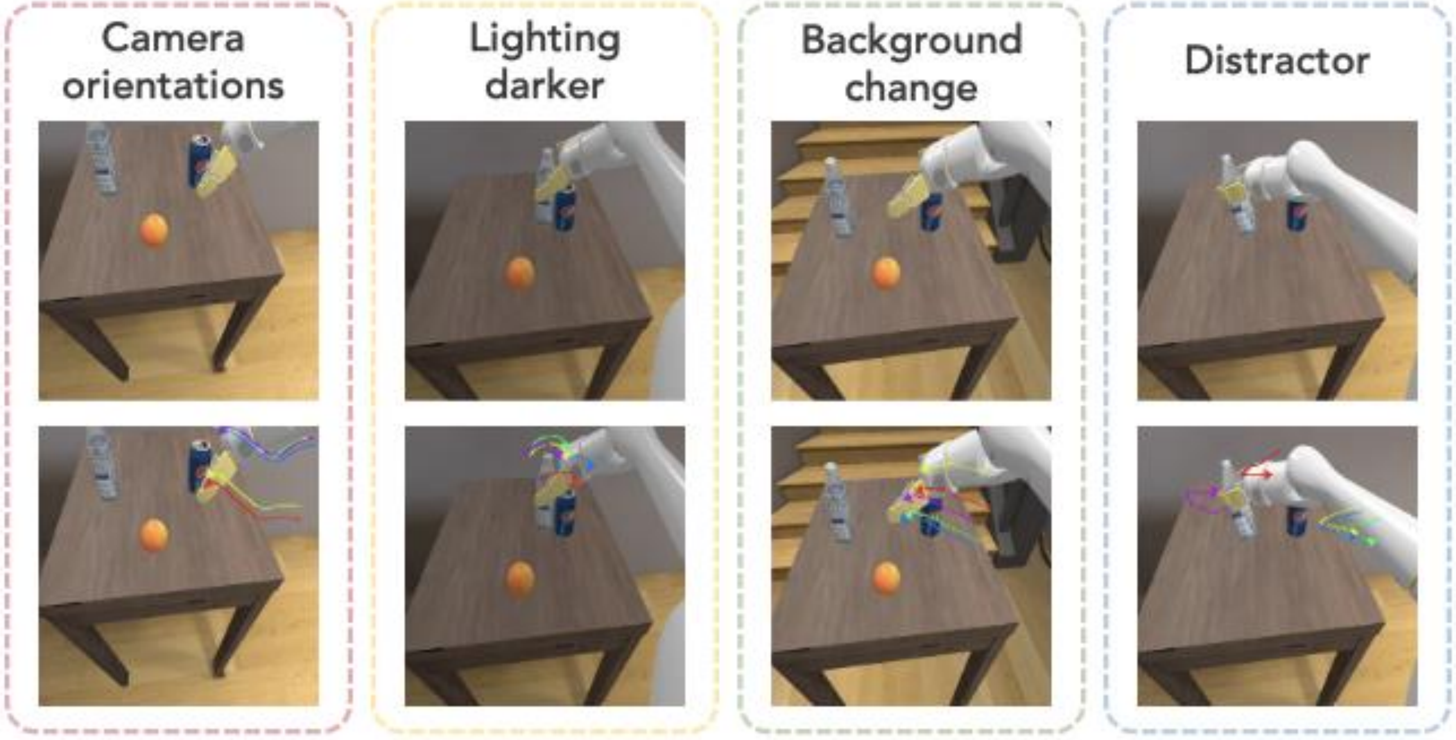
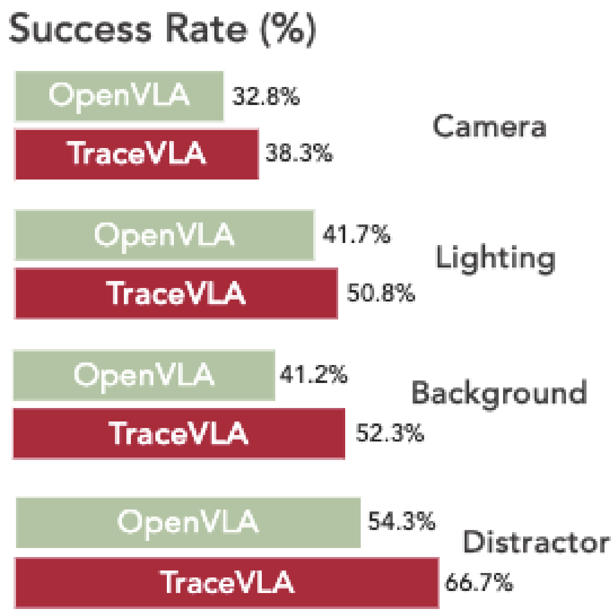
TraceVLA Improves OpenVLA Robustness/Generalization





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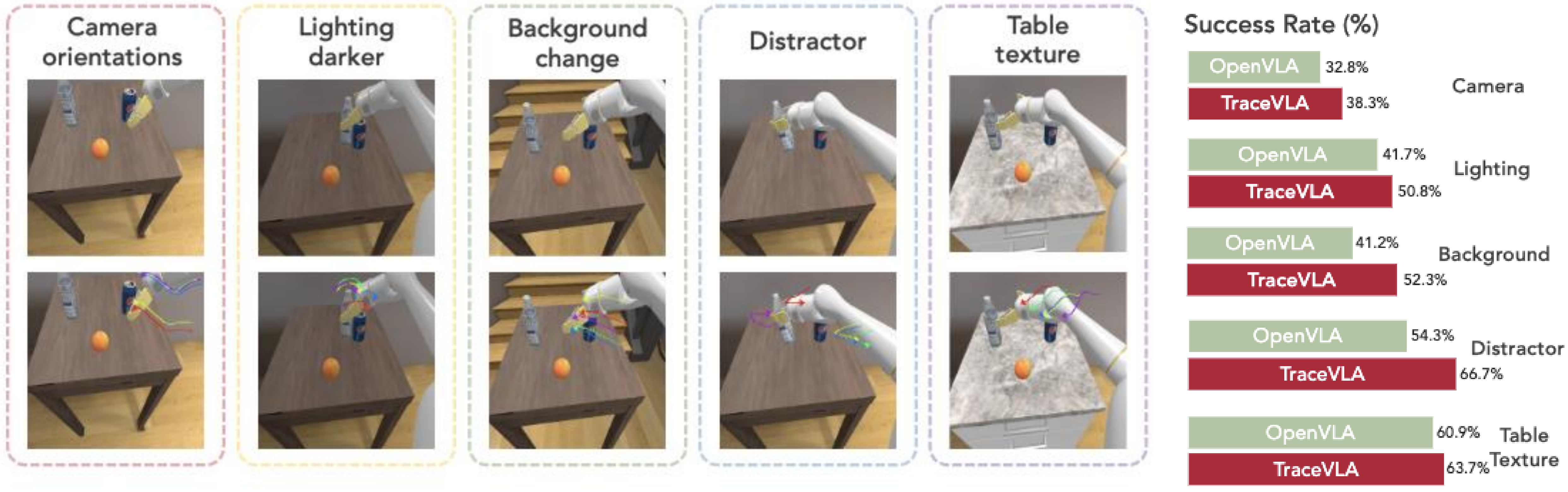
TraceVLA Improves OpenVLA Robustness/Generalization



Zheng, Liang, Huang, Gao, Daumé III, Kolobov, H., Yang. “TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies.” ICLR 2025.

Real-world WidowX-250 Robotic Manipulation Experiments

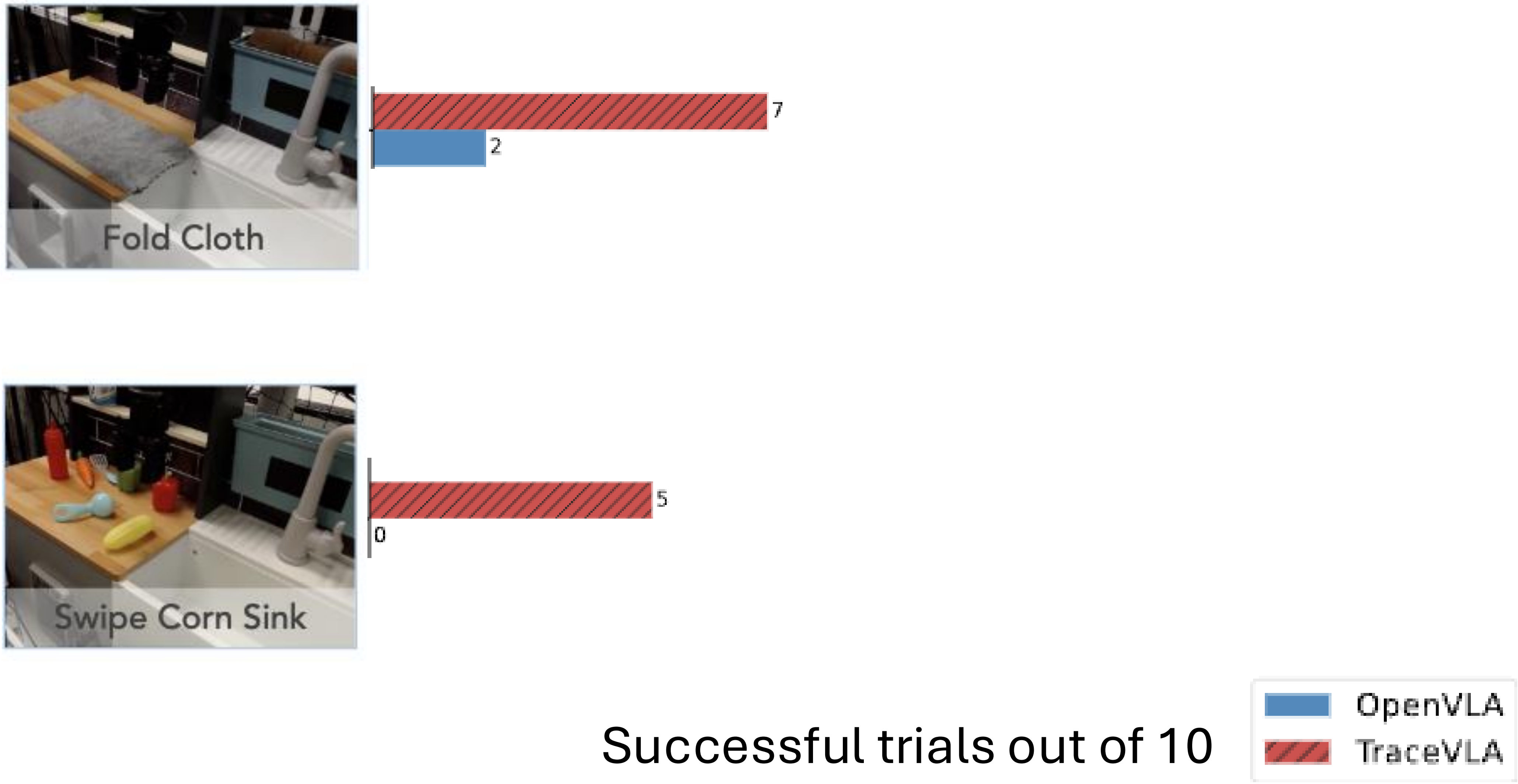
Zheng, Liang, Huang, Gao, Daumé III, Kolobov, H., Yang. "TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies." ICLR 2025.

Real-world WidowX-250 Robotic Manipulation Experiments

Successful trials out of 10

Zheng, Liang, Huang, Gao, Daumé III, Kolobov, H., Yang. “TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies.” ICLR 2025.

Real-world WidowX-250 Robotic Manipulation Experiments



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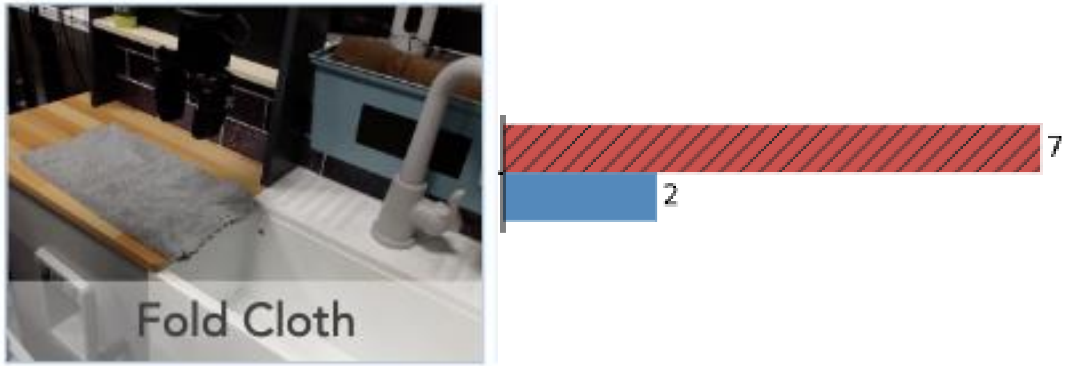
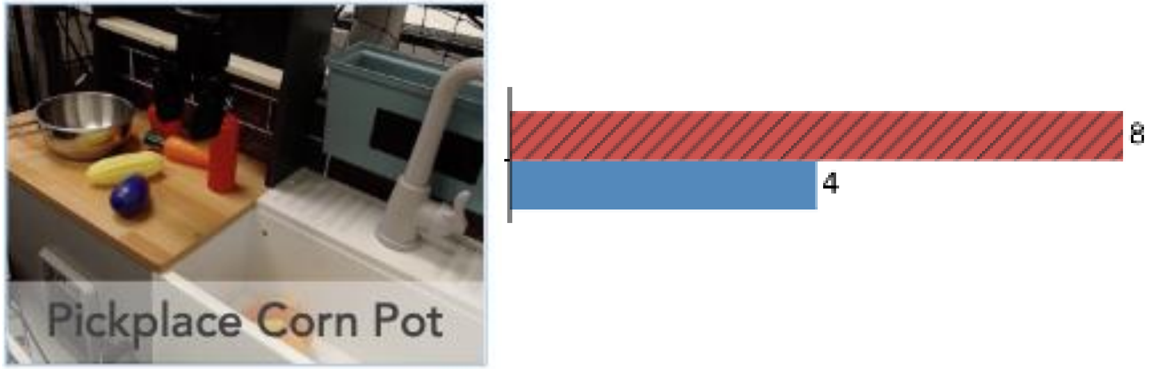
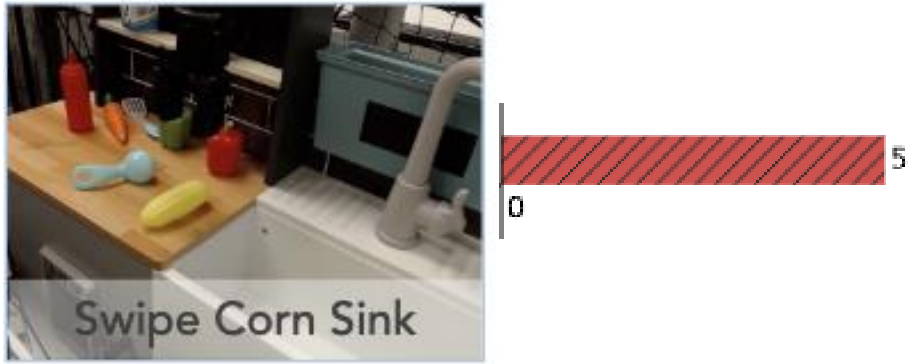
Real-world WidowX-250 Robotic Manipulation Experiments







Successful trials out of 10

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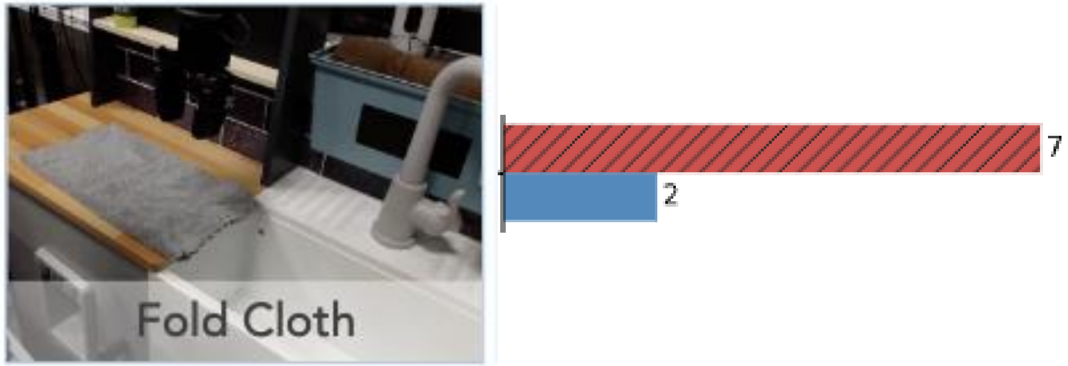
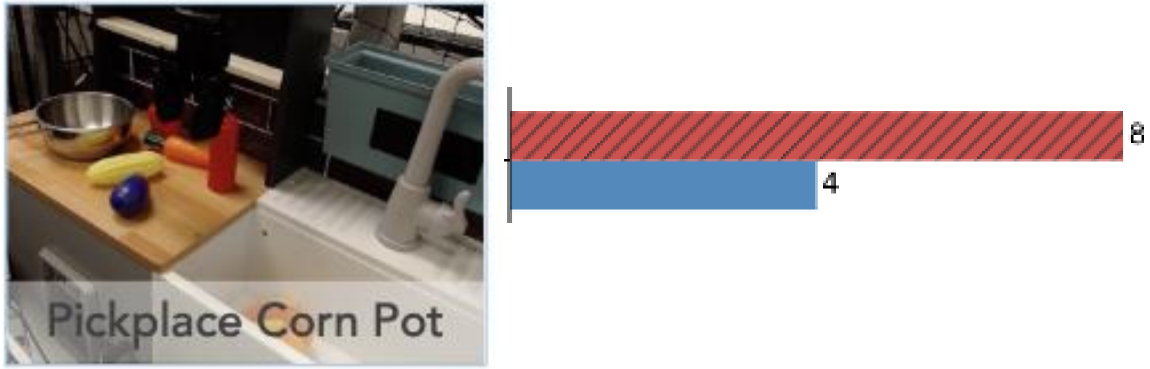
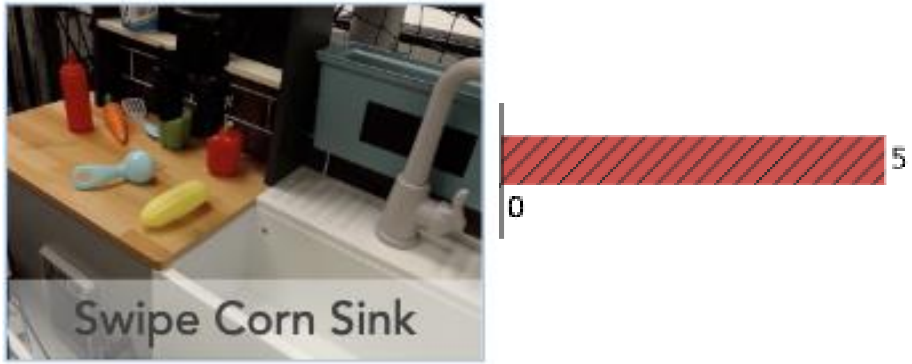
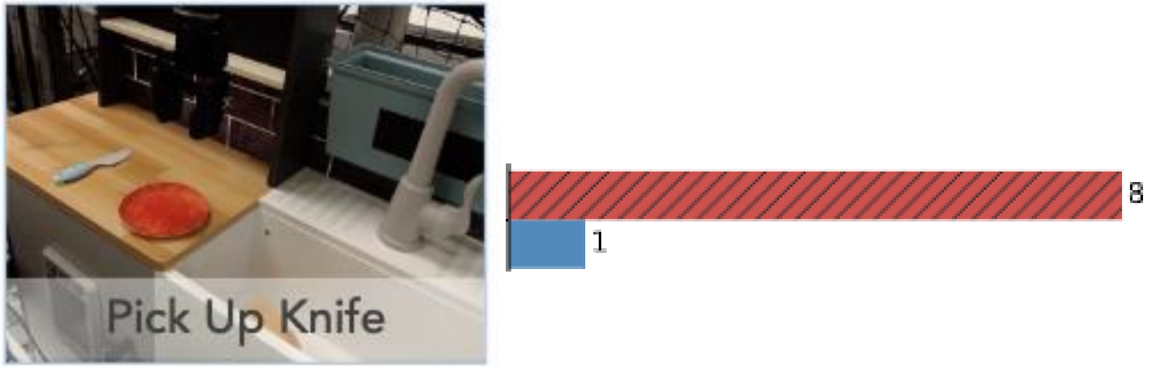
Real-world WidowX-250 Robotic Manipulation Experiments



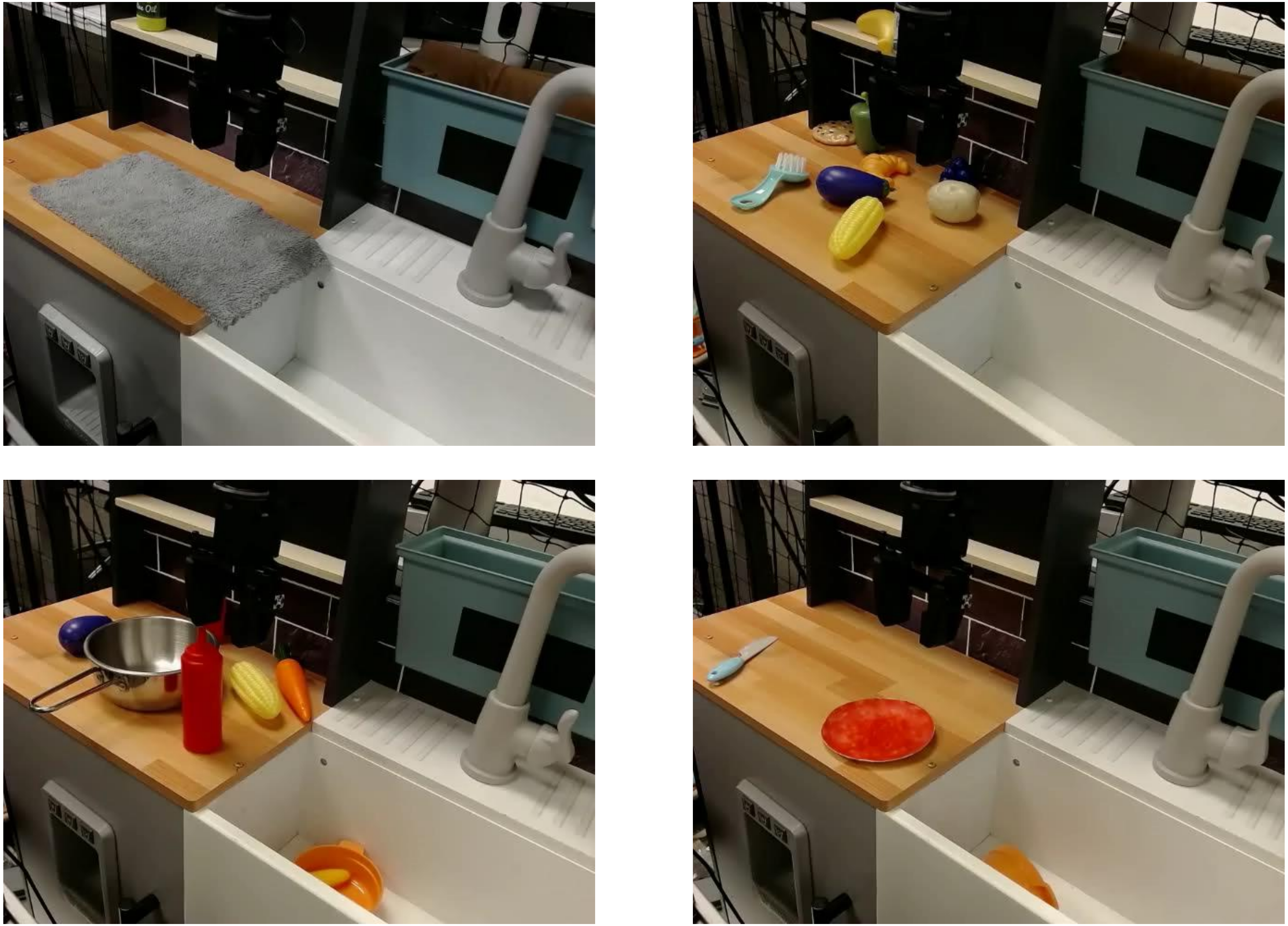






Successful trials out of 10

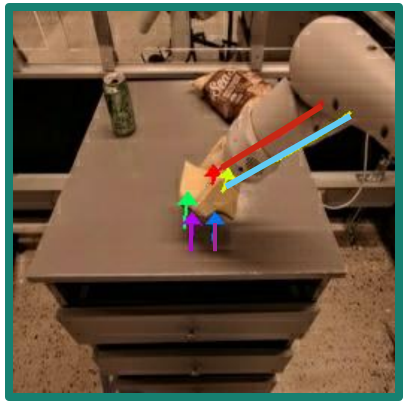
Zheng, Liang, Huang, Gao, Daumé III, Kolobov, H., Yang. “TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies.” ICLR 2025.



Why Visual Trace Helps?

Original Image

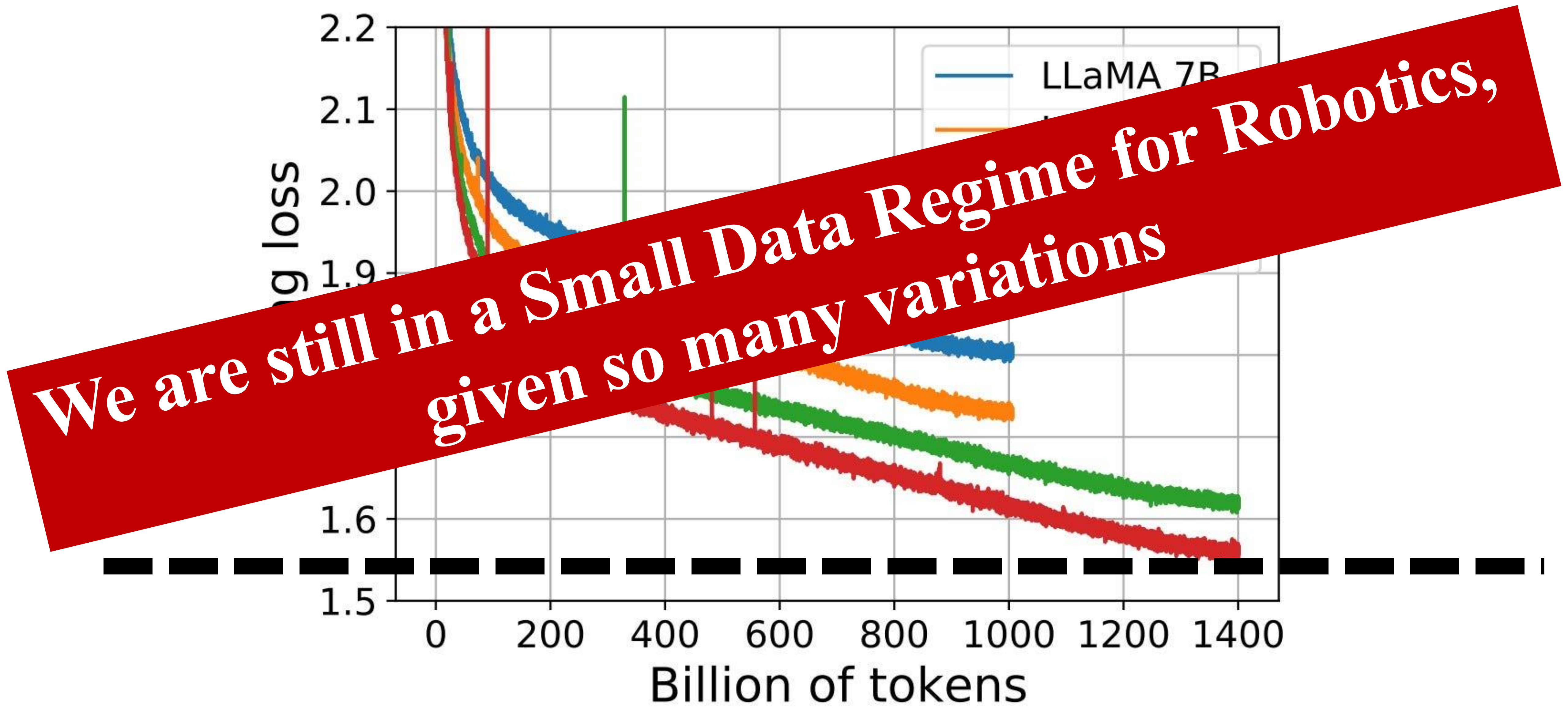
Visual Trace Prompting



Historical information

Compact representation

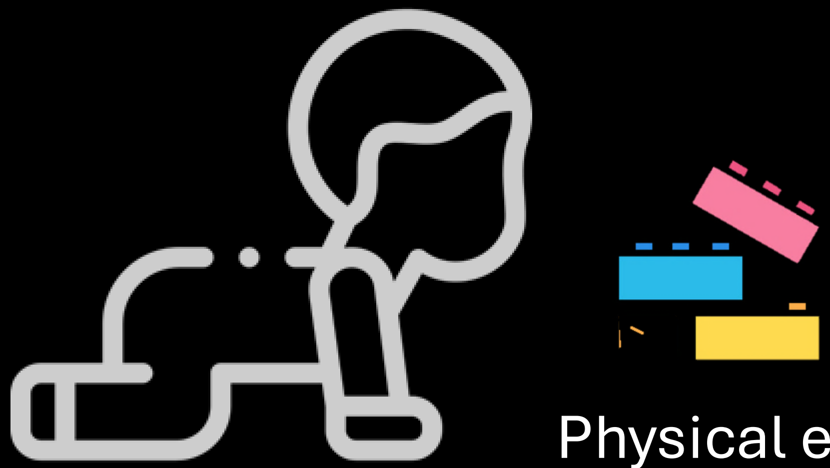
Spatial calibration


We are still in a Small Data Regime for Robotics,
given so many variations
LLaMA 7B
2.2
2.1
2.0
1.9
1.6
1.5
0
200
400
600
800
1000
1200
1400
Billion of tokens

How to solve the small data problem?

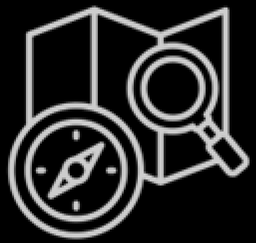
Can the model collect data by itself?

Exploration!

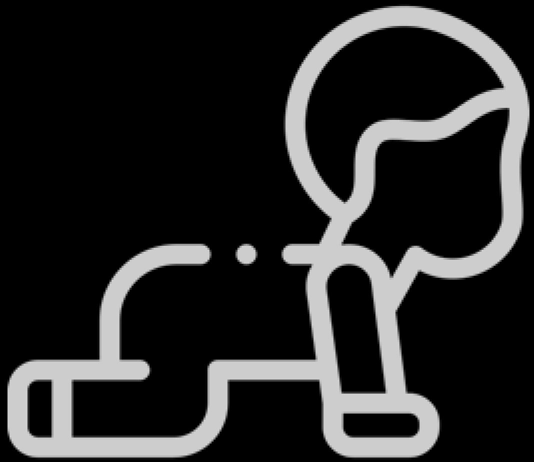


Physical env

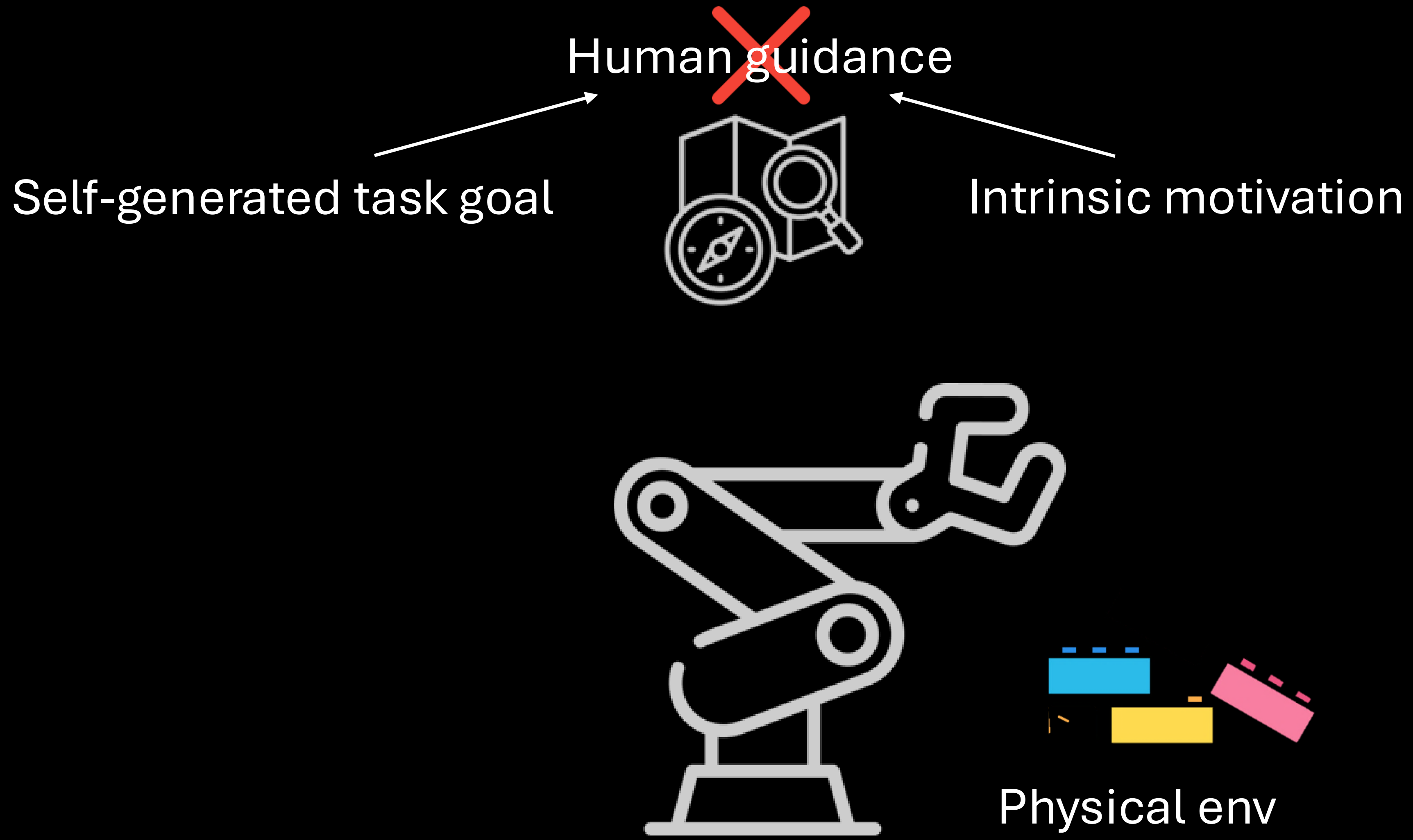
Self-generated task goal



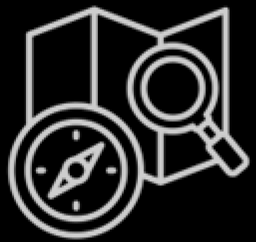
Intrinsic motivation



Physical env


Human guidance
Self-generated task goal
Intrinsic motivation
Physical env

Exploration RL



~~Self-generated task goal~~

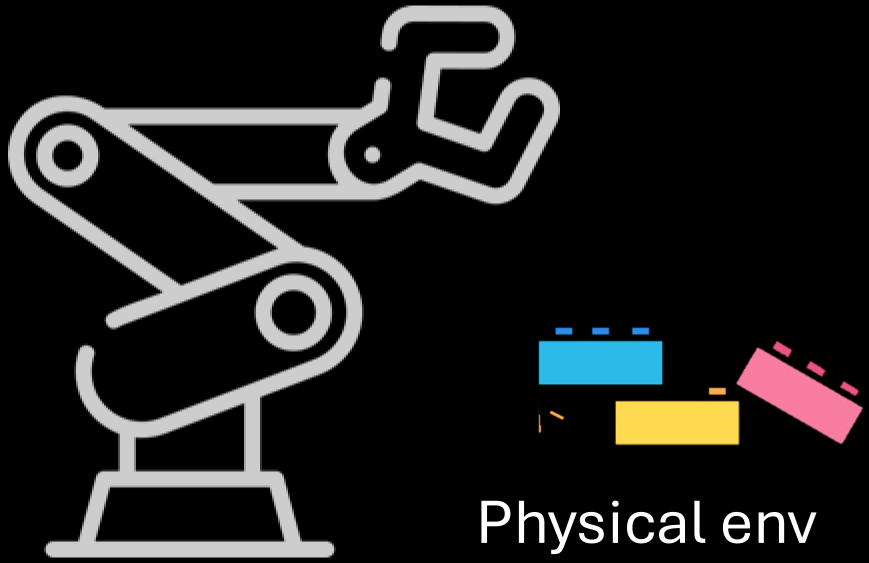
Goal sampling methods

Skew-fit (V. H. Pong et al, 2020), MEGA (S. Pitis et al, 2020)

~~Intrinsic motivation~~

Train an intrinsic reward model

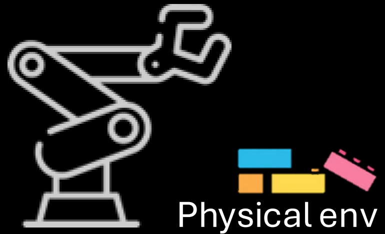
ICM (D. Pathak et al, 2017), RE3 (Y. Seo et al, 2021)

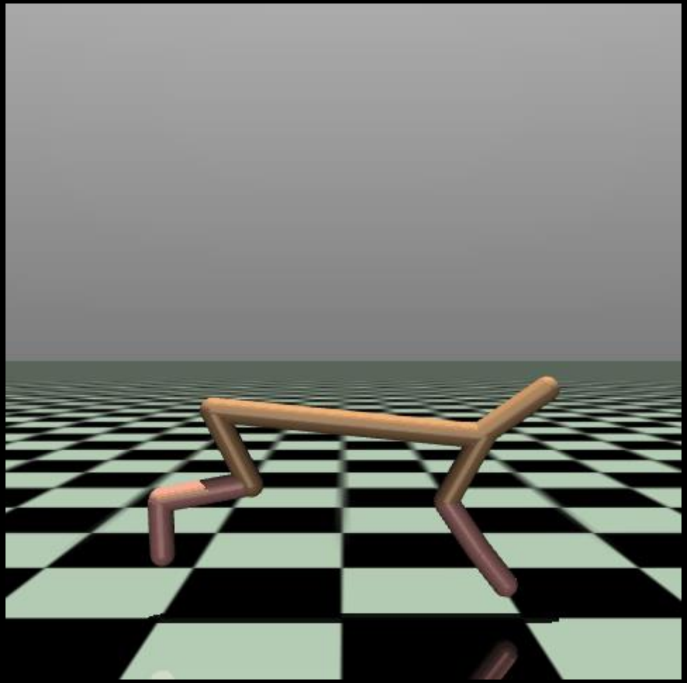


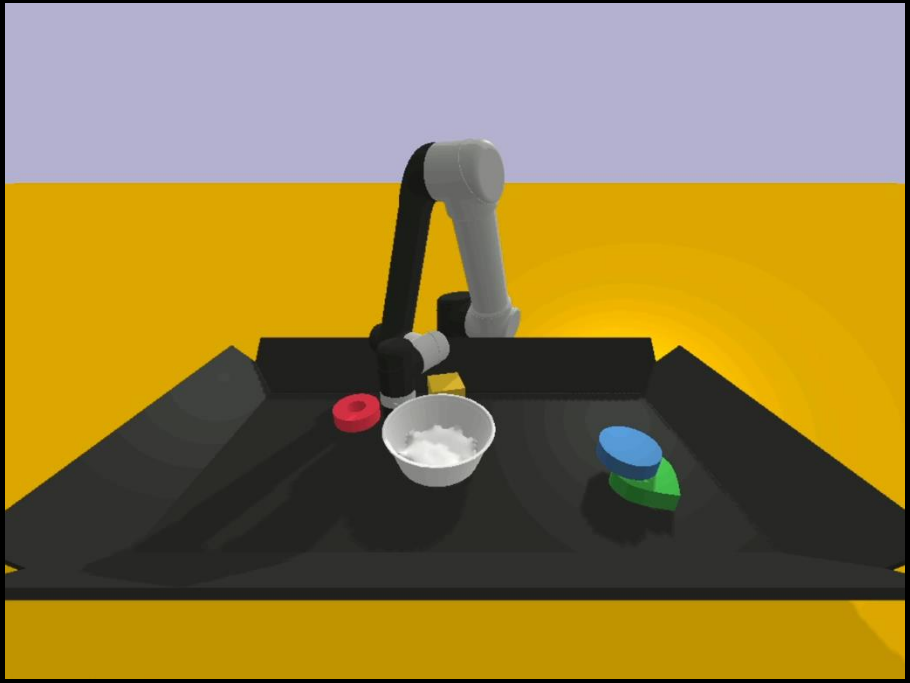
Physical env

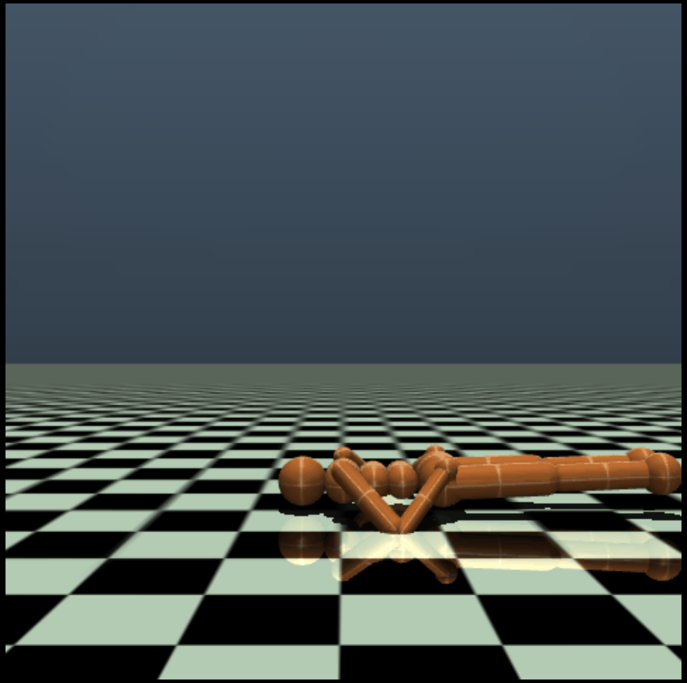
Works with minimum human effort!

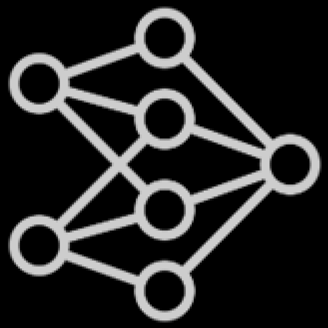
Exploration RL



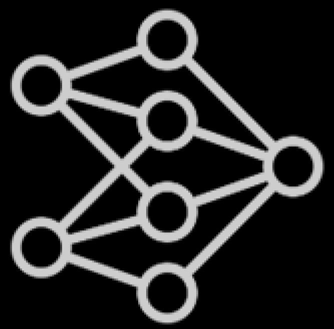






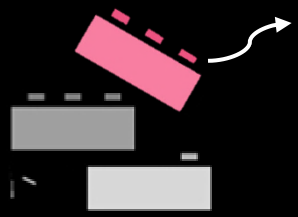


VLM

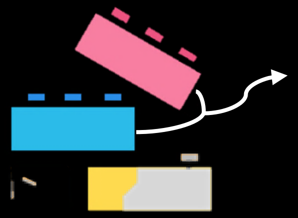


VLM

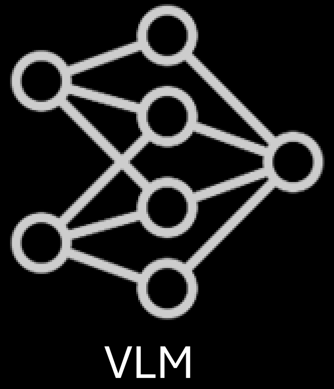
Object semantics



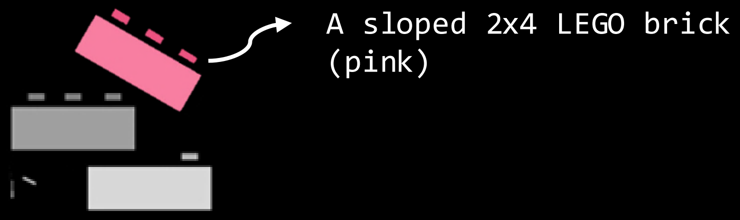
Spatial relations



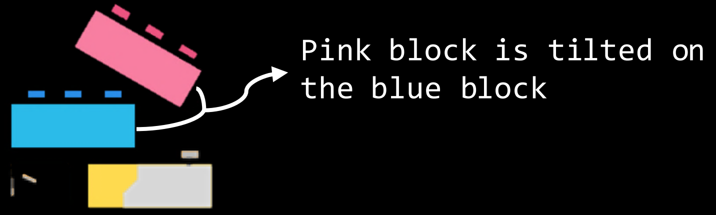
Potential outcomes



Object semantics



Spatial relations



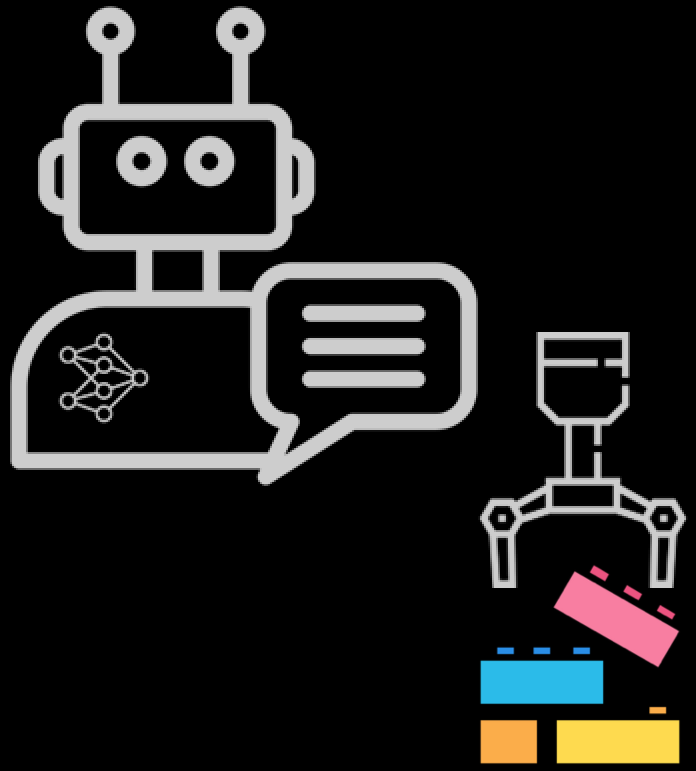
Potential outcomes

Object semantics

Spatial relations

Potential outcomes

A VLM-powered AI agent



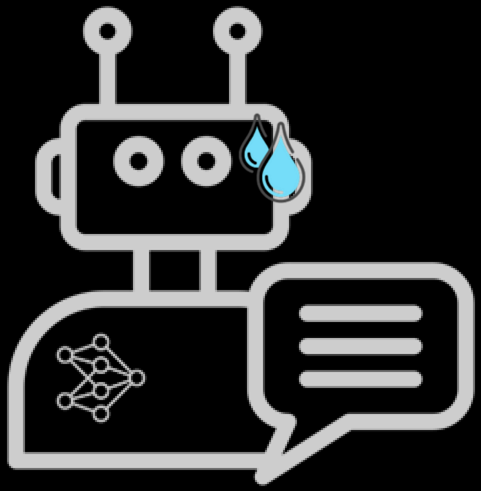
Object semantics

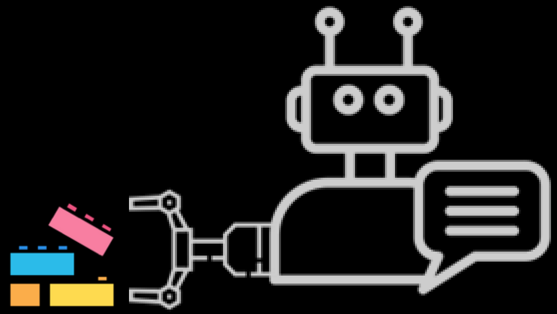
Spatial relations

capable of semantic understanding and planning,
but disconnected from physical experience

Potential outcomes

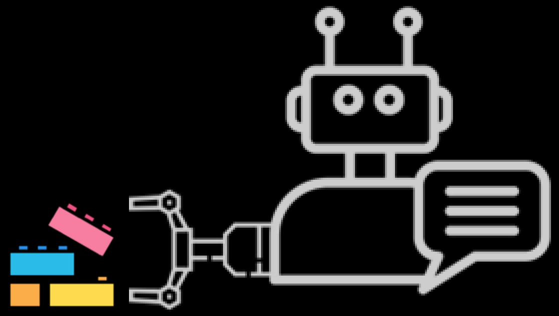
A VLM-powered AI agent





Imagine, Verify, Execute

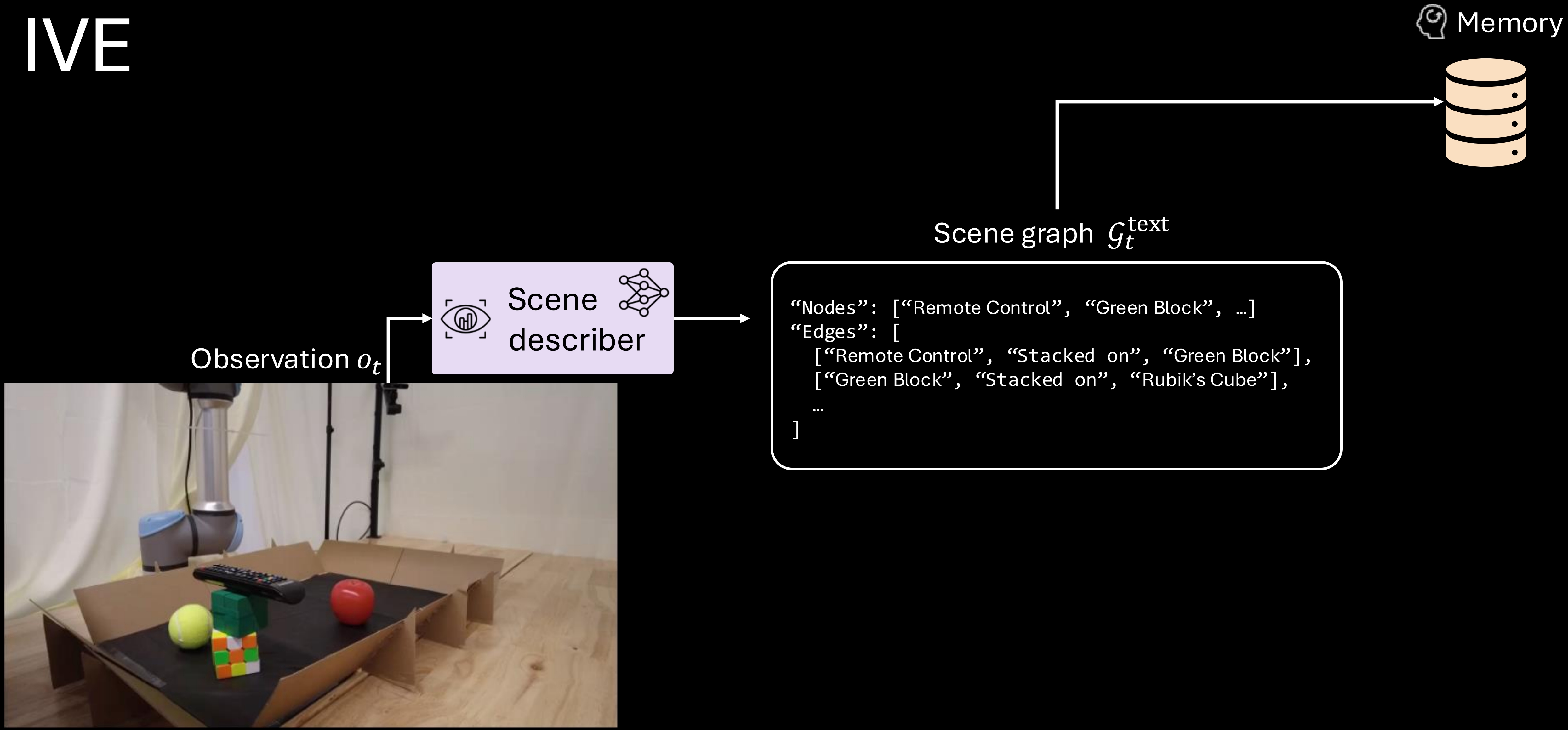
Lee, Ekpo, Liu, H*, Shrivastava*, Huang*. Imagine, Verify, Execute: Memory-Guided Agentic Exploration with Vision-Language Models. CoRL 2025.



I V E

Lee, Ekpo, Liu, H*, Shrivastava*, Huang*. Imagine, Verify, Execute: Memory-Guided Agentic Exploration with Vision-Language Models. CoRL 2025.

IVE


Memory
Scene graph $\mathcal{G}_t^{\text{text}}$
"Nodes": ["Remote Control", "Green Block", …]
"Edges": [
["Remote Control", "Stacked on", "Green Block"],
["Green Block", "Stacked on", "Rubik's Cube"],
…
]
Scene describer
Observation o_t

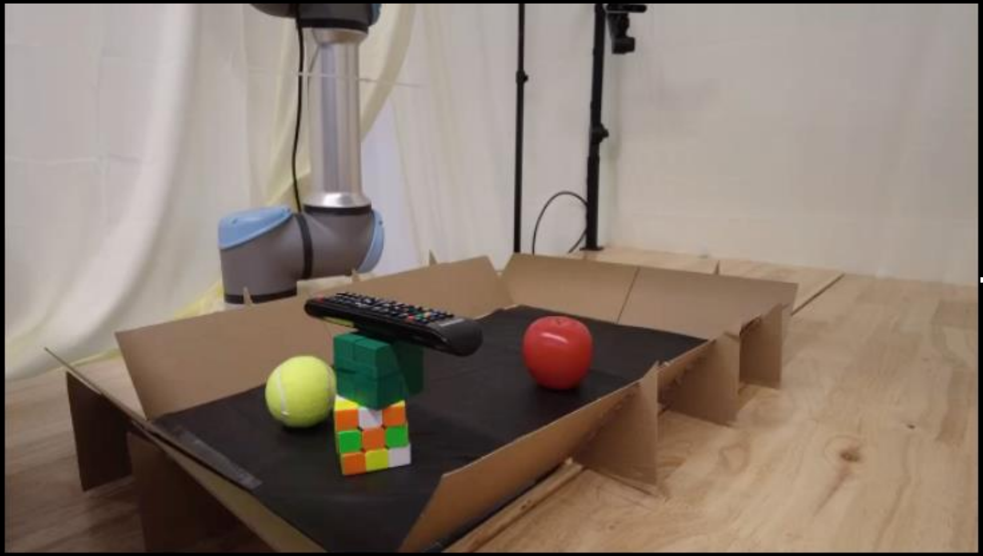
IVE

Memory

Scene graph $\mathcal{G}_t^{\text{text}}$

```
"Nodes": ["Remote Control", "Green Block", …]
"Edges": [
  ["Remote Control", "Stacked on", "Green Block"],
  ["Green Block", "Stacked on", "Rubik's Cube"],
  …
]
```

Observation o_t



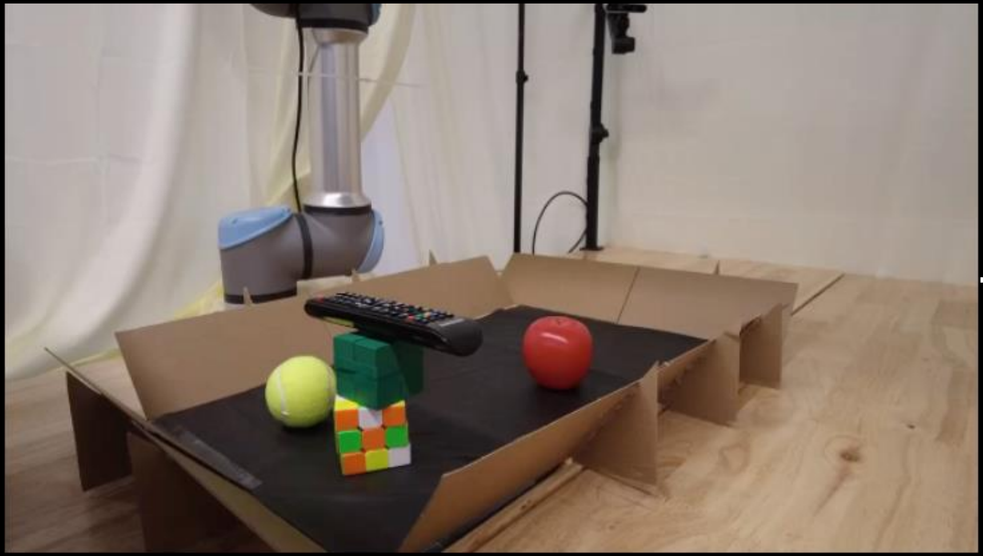
Explorer

IVE

Scene graph $\mathcal{G}_t^{\text{text}}$

```
"Nodes": ["Remote Control", "Green Block", …]
"Edges": [
  ["Remote Control", "Stacked on", "Green Block"],
  ["Green Block", "Stacked on", "Rubik's Cube"],
  …
]
```

Observation o_t



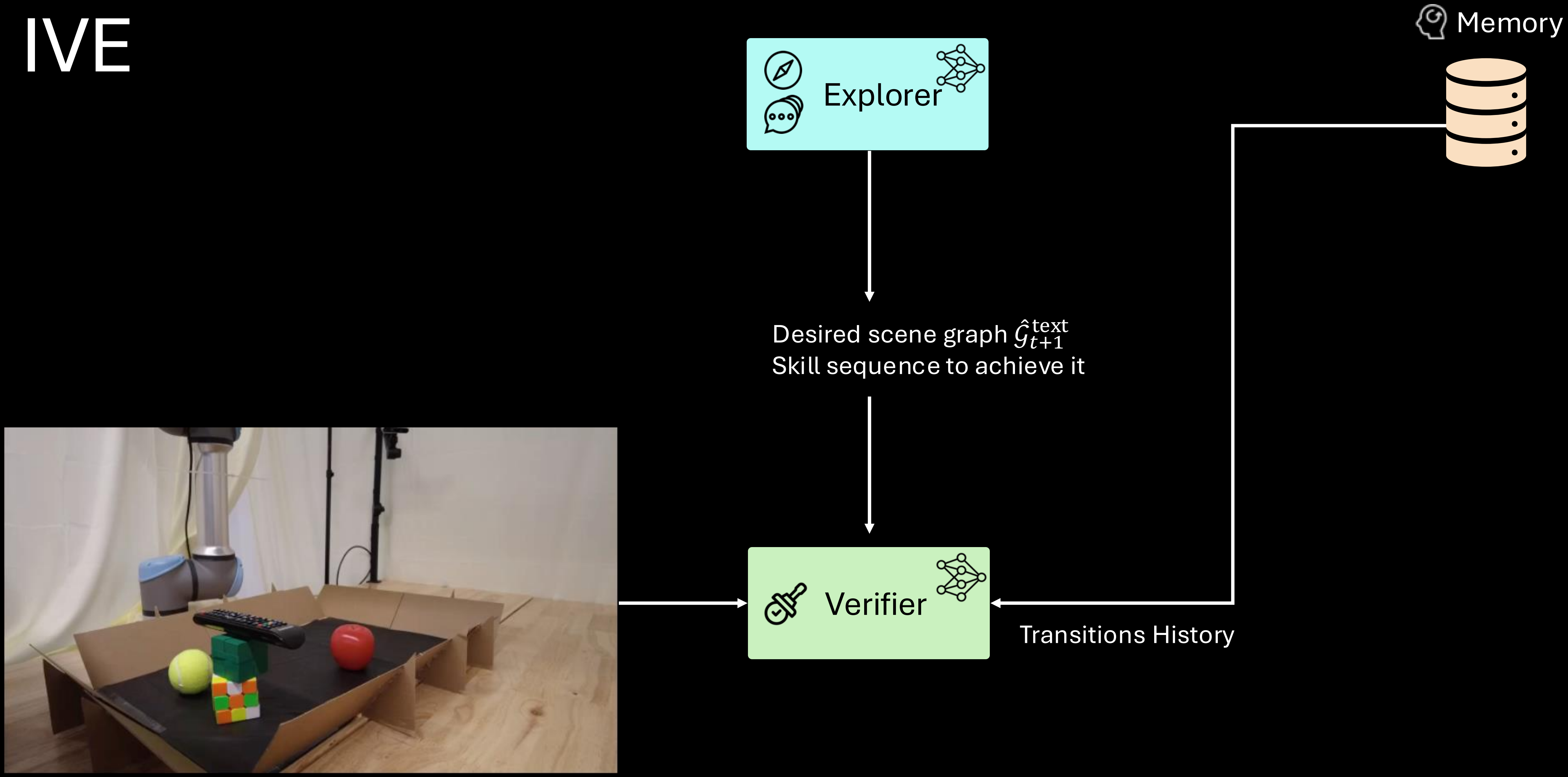
Relevant scene graphs to $\mathcal{G}_t^{\text{text}}$

Memory

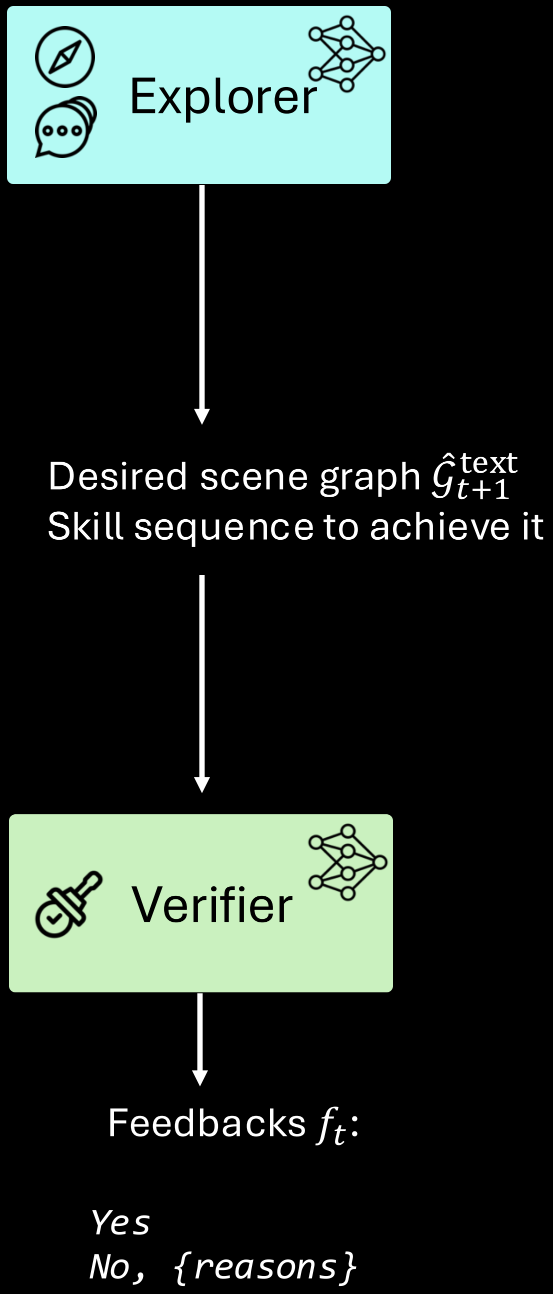
Explorer

Desired scene graph $\hat{\mathcal{G}}_{t+1}^{\text{text}}$
Skill sequence to achieve it

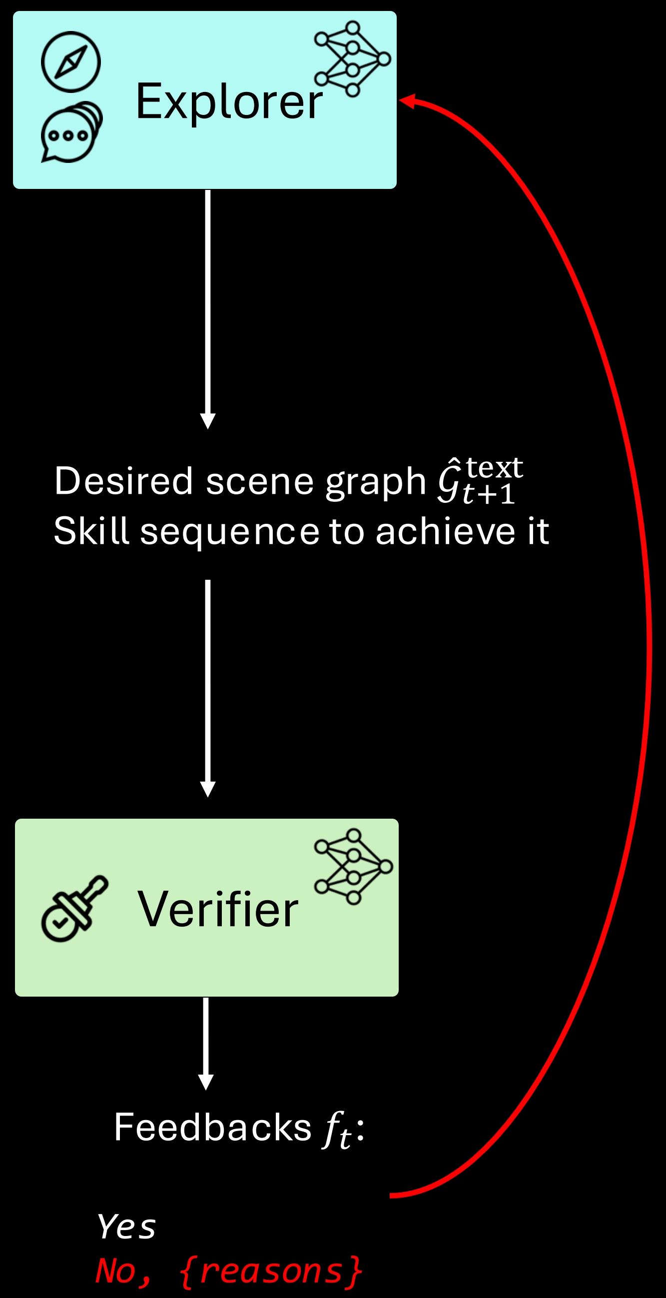
IVE


Memory
Explorer
Desired scene graph $\hat{\mathcal{G}}_{t+1}^{\text{text}}$
Skill sequence to achieve it
Verifier
Transitions History

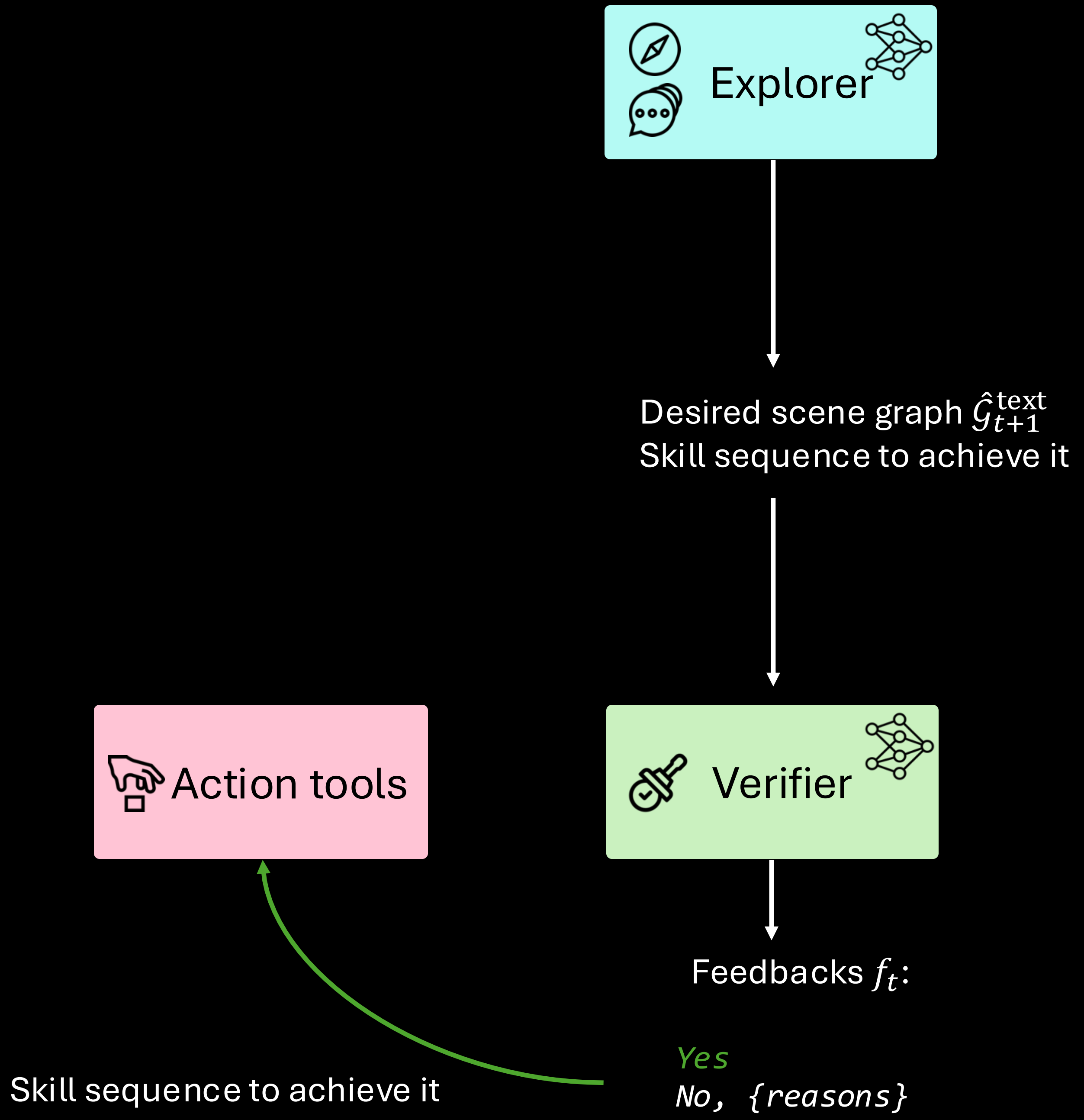
IVE


Explorer
Desired scene graph $\hat{\mathcal{G}}_{t+1}^{\text{text}}$
Skill sequence to achieve it
Verifier
Feedbacks f_t:
Yes
No, {reasons}

IVE

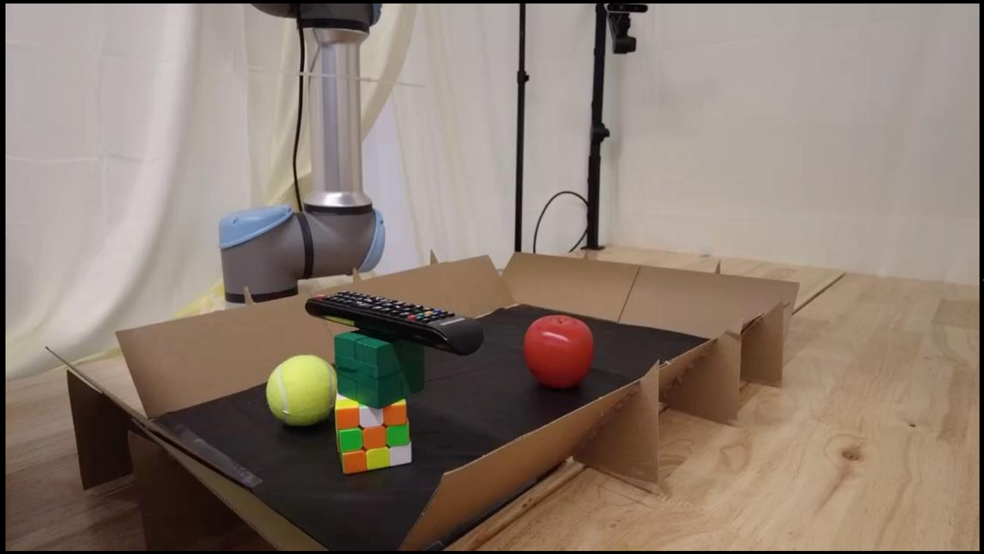

Explorer
Desired scene graph $\hat{\mathcal{G}}_{t+1}^{\text{text}}$
Skill sequence to achieve it
Verifier
Feedbacks f_t:
Yes
No, {reasons}

IVE


Explorer
Desired scene graph $\hat{\mathcal{G}}_{t+1}^{\text{text}}$
Skill sequence to achieve it
Action tools
Verifier
Feedbacks f_t:
Yes
No, {reasons}
Skill sequence to achieve it

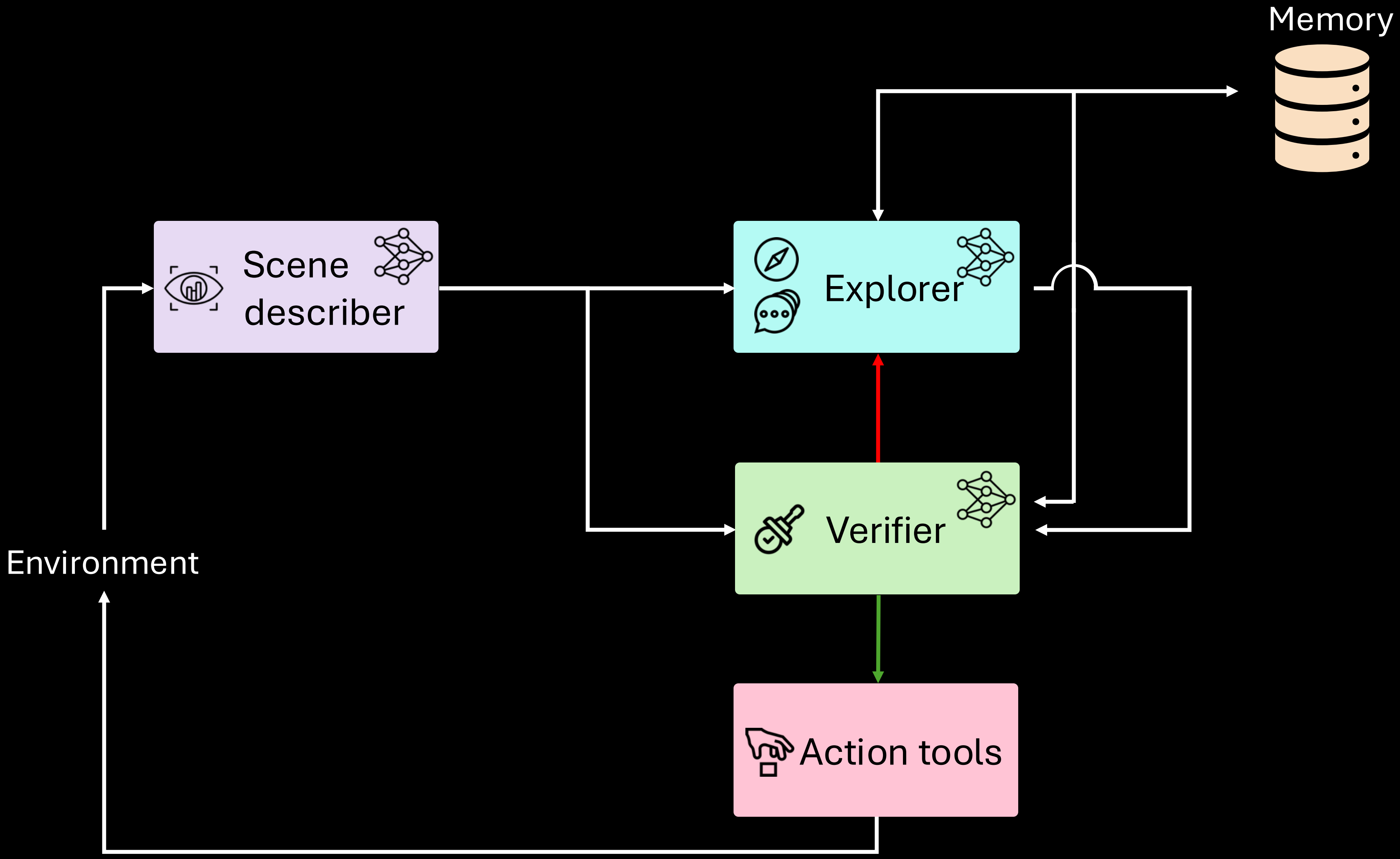
IVE

Observation o_{t+1}



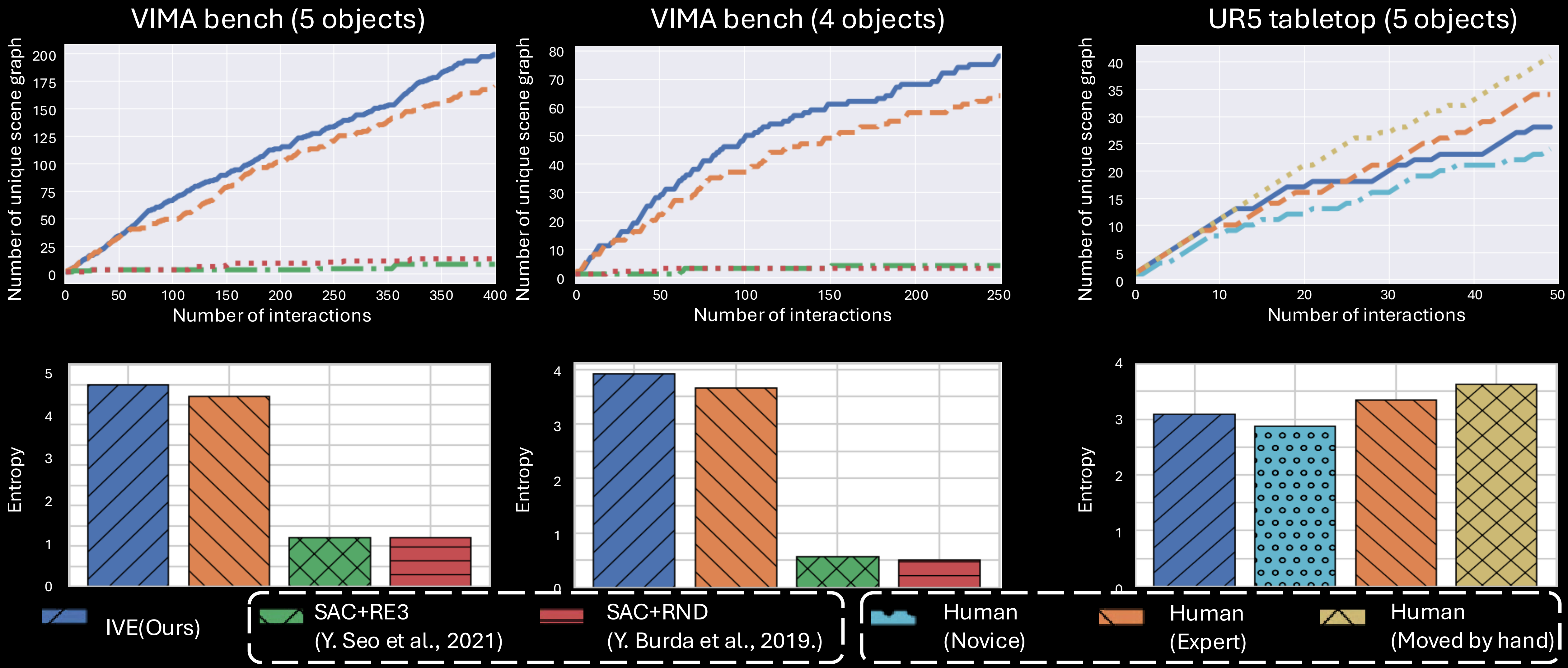
Action tools

IVE

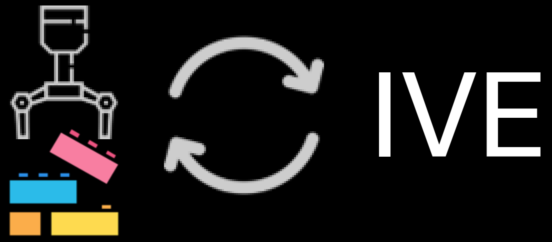


Memory
Scene describer
Explorer
Verifier
Action tools
Environment

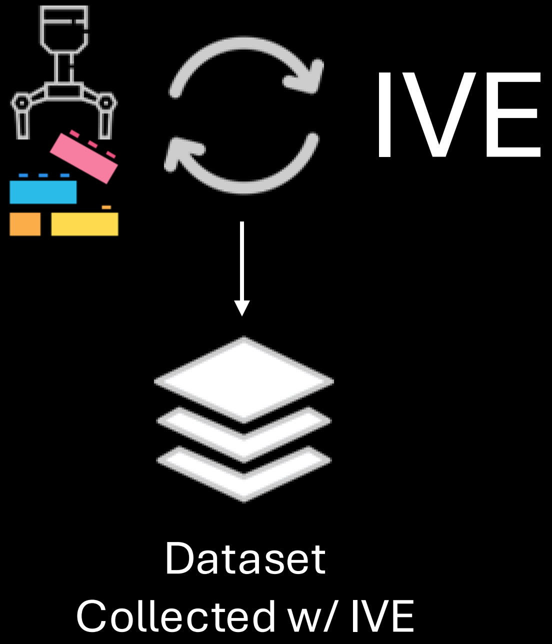
IVE – Exploration Scores



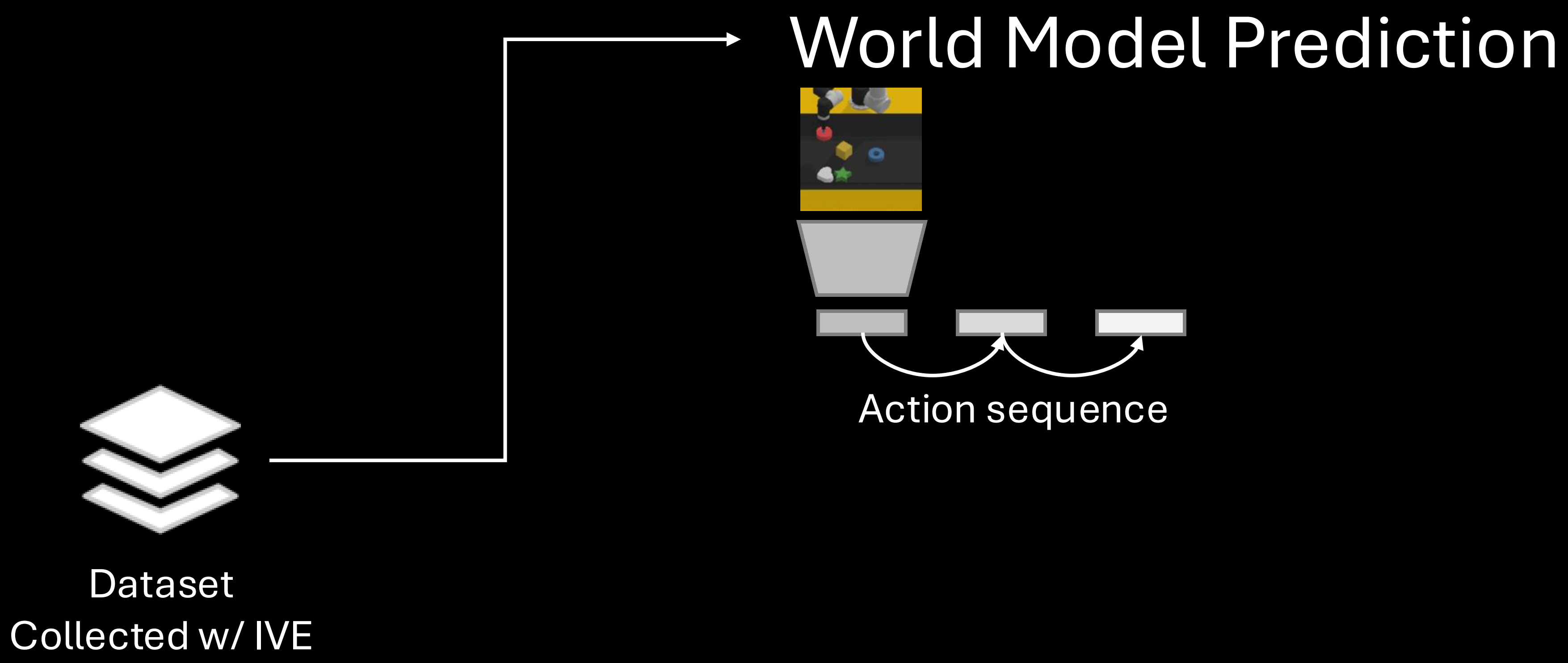
IVE – Downstream tasks



IVE – Downstream tasks

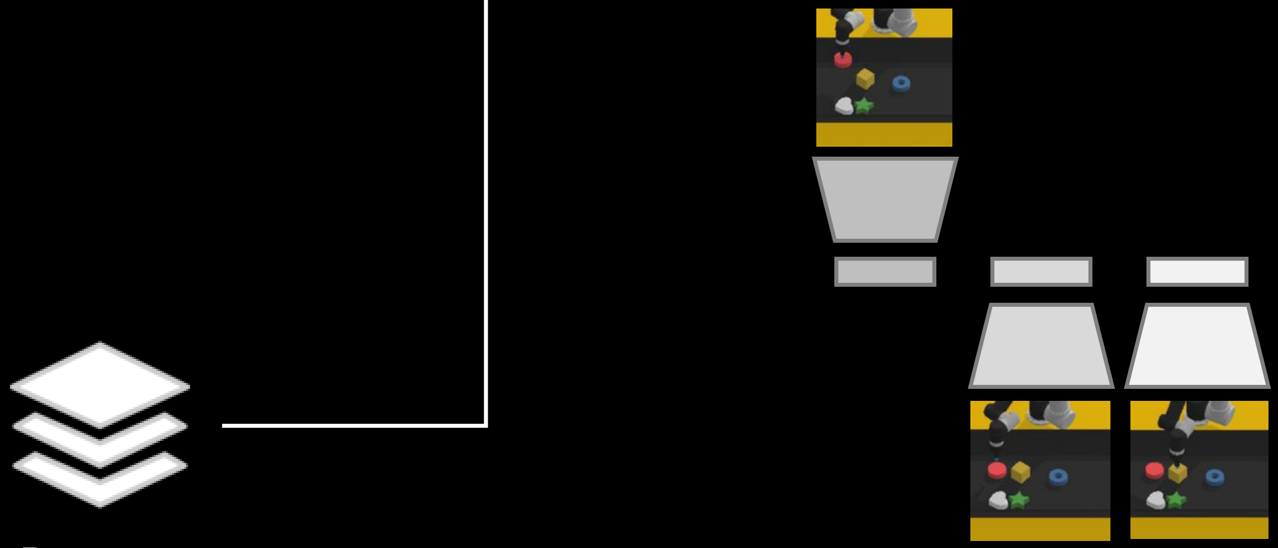


IVE – Downstream tasks



IVE – Downstream tasks

World Model Prediction



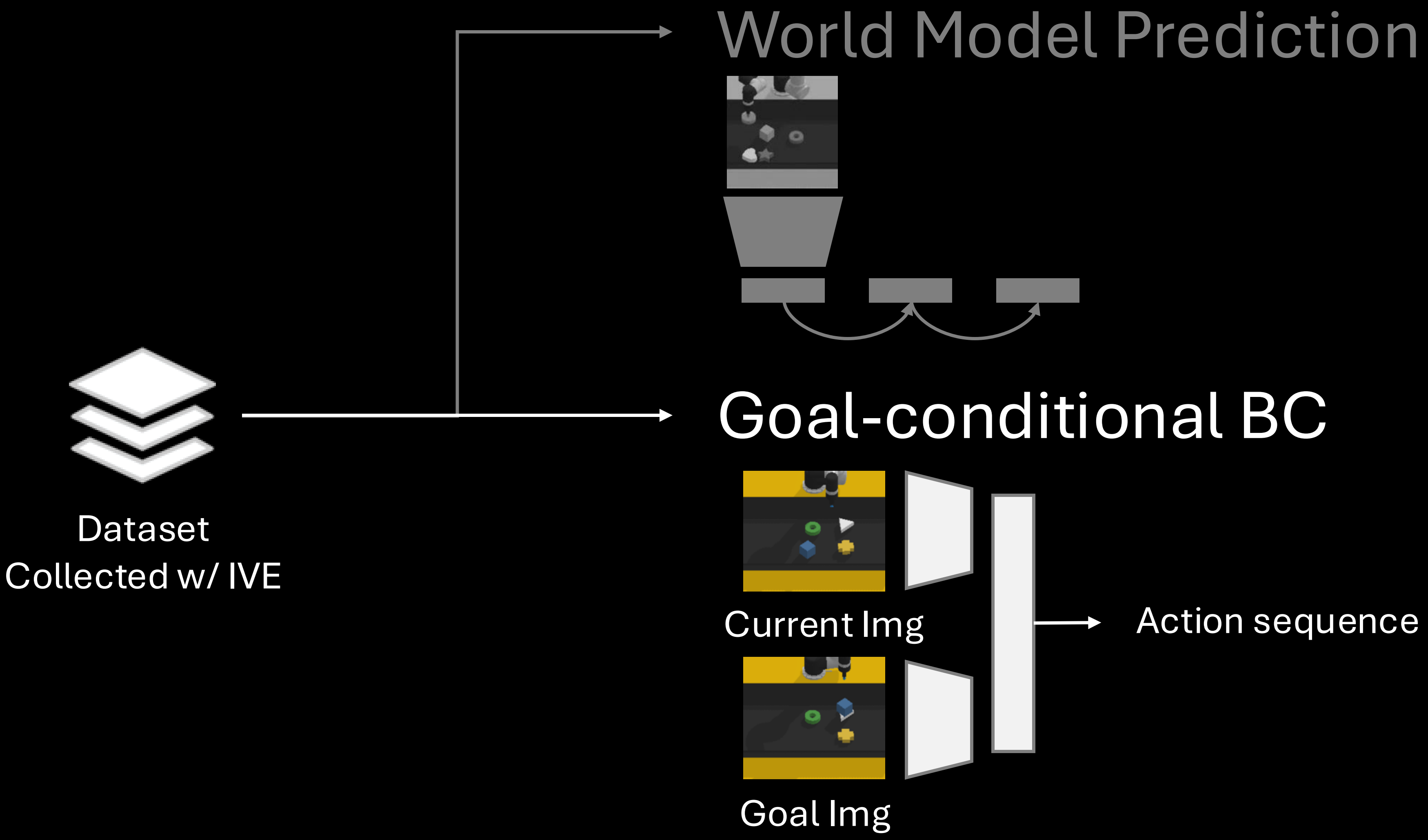
Dataset
Collected w/ IVE

IVE – Downstream tasks

World Model Prediction

	VIMA Bench (5 objects)		VIMA Bench (4 objects)	
	SSIM(↑)	LPIPS(↓)	SSIM(↑)	LPIPS(↓)
SAC+RND (Y. Burda et al., 2019.)	0.812 ± 0.039	0.198 ± 0.060	0.855 ± 0.036	0.168 ± 0.061
SAC+RE3 (Y. Seo et al., 2021)	0.814 ± 0.040	0.199 ± 0.057	0.850 ± 0.034	0.168 ± 0.059
IVE(Ours)	**0.837 ± 0.032**	0.129 ± 0.044	0.853 ± 0.032	0.160 ± 0.058
Human	0.833 ± 0.032	**0.126 ± 0.042**	**0.862 ± 0.027**	**0.139 ± 0.047**

IVE – Downstream tasks



IVE – Downstream tasks

Goal-conditional BC

	VIMA Bench (5 objects)	VIMA Bench (4 objects)
SAC+RND (Y. Burda et al., 2019.)	8.33%	0.00%
SAC+RE3 (Y. Seo et al., 2021)	0.00%	0.00%
IVE(Ours)	**58.33%**	**41.67%**
Human	50.00%	33.33%

Now

The future?

Deal with the "small-data" problems via

World-Models

Low-quality Cross-Embodiment Alignment

In-the-wild Behavioral Data

Synthetic Data and Self-Improvement

Blend world-model objectives ➢ Smart [ICLR23] ➢ Taco [NeurIPS23] ➢ Premier-TACO [ICML24] into VLAs

Blend Self-improving RLAIF ➢ SAIL [ICMLw24] ➢ SIMA [NAACL25] ➢ VisVM [ICLRw25] into VLAs

Enhance the brain VLM via

Enhancing spatial-temporal reasoning capabilities

Enabling Function and Interaction Aware reasoning capabilities

Enable function and interaction aware scene-graph for autonomous exploration & planning ➢ IVE [corl25]

1. **[TACO]** Zheng, Ruijie, Xiyao Wang, Yanchao Sun, Shuang Ma, Jieyu Zhao, Huazhe Xu, Hal Daume III, and Furong Huang. “TACO: Temporal Latent Action-Driven Contrastive Loss for Visual Reinforcement Learning.” The Thirty-seventh Annual Conference on Neural Information Processing Systems (NeurIPS) 2023.
2. **[Premier-TACO]** Zheng, Ruijie, Yongyuan Liang, Xiyao Wang, Shuang Ma, Hal Daume III, Huazhe Xu, John Langford, Praveen Palanisamy, Kalyan Shankar Basu, and Furong Huang. “Premier-TACO is a Few-Shot Policy Learner: Pretraining Multitask Representation via Temporal Action- Driven Contrastive Loss.” Proceedings of the 41st International Conference on Machine Learning (ICML) 2024.
3. **[FLARE]** Ruijie Zheng, Jing Wang, Scott Reed, Johan Bjorck, Yu Fang, Fengyuan Hu, Joel Jang, Kaushil Kundalia, Zongyu Lin, Loic Magne, Avnish Narayan, You Liang Tan, Guanzhi Wang, Qi Wang, Jiannan Xiang, Yinzhen Xu, Seonghyeon Ye, Jan Kautz, Furong Huang, Yuke Zhu, Linxi Fan. “FLARE: Robot Learning with Implicit World Modeling.” CoRL 2025.
4. **[TraceVLA]** Ruijie Zheng, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé III, Andrey Kolobov, Furong Huang, Jianwei Yang. “TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies.” The Thirteenth International Conference on Learning Representations (ICLR), 2025.
5. **[IVE]** Seungjae Lee, Daniel Ekpo, Haowen Liu, Furong Huang, Abhinav Shrivastava, Jia-Bin Huang. “Imagine, Verify, Execute: Memory-Guided Agentic Exploration with Vision-Language Models.” CoRL 2025.